A New Approach to Fiscal Multipliers: Time Variation and Highly Disaggregated Shocks

Valeria Gargiulo* Atsushi Inoue† Barbara Rossi‡

November 11, 2026

Abstract

What are the effects of fiscal shocks on the economy, and do they change over time? We show that recent advances in data collection at very disaggregated, high-frequency levels can help answer this question. This paper makes two main contributions to the literature. The first contribution is the construction of a new dataset of highly disaggregated U.S. government spending shocks, available at the most disaggregated county level at several frequencies (including monthly, quarterly, and yearly), as well as at daily frequencies at the aggregate national level. The second contribution is to shed light on the time-varying effects of fiscal policy shocks at a highly disaggregated level by leveraging the richness of the new dataset and a novel econometric approach. We find that the COVID-19 pandemic was associated with unprecedented challenges in the ability of governments to stimulate the economy: economic variables significantly respond to government spending shocks before the pandemic, while there are no significant responses during the pandemic. The fiscal stimulus that occurred before the pandemic had persistent effects over time, suggesting that the recession in those years could have been even more severe without it.

J.E.L. Codes: C22, C26, C32, C36, C53

Keywords: Time Variation, Panel Local Projections, Instability, Fiscal Policy.

Acknowledgments: We thank E. Castelnuovo, D. Debortoli, J. Galí, D. Giannone as well as seminar participants at the Barcelona Summer Forum, Bank of Italy, the Dolomiti Macro Meetings, the 5th WE_ARE_IN Macroeconomics and Finance conference, the EC^2 conference and CREI for comments.

*Department of Economics, Universitat Pompeu Fabra. E-mail: valeria.gargiulo@upf.edu.

†Department of Economics, Vanderbilt University. 2301 Vanderbilt Place, Nashville, TN 37235, USA. E-mail: atsushi.inoue@vanderbilt.edu.

‡European University Institute and ICREA–Universitat Pompeu Fabra (on leave), Barcelona School of Economics and CREI (on leave). E-mail: barbara.rossi@eui.eu.

1 Introduction

What are the effects of fiscal shocks on the economy, and do they change over time? We build on recent advances in data collection at very disaggregated geographical and high-frequency levels to shed light on this question.[1]

This paper makes two contributions to the literature. Our first contribution is the construction of a novel dataset of highly disaggregated U.S. government spending shocks. The disaggregation is two-fold: at the geographical level, focusing on county-level data; and at the frequency level, focusing on frequencies as high as daily. We construct a dataset of daily defense procurement contracts available at the county level. From this database, we identify two fiscal shocks, each exploiting a different dimension of disaggregation: a county-level shock, which leverages geographical disaggregation and is available at the monthly, quarterly, and yearly frequencies; and a daily shock, which leverages high-frequency disaggregation and is available at the aggregate national level. Our second contribution is to use these novel shocks to shed light on the time-varying effects of fiscal policy. We perform two separate analyses: the main analysis estimates time-varying county-level fiscal multipliers exploiting information from the county-level shock and using a novel econometric approach; the second analysis studies the effects of the daily national shock on real activity at the highest available frequency.

Our first contribution, namely the novel dataset of fiscal policy shocks, contributes to the literature in several ways. While the database of defense procurement contracts we utilize has already been considered in the literature, a first strand of the literature has mainly exploited its aggregate features. This literature shows that defense procurement data may have appealing features in quarterly, aggregate U.S. data. For example, Briganti and Sellemi (2023) show that *aggregate* defense procurement data track the timing of government spending in aggregate U.S. data more accurately than government spending based on NIPA data, due to the effect on business inventories[2]. Furthermore, Briganti (2023) shows that shocks estimated from *aggregate* procurement data are a valid instrument for aggregate government spending because firm-level production occurs only after the procurement contract has been awarded. Unlike this literature, we use procurement contracts to identify *disaggregated* fiscal shocks, both geographically (at the county level), and in time (at the daily frequency).

A second strand of the literature has instead focused on exploiting the cross-sectional richness of the

[1] See Ramey (2011) for a recent, broad overview of the literature on the effects of fiscal policy shocks.

[2] Briganti and Sellemi (2023) find that the immediate government spending increase during a military build-up is due to an increase in inventories, rather than actual production. NIPA accounting includes inventories, while procurement data record when the contract is signed, before production starts; thus, the latter are more accurate measures of government spending. Briganti and Sellemi (2023) also find a delayed effect of government spending shocks when using procurement contracts data.

defense procurement contracts dataset. For example, Nakamura and Steinsson (2014) measure the effects of fiscal policy at the state level. In contrast, we focus on more disaggregated data at the county level; importantly, we exploit the geographic granularity of our data to study whether and how the multiplier varies over time. Auerbach et al. (2020) measure the effects of increases in local defense spending at the city level, and Auerbach et al. (2022) at the state level during the COVID-19 recession. Relative to this literature, we work at the county level, allow the multiplier to vary year by year, and identify a fiscal policy shock rather than using raw increases in government spending. Identifying the shock at the county level also changes the source of the identifying variation: whereas the shift-share design of Nakamura and Steinsson (2014) exploits states' heterogeneous exposure to an aggregate shock, our shock identifies changes in local county-level spending that are orthogonal to lagged national spending and output, thus isolating idiosyncratic county-level procurement innovations. Cruicini and Vu (2021) instead study how common economic shocks and local, county-specific shocks affect county-level wage income dynamics during a specific fiscal shock episode, namely the American Recovery and Reinvestment Act; relative to them, we study a broader question, namely what are the effects of fiscal shocks in general and how the fiscal multiplier varies over time.

A third strand of the literature focuses on estimating the effects of government spending in high-frequency data and parallels contributions in monetary policy analysis (e.g. Gurkaynak and Wright (2013) and Buda et al. (2025), among others). Like this literature, our daily shock is defined at the aggregate, national level. We are not the first to exploit the daily frequency feature of procurement data: Auerbach and Gorodnichenko (2016) study the effects of daily defense contract announcements on exchange rates and other financial variables, and Auerbach et al. (2020) on credit markets. We instead use the universe of daily contract obligations to *identify* a fiscal policy shock, removing the institutional end-of-fiscal-year surge that generic seasonal adjustment leaves in the data; and we validate the econometric properties of our shock by showing that it is uncorrelated with identified monetary policy and central bank information shocks.

Our second contribution to the literature concerns the analysis of time-varying effects of fiscal policy (e.g., Ramey and Zubairy (2018) focus on state-dependent models). In our main analysis, we focus on the effects of government spending at the most disaggregated geographical level, using the panel of county-level fiscal shocks estimated from our novel dataset. The rich cross-sectional dimension of our dataset allows us to shed light on time variation even in very short time periods, such as the COVID-19 pandemic. Indeed, we find positive and statistically significant multipliers for the fiscal shocks of 2018 and 2019 – the one-year cumulative multiplier of the 2019 shock is 0.52 – whose effects on output persist into the pandemic years,

while multipliers for shocks occurring during the pandemic are essentially zero, and those for the 2022 shock are positive but imprecisely estimated. Unemployment tells the same story: the 2018 shock lowers county unemployment significantly one and three years ahead, whereas the pandemic-year shocks have no significant effect. This evidence suggests that the economic slowdown during the pandemic could have been even more severe without the fiscal stimulus over the previous years. Two additional findings underscore the importance of allowing for time variation. First, a constant-parameter panel local projection estimated on the same data delivers responses that are small and insignificant at every horizon, because it averages the positive pre-pandemic responses with the zero or negative responses during the pandemic. Second, the strength of our instrument is itself time-varying: the test of Lewis and Mertens (2022) rejects weak identification in the pre-pandemic years but not in 2020–2022: this implies that the small responses during the pandemic should be interpreted with caution, as they are consistent either with a genuinely small multiplier and with a low signal-to-noise ratio of procurement shocks during those years.

Our estimation relies on a novel Time-Varying Parameter Panel Local Projection (TVP-PLP) estimator. Since the seminal paper Jordà (2005), local projections (LP) have become a widely used empirical tool for estimating the response of economic variables to economic shocks. The recent literature has emphasized the versatility of local projections in estimating panel regressions (Jordà and Taylor, 2025) and, in parallel, several advances have been made in understanding the econometric properties of local projections in standard regressions (see Stock and Watson (2018), Plagborg-Møller and Wolf (2021), Plagborg-Møller and Wolf (2022)) and, more recently, in non-linear models (Gonçalves et al. (2024a), Gonçalves et al. (2024b)) as well as panel models (Dube et al. (2023)). Time variation can be seen as a source of non-linearity; Gonçalves et al. (2024b), Gonçalves et al. (2021), Gonçalves et al. (2024a) and David et al. (2026) point out difficulties in the estimation of panel local projections in the presence of non-linearities due to state-dependence; our approach does not suffer from their criticisms because it splices the panel dimension at each point in time. More broadly, the panel dimension may offer several advantages when estimating local projections, such as the additional information available from a large cross-section of data over time. The goal of this paper is to take advantage of the versatility of local projections in panels and to exploit cross-sectional variation to identify potentially time-varying local projection estimates. Although time variation in the parameters has been considered in local projections (e.g. Inoue et al. (2024)) and instrumental variable estimation (e.g. Inoue et al. (2025)), to our knowledge this is the first contribution to use the rich information in large cross-sectional panel datasets to identify and estimate time-varying parameters. Our estimator has several advantages, including the ability to estimate general forms of time variation and strong performance even

when the time-series dimension is small, provided the cross-section dimension is sufficiently large.

In addition, our paper relates to the growing literature on estimating panel local projections, including Almuzara and Sancibrian (2024), Cloyne et al. (2020) and Dube et al. (2023). Unlike Dube et al. (2023), which uses difference-in-differences to estimate causal effects, we focus on impulse responses estimated in standard panel local projections. Almuzara and Sancibrian (2024) estimate the effects of macro (time series) shocks on micro (panel) outcomes in a constant parameter environment, while we construct micro (panel) shocks and allow for time variation.[3] Cloyne et al. (2020) introduce time variation in panel local projections via interaction terms. In their approach, any time variation is linked to a known source, whereas our approach does not require such knowledge. Time-varying cross-section regressions similar to the ones we consider also appear in the literature on international risk sharing (e.g. Giannone and Reichlin (2006), Sørensen et al. (2007)); our paper provides a theoretical foundation for such approaches. Finally, alternative approaches to study the heterogeneity of fiscal responses at a disaggregated geographical level include modeling them using group structures, as in Florens and Simoni (2025), Bianco et al. (2026) or Inoue et al. (2026); relative to these contributions, our objective is to use the cross-section dimension to estimate time variation in a robust fashion, rather than identifying group structures.

Our treatment of aggregate conditions is best understood in light of the literature on what cross-sectional estimates identify. When the researcher is interested in estimating the aggregate fiscal multiplier, the challenge is that cross-sectional designs deliver a *relative* multiplier rather than the *aggregate* one, because time effects difference out the aggregate general-equilibrium response: this issue is the so-called "missing intercept" problem discussed by Wolf (2023); Guren et al. (2021) and Chodorow-Reich (2019) discuss how relative and aggregate fiscal multipliers are related, and Matthes et al. (2025) propose to recover the aggregate multiplier by combining cross-sectional estimates with an identified aggregate shock. Our objective is complementary: Rather than recovering the missing intercept, we focus on identifying the relative multiplier from idiosyncratic county variation alone. When the researcher is interested in estimating the relative fiscal multiplier, the challenge instead is that, as Donaldson (2026) shows, cross-sectional designs that exploit heterogeneous exposure to aggregate shocks fail to identify partial-equilibrium elasticities when exposure to the shock of interest is correlated with exposure to other aggregate variables that move in general equilibrium; this issue arises because heterogeneous exposure survives time fixed effects. We purge all the variables entering the estimation of a common component with county-specific loadings, estimated over the full sample, exactly to

[3]We also relate to the literature on estimating time-varying local projections, in particular the works by Ruisi (2019) and Inoue et al. (2024). Ruisi (2019) estimates time-varying local projections via Bayesian methods while Inoue et al. (2024) estimate time-varying local projections via frequentist methods with optimal properties. While we also allow our parameters to vary over time, unlike these works, we focus on panel local projections.

avoid the Donaldson (2026) problem. This is where identifying the shock at the county level in our approach pays off: our disaggregated instrument carries virtually no common component to begin with.

Our second empirical analysis focuses instead on the high-frequency effects of aggregate fiscal policy. We study the effects of fiscal policy on real business conditions, as measured by the daily index of Aruoba et al. (2009). We show that daily data can uncover interesting time variation in the responses to fiscal shocks even in aggregate, national data. Over the full 2007–2024 sample, the response is positive but insignificant; in the 2017–2024 sub-sample, it is significant between roughly one and a half and three months after the shock. When looking at the two sub-samples characterized by high unemployment, as in Ramey and Zubairy (2018), namely the Great Financial Crisis and COVID-19, we observe different responses to a fiscal stimulus: during the financial crisis the response is significant only between two weeks and about one month after the shock, whereas during the pandemic it becomes significant about ten days after the shock and remains so over the whole horizon we consider. These results suggest that a state-dependent model is not suited to study a period characterized by a shock unrelated to the state of the economy, such as COVID-19. We also show that using our identified shock, rather than raw daily changes in government spending, matters: the variation that a regression on raw daily spending exploits is predictable and dominated by the end-of-the-year surge; using raw daily changes rather than our identified shock leaves the sign and timing of the estimated responses unchanged but more than halves their magnitude and makes them imprecisely estimated.

The paper is organized as follows. The next section discusses the novel county-level fiscal spending dataset, available at several frequencies, and introduces our two identified fiscal shocks: the yearly county-level shock and the daily national shock. Section 3 then describes the empirical methodology using a simple motivating example. Section 4 presents our main application, the time variation in U.S. county-level fiscal multipliers for both output and unemployment, while Section 5 presents the second application, the effects of the daily national shock at high frequencies. Section 6 concludes. The appendix contains detailed results for alternative empirical specifications, as well as the theoretical background of the methodology, including several Monte Carlo simulations.

2 Disaggregated Fiscal Policy Shock Datasets

In this section, we focus on our first contribution, namely our novel dataset of fiscal spending shocks at the county level. The fiscal shock is constructed at the monthly, quarterly, and yearly frequencies at the most geographically disaggregated (county) level, as well as at the daily frequency at the aggregate national level.

2.1 Department of Defense Procurement Data Description

The goal of this paper is to analyze the effects of fiscal spending shocks at the most disaggregated level. Therefore, in order to identify our novel county-level military spending shock, we first need to construct a panel dataset containing information about county-level military spending.

We collect data on military spending from the Federal Procurement Data System (FPDS), which is an official source of the US government. The FPDS database contains detailed information on contracts awarded by the U.S. Department of Defense (DoD) to private firms,[4] such as: the date in which the contract was signed; its dollar value and the principal place of performance of the action agreed in the contract;[5] the estimated/actual completion date; as well as details about the non-governmental party involved. The FPDS also contains a column associated to "Contract Modifications", meaning a contract that modifies the terms of a pre-existing one, which add considerable noise to the data. In our analysis, we only select contracts that are not modifications, both in order to eliminate the noise and to ensure that we do not account for contracts outside the time period we consider. Therefore, we exclude situations in which the original contract is not in our time sample but a later modification to the same contract is, due to its timing. By doing so, we also make sure that the entries that we consider only refer to non-negative transactions.

We are not the first to use DoD contracts to measure shocks to fiscal spending, although our dataset is unique because it collects information on DoD spending at the most disaggregated level and at the highest frequency. As discussed in Briganti (2023) and Briganti and Sellemi (2023), there are several advantages in using defense contracts to measure fiscal spending at the national, aggregate level. Namely, defense contracts measure fiscal spending at the time the fiscal expense is committed, whereas traditional measures (based on the National Income and Product Accounts) record defense spending only after the payment has occurred, which happens after delivery, and hence potentially a long time after the contract has been signed, or by tapping into inventories (rather than fiscal spending). For these reasons, they can be considered a better and more timely measure of fiscal spending, and hence produce a better instrument. However, differently from them, we focus on highly disaggregated data. The strength of the estimated shock as an instrument is assessed, year by year and horizon by horizon, with the test of Lewis and Mertens (2022) in Section 4.4 for our main specification.[6] Following the above reasoning, we refer to the date in which the contracts have been signed to retrieve the correct timing. This is especially important in the construction of the daily national

[4]The dataset starts on January 1st, 1967 with the first contract signed. Note that for the year 1967 this is the only recorded contract, with no other entries (or entries with no dollar value) until the year 1970.

[5]While it is possible that the execution of the work is distributed across different locations, the FPDS only provides information about the location in which the majority of it was performed.

[6]See Appendix B.3 for robustness analyses.

spending shock, discussed in the following sub-section.

To verify the validity of the dataset, we aggregate our data at the yearly frequency and compare the yearly total dollar size of the contracts awarded from the FPDS with the aggregated yearly spending reported by Nakamura and Steinsson (2014)[7]. The analysis shows that the FPDS is not as exhaustive as the Nakamura and Steinsson's (2014) data for the first few years of their sample. The total dollar amount of the transactions is smaller in the former, but it starts to be of comparable magnitude in 1982. However, from the early 2000s, the total FPDS awarded dollars exceeded those reported in Nakamura and Steinsson (2014). It seems that the source they used has become outdated and that the same type of reports are now only accessible in the FPDS. For these reasons, we collected data from the FPDS starting in 1982. The data collection is performed manually.[8]

Importantly, to identify the location of the fiscal shock, which is crucial at the level of disaggregation (county-level) we focus on, we refer to the "principal place of performance", which is not necessarily the legal address of the private company providing the service to the DoD, but rather where the majority of the work took place. The FPDS dataset provides such information; in particular, it displays the associated state codes throughout the sample, and the county name starting in mid 2004.

Our goal is to construct a dataset that is balanced over the estimation period, and characterized by the largest possible cross-sectional dimension at the county level. Therefore, we verify at which point in time the FPDS starts reporting the variable "Principal Place of Performance: County" in a consistent manner for the vast majority of contracts awarded. We perform this exercise by randomly selecting time periods for every year starting in 2000, checking the number of contracts awarded over that period, and computing the fraction of those that display the information we are interested in. We observe a consistent reporting of the variable "Principal Place of Performance: County" starting in 2007, which is the year we choose as the starting point of our sample. The data thus collected are then used in our empirical analysis for the construction of: (i) the daily national spending shock; (ii) the county-level spending shock; and (iii) as the government spending (panel) variable.

Our approach is different from Nakamura and Steinsson (2014), who use state-level military spending in the US to compute a measure of the fiscal multiplier. In particular, they collect military spending data from the electronic database of DD-350 forms available from the US Department of Defense. These are forms that document military purchases of any type and provide information on the location where the majority

[7]Their national spending is computed simply as the sum of the state-level spending.

[8]The contract data can be downloaded via spreadsheets with a maximum number of 150000 rows each; therefore, the time span needs to be selected such that the number of contracts awarded in it is at most 150000.

of the work was performed (state-level). However, the data they use are only available annually; it only covers the period 1966-2006; and it is available at the state-level. Unlike their measure, ours is available at the county-level and at higher frequencies than annual.

We also differ from Auerbach et al. (2020) and Auerbach et al. (2022), who work at the level of core-based statistical areas. We focus instead on counties as our cross-sectional unit, for three reasons. Counties partition the entire territory, whereas statistical areas cover metropolitan regions only, and a sizable share of defense procurement is performed at installations and plants located in non-metropolitan counties. Counties are also the finest geographical unit at which the FPDS place-of-performance variable and our outcome variables – BEA county GDP and county unemployment – are consistently available at the yearly frequency, allowing the construction of a balanced panel; in addition, county boundaries are fixed over our sample, whereas statistical-area definitions are periodically revised. Finally, our estimator identifies the multiplier from a single cross-section in each year, so the size of the cross-section matters directly for precision: we have 2,158 counties in every year, against fewer than four hundred statistical areas. The cost is that a county is smaller than a local labor market, which gives more scope for spillovers across neighboring counties; the relative multiplier we estimate nets out the part of the response that spills over to other counties.

2.2 The County-level Fiscal Spending Shock

The dataset discussed above contains an abundance of information, both in terms of frequency and geographical disaggregation. In fact, the highest frequency is daily, and the highest level of disaggregation is at the county-level, which we exploit in what follows.

We estimate the county-level fiscal policy shock ($\varepsilon_{i,t}^{F,X}$) as the residual of the following regression:

$$G_{i,t} = \alpha_{g,i} + \sum_{j=1}^{p_i^*} \beta_j G_{i,t-j} + \gamma_i G_{t-1} + \delta_i X_{t-1} + \varepsilon_{i,t}^{F,X} \tag{1}$$

where $G_{i,t}$ is the real, per-capita fiscal expenditure at the county level and G_{t-1} is the real, per-capita aggregate fiscal expenditure at the national level calculated as $G_{t-1} = \sum_{i=1}^{n} G_{i,t-1}$. The real, per-capita variables are obtained by dividing the original series by the national GDP deflator, and by the county and national population, depending on the nature of the series. That is, we estimate the fiscal shock to real, per capita, defense spending. The county-level population data are from the Bureau of Economic Analysis (BEA): Their population series is only available annually, covering the entire time span also covered by the

county-level DoD spending variable mentioned above.[9] The GDP deflator is the national-level series from FRED. The fiscal shock in equation (1) is estimated over 2008–2023, with the starting year determined by the availability of the data in the FPDS dataset, as discussed in Section 2.1. In the regression, we also control for real, per-capita aggregate GDP, denoted by X_t, downloaded from FRED. The county-specific number of lags in the autoregressive specification (p_i^*) is selected by the Bayesian Information Criterion.

Note that the regression above can be estimated at the highest possible frequency, that is, daily, thus producing a daily series of county-specific fiscal policy shocks. However, in many macroeconomic applications, we are interested in the effects of policy shocks at the monthly or quarterly frequency; therefore, we construct monthly, quarterly, and yearly times series of the same county-level fiscal shocks. To improve the precision of our estimated shocks and avoid aggregation biases, each of these shocks is built separately starting from the raw FPDS data: we obtain $G_{i,t}$ by aggregating the contract dollar value by county and frequency of choice. For the analysis in Section 4, we focus on the yearly shock to match the data availability of other variables of interest at the county level, such as the GDP.

Identifying the fiscal shock at such a disaggregated level turns out to be extremely useful for our analysis of time-varying fiscal multipliers. Having a panel of shocks not only allows our estimates to vary in time by exploiting the cross-sectional dimension, but also implies that we avoid a widespread identification challenge associated with estimating fiscal multipliers in a panel setup, recently pointed out by Donaldson (2026). Donaldson (2026) highlights that the common approach of estimating elasticities via cross-sectional designs that exploit heterogeneous exposure to aggregate shocks while controlling for time fixed effects may lead to biased estimates. In our context, using an aggregate fiscal shock interacted with county exposures would expose cross-sectional fiscal multiplier estimates to this bias. Our disaggregated shocks mitigate this problem, because equation (1) nets out, county by county, the response of local spending to lagged national spending and output. It does not, however, eliminate it by construction: a contemporaneous common component of county spending may remain in the residual, and heterogeneous county exposure to the general-equilibrium effects associated with it could still bias the estimates. We address this directly in Section 4.2, where we remove a common component with county-specific loadings from all the variables entering the estimation, as well as in Appendix B.4, where we state the condition under which the design is robust to the critique of Donaldson (2026) and show that purging the instrument alone leaves the estimates unchanged.

The shock identification we follow in eq. (1) is inspired by the residual-based approach of Blanchard and Perotti (2002) and Briganti (2023). We differ from their approach mainly because of the granularity of our

[9] We compute total national population by summing county-level population.

data. In fact, we estimate our fiscal policy shock using a univariate equation at the county-level, including aggregate fiscal spending at the national level as a control variable. Our estimation differs from Blanchard and Perotti (2002), who estimate the fiscal policy shock using a recursively-identified VAR, where government spending is ordered before GDP. The reasons why we use a univariate regression instead of a VAR are two: first, we want to fully exploit the granularity of our data and produce a county-level fiscal shock, since other macroeconomic variables are not available at the county-level; second, Blanchard and Perotti (2002) require quarterly data to ensure that their VAR-based identifying restriction holds, but county-level data are only available annually.

2.3 The Daily National Spending Shock

In this sub-section, we construct a novel time series of daily national government spending shocks. A natural starting point is to define the shock as the residual of an autoregression of the daily national military spending series. Federal procurement, however, has a pronounced end-of-fiscal-year seasonality: because the U.S. federal fiscal year ends on September 30, federal fiscal agencies whose budget is expiring in the final days of September have the incentive to spend any remaining budget, thus generating a large and predictable "use-it-or-lose-it" surge.[10] A plain autoregression cannot absorb this calendar pattern, so the surge would pass into the residual and contaminate the shock with an anticipated component. We therefore construct the shock as the residual of an autoregression of the daily national military spending series (which we label g_t^F below) that also controls for a full set of end-of-fiscal-year daily dummy indicators, where the number of autoregressive lags is determined by the Bayesian Information Criterion (BIC) and equals 28:[11]

$$g_t^F = \alpha_g + \sum_{j=1}^{28} \phi_j g_{t-j}^F + \sum_{d=19}^{30} \psi_d \, D_{d,t} + \epsilon_t^F, \tag{2}$$

where $D_{d,t}$ is an indicator equal to one when date t is the d-th day of September (for $d = 19, \ldots, 30$) and zero otherwise. By construction, the resulting shock is orthogonal to the average end-of-fiscal-year seasonality: a regression of the shock on the dummy indicators has an R^2 of essentially zero, versus 0.14 for the residual of the plain autoregression; thus, no predictable end-of-the-year component remains.[12] The

[10] This end-of-fiscal-year spending surge is well documented (see Liebman and Mahoney, 2017). In our data, obligations over September 19–30 account for roughly 11% of annual defense obligations, against about 3% under a uniform calendar, and the daily total rises monotonically to a peak on September 30.

[11] The maximum number of lags in the BIC is 30. The BIC selects 28 lags both with and without the dummy indicators.

[12] Equivalently, the mean of the plain autoregression residual over September 19–30 is +\$0.84 billion, whereas the mean of ϵ_t^F over the same window is zero by construction. We show in Appendix B.1 that a coarser control based on a single end-of-fiscal-year dummy leaves part of this seasonality in the shock and yields less precise estimates; the day-specific control removes it entirely.

procedure discussed above ensures that the shocks we identify reflect genuinely unanticipated fiscal variation rather than the routine, calendar-driven end-of-the-fiscal-year surge.

The shock is depicted in Figure 1. The shock still includes very large events; the highest spike is on 29/09/2008, recorded during the financial crisis of 2007–2009, only a couple of weeks after the bankruptcy of Lehman Brothers. Removing the average end-of-the-year profile leaves such genuinely anomalous days largely intact (the September 2008 spike, a deviation from the typical end-of-fiscal-year level, remains the largest in the sample) while stripping out the routine seasonal component.

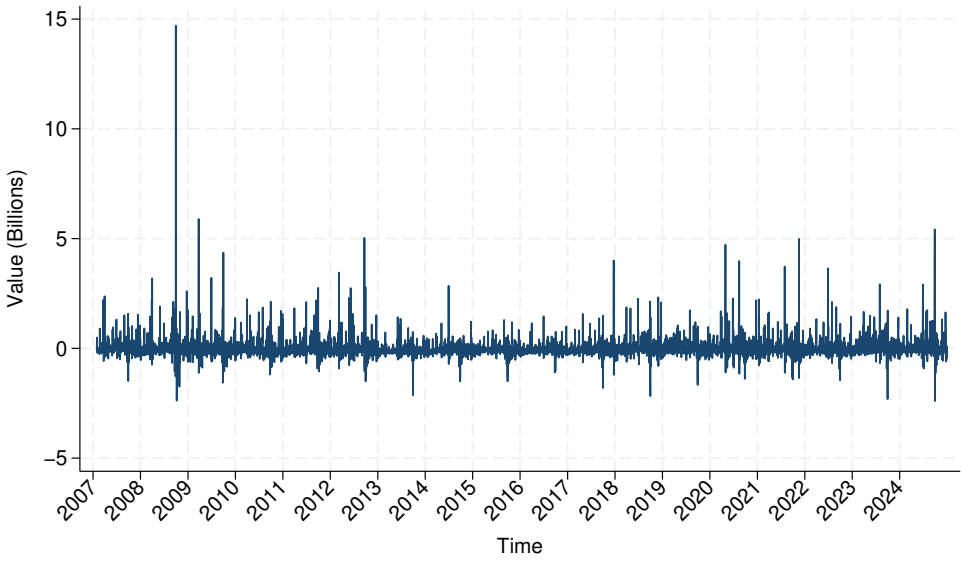


Figure 1: Daily national fiscal spending shock: 29/1/2007-31/12/2024.

We test the shock validity by estimating its correlation with other existing shock measures, in particular the monetary policy shock series provided by Jarociński and Karadi (2020) as well as their central bank information shock. If our shock is indeed a fiscal shock, it should be uncorrelated with any other economic shock. We verify that is the case by first aggregating our series to the same monthly frequency as in Jarociński and Karadi (2020), and then checking the pairwise correlations (and their significance) with all the shock series identified in their paper.[13] We find no significant correlation with any of these shocks.

[13]In unreported results, we consider both the monetary policy shock and Central Bank information shocks, estimated either via their poor man's approach and their fully Bayesian one.

3 Empirical Methodology

This section provides intuition behind the empirical method used in the paper using a simple, motivating example. Consider the following standard panel local projection regression (Jordà, 2020):

$$y_{i,t+h}^{(2)} = \alpha_i + \beta_h y_{i,t}^{(1)} + \gamma_h' \mathbf{w}_{i,t} + v_{i,t+h}, \tag{3}$$

where $i = 1, 2, ..., N$ denotes cross-sectional units, and $t = 1, 2, ..., T$ denotes time periods; h is the horizon; α_i is a unit-fixed effect; β_h is the response coefficient of interest; $y_{i,t}^{(1)}$ and $y_{i,t}^{(2)}$ are scalar variables for each cross section unit i observed at time t;[14] $\mathbf{w}_{i,t}$ is a vector of exogenous and pre-determined variables;[15] and $v_{i,t+h}$ is the regression residual. Note that in the standard panel local projection regression above the parameters are constant.

The main objective of this paper is to estimate a time-varying response of $y_{i,t+h}^{(2)}$ to a fiscal policy shock, $\epsilon_{i,t}^{(1)}$. We consider a panel local projection regression similar to eq. (3) except that we allow for time-varying parameters:

$$y_{i,t+h}^{(2)} = \alpha_i + \beta_{h,t+h} y_{i,t}^{(1)} + \gamma_{h,t+h}' \mathbf{w}_{i,t} + v_{i,t+h}. \tag{4}$$

Here and in the rest of the paper, $y_{i,t}^{(1)}$ denotes government spending, while $y_{i,t}^{(2)}$ denotes the macroeconomic outcome of interest, such as GDP or unemployment. Note that $\beta_{h,t+h}$ measures the effect on the dependent variable of an exogenous increase in $y_{i,t}^{(1)}$ at time t after h periods; that is, the time of the exogenous increase is obtained as the difference of the second and the first subscripts in $\beta_{h,t+h}$.

Local projections are challenging to estimate in short-panels due to the typical small sample *time* dimension available for *each* cross-section. However, cross-sectional information brings more power to estimate parameters at each point in time: we exploit exactly that dimension to identify time-varying effects.

Let $\tilde{x}$ denote least squares residuals from a regression of x on the conditioning variables $\mathbf{w}_{i,t}$. Without loss of generality, by the Frisch-Waugh-Lovell theorem, the researcher can equivalently estimate the parameter $\beta_{h,t+h}$ from the following regression:

$$\tilde{y}_{i,t+h}^{(2)} = \tilde{\alpha}_i + \beta_{h,t+h} \tilde{y}_{i,t}^{(1)} + \tilde{v}_{i,t+h}, \tag{5}$$

[14] The assumption that the regressor is a scalar allows us to focus on a simple motivating example. We consider the more general case with multiple instruments in the Appendix.

[15] Note that $\mathbf{w}_{i,t}$ may typically include one; in our time-varying specification, the intercept captures time fixed effects.

where $\tilde{y}_{i,t}^{(1)}$ is the pooled least square residual from regressing $y_{i,t}^{(1)}$ on $\mathbf{w}_{i,t}$.

We assume that only short panel data are available, i.e., T is fixed; however, the size of the cross-section dimension N is large. This assumption is realistic in practice[16].

We use external instrumental variables to identify structural responses (Stock and Watson (2018), Mertens and Ravn (2013)).[17] For simplicity, in this motivating example, we will assume that the instrument is scalar. Let $z_{i,t}$ denote an instrumental variable that satisfies the following assumptions: (i) $E(z_{i,t}\epsilon_{i,t}^{(1)}) \neq 0$ (relevance); (ii) $E(z_{i,t}\epsilon_{i,t}^{(k)}) = 0$ for $k \neq 1$ (contemporaneous exogeneity); (iii) $E(z_{i,t}\epsilon_{i,s}^{(k)}) = 0$ for every $t \neq s$ and every k (lead and lag exogeneity); (iv) $E(z_{i,t}c_i) = 0$ for every t (orthogonality); where $\epsilon_{i,t}^{(k)}, k = 1, ..., m$ are the underlying shocks generating the observable variables in the true moving average data generating process and $\epsilon_{i,t}^{(1)}$ is the fiscal policy shock of interest.[18]

The time-varying estimator of $\beta_{h,t+h}$ that we use is a cross-sectional instrumental variable estimator based on equation (5). In our scalar instrument case, our proposed Time-Varying Parameter Local Projection Instrumental Variable (TVP-LP-IV) estimator will be based on the cross-sectional average at each point in time t, $t = 1, ..., T$ as follows:[19]

$$\hat{\beta}_{h,t+h} = \frac{\frac{1}{N}\sum_{i=1}^{N} \tilde{z}_{i,t}\Delta\tilde{y}_{i,t+h}^{(2)}}{\frac{1}{N}\sum_{i=1}^{N} \tilde{z}_{i,t}\Delta\tilde{y}_{i,t}^{(1)}}, \tag{8}$$

where $\Delta\tilde{y}_{i,t}$ denotes $\tilde{y}_{i,t} - \tilde{y}_{i,t-1}$, which eliminates the unobserved heterogeneity. Note that, in accordance with current practice, our empirical analysis is performed using long differences (as, for example, eq. (21) below) and cumulative regressions (as, for example, the multiplier regressions in eq. (19) below), in which case the differencing approach is already implemented by construction.[20]

Note that, since the estimation exploits the cross-sectional dimension, the parameter process can be very general as long as we assume that its time evolution is deterministic. We refer the reader to Appendix C for more details about the theoretical setup and to Appendix D for the analysis of the estimator's performance

[16] For example, N=175 and T=50 in Dube et al. (2023); in our empirical application, T = 6 and N = 2158.

[17] As discussed in Section 2.2, the IV estimator is required because we cannot condition on infinitely many lags.

[18] The moving average data generating process is

$$y_{i,t}^{(1)} = \mu_1 + c_i^{(1)} + \epsilon_{i,t}^{(1)} + \Theta_{0,t}^{(1,2)}\epsilon_{i,t}^{(2)} + ... + \Theta_{h,t}^{(1,1)}\epsilon_{i,t-h}^{(1)} + \Theta_{h,t}^{(1,2)}\epsilon_{i,t-h}^{(2)} + ..., \tag{6}$$

$$y_{i,t+h}^{(2)} = \mu_2 + c_i^{(2)} + \Theta_{0,t+h}^{(2,1)}\epsilon_{i,t+h}^{(1)} + \Theta_{0,t+h}^{(2,2)}\epsilon_{i,t+h}^{(2)} + ... + \beta_{h,t+h}\epsilon_{i,t}^{(1)} + \Theta_{h,t+h}^{(2,2)}\epsilon_{i,t}^{(2)} + ... \tag{7}$$

where the shocks $\epsilon_{i,t}^{(k)}$ are martingale difference sequences over time, with zero mean and variance σ_k^2; are independent across i, for $k = 1, ..., m$; and $c_i^{(1)}$ and $c_i^{(2)}$ are the unobserved heterogeneity parameters. Note that we normalized $\Theta_{0,t}^{(1,1)} = 1$. This allows us to measure the response to a shock that implies a unitary increase in the variable $y_{i,t}^{(1)}$ – see the appendix for details.

[19] Note that, under conditional homoskedasticity, the above estimator can equivalently be obtained by two-stage least squares.

[20] In particular, with the long differencing, the coefficient of interest is estimated directly as $\beta_{h,t+h}$. This is because the long difference is usually defined as: $\Delta_h y_{i,t+h} = y_{i,t+h} - y_{i,t-1}$, and the outcome variable at $(t-1)$ is not affected by the shock that happens at time t.

in small samples.[21]

4 Time-varying County-level Fiscal Multipliers

This section focuses on estimating the effects of fiscal policy shocks and the fiscal multiplier based on county-level data and allowing for time variation. We first explain how we transform the data and how we purge them from common factors, which is what allows us to identify a relative multiplier from idiosyncratic county-level variation; then we discuss how we construct the time-varying fiscal multipliers; and finally we discuss the empirical results.

4.1 Data Transformation

Typically, one would use logarithms to estimate elasticities, which then need to be converted to a dollar equivalent. This is normally done by multiplying the elasticity by the average ratio of GDP to government spending in the sample. However, this approach may produce artificially high/low results depending on the sample. We follow Hall (2009) and Barro and Redlick (2011) and convert GDP and government spending changes to the same units prior to estimation. The transformed variables we use in our analysis, therefore, are as follows:

$$x_{i,t+h} = \frac{X_{i,t+h} - X_{i,t-1}}{X_{i,t-1}}, \qquad g_{i,t+h} = \frac{G_{i,t+h} - G_{i,t-1}}{X_{i,t-1}}, \tag{9}$$

where $X_{i,t}$ is real, per capita, county-level GDP available annually from the BEA,[22] and $G_{i,t}$ is real, per capita, county-level DoD spending; *t=1,2,..,T* and *i=1,2,...,N*, where T is the sample size of the time series dimension and N is that of the cross section. Our analysis focuses on shocks available in the period 2018–2023, so that the sample for the estimation starts in 2017 and, given the lag existing in the transformation above, the effective panel has $T = 6$ and $N = 2{,}158$ counties in each year. The choice of the sample period is guided by our focus on the COVID-19 pandemic period, because it is exactly the kind of large, abrupt and exogenous change in the environment that a time-varying analysis is designed to capture. In our main specification $X_{i,t}$, $G_{i,t}$ and $\varepsilon^{F,X}_{i,t}$ are further purged as discussed in Section 4.2. The results for the specification using the raw variables are reported in Appendix B.3.

[21] As we discuss in the appendix, examples include: the one-time structural break case ($\beta_{h,t+h} = \beta_{h,1}\delta(t)+\beta_{h,2}(1-\delta(t))+\eta_{h,t}$, $\delta(t) = 1(t < \tau)$, where the time of the break, τ, is known; the random walk, as in Primiceri (2005) and Cogley and Sargent (2005) ($\beta_{h,t+h} = \beta_{h,t+h-1} + \eta_{h,t},$), where the parameter path is deterministic once it has been realized; smooth time variation, where $\beta_{h,t+h} = \beta_{h,0} + \beta_{h,1}t + \beta_{h,2}t^2$; etc.

[22] County-level GDP is available from 2007. We use the full sample to estimate the shock of Section 2.2 and the common factors of Section 4.2.

4.2 Identifying the Relative Multiplier from Idiosyncratic Variation

Due to our use of disaggregated county-level data, as well as our novel county-level fiscal shock, the multipliers we estimate measure the effect that an increase in government spending in one county relative to another has on relative output, as in Nakamura and Steinsson (2014). As they argue, this is the effect of government spending in a monetary and fiscal union such as the United States, which they refer to as the (small) open economy relative multiplier (Nakamura and Steinsson (2014), p. 780). Nakamura and Steinsson (2014) identify the shock with a shift-share design: state military spending is instrumented by the state's exposure to national military spending, so that the identifying variation is, by construction, heterogeneous exposure to an aggregate shock. The exogeneity of that variation rests on two arguments, which carry over to our setting: military spending is driven mostly by international events, and an increase in procurement contracts in one county relative to another is not decided with the intention of favoring that county's economy.

The open economy multiplier is a *relative* one because, as in Nakamura and Steinsson (2014), the time fixed effects difference out aggregate shocks and aggregate policy – e.g. monetary and tax responses, movements in national prices – that affect all cross-sectional units at the same time: this is the "missing intercept" of the cross-regional literature, and it is what makes the estimate independent of aggregate policy. Our time-varying framework shares this property. Because the multiplier is estimated year by year from the cross-section of counties (Section 4.3), and every cross-sectional regression contains a constant, a time fixed effect for each year is included by construction, and what we estimate is a relative multiplier in exactly their sense.

What a time fixed effect does not absorb, in their design, is the *heterogeneous* exposure to aggregate conditions. If county i responds to an aggregate cycle f_t with a loading λ_i that differs across counties, the term $\lambda_i f_t$ survives the time effect, and if λ_i is correlated with the county's exposure to the fiscal shock then the estimate is biased, as Donaldson (2026) shows. For a shift-share design the problem is intrinsic: heterogeneous exposure to the aggregate shock *is* the identifying variation, so the bias cannot be eliminated without removing the identification itself. Our county-level shocks identify the same relative multiplier from the opposite source of variation – idiosyncratic county-level procurement innovations, which by construction do not load on aggregate shocks. In addition, cleaning the variables from common factors, as described further below, avoids the Donaldson (2026) bias, because it removes from every variable, including the instrument, any residual common component with county-specific loadings. To clean the variables from common factors, we proceed as follows: We first extract latent common factors from our key economic variables and then use the residuals of county-by-county regressions of those variables on the factors. The decomposition

below shows that the resulting estimator isolates the idiosyncratic component of the multiplier: a relative multiplier in the sense of Nakamura and Steinsson (2014), cleaned of the heterogeneous exposure term that contaminates the shift-share estimate.

To clarify this point further, consider the following model:

$$y_{i,t}^{(1)} = \lambda_{1,i} f_t + e_{i,t}^{(1)}, \tag{10}$$

$$y_{i,t}^{(2)} = \lambda_{2,i} f_t + e_{i,t}^{(2)}, \tag{11}$$

$$z_{i,t} = \lambda_{z,i} f_t + e_{i,t}^{(z)}, \tag{12}$$

where f_t, $e_{i,t}^{(1)}$, $e_{i,t}^{(2)}$ and $e_{i,t}^{(z)}$ have zero means with variance-covariance matrix:

$$\begin{bmatrix} \sigma_f^2 & 0 & 0 & 0 \\ 0 & \sigma_{11} & \sigma_{12} & \sigma_{1z} \\ 0 & \sigma_{21} & \sigma_{22} & \sigma_{2z} \\ 0 & \sigma_{z1} & \sigma_{z2} & \sigma_{zz} \end{bmatrix}. \tag{13}$$

The population value of the instrumental variable (IV) estimator from regressing $y_{i,t}^{(2)}$ on $y_{i,t}^{(1)}$ with $z_{i,t}$ as an instrument is:

$$\frac{\lim_{N\to\infty} \frac{1}{N} \sum_{i=1}^{N} \lambda_{z,i}\lambda_{2,i}\sigma_f^2 + \sigma_{z2}}{\lim_{N\to\infty} \frac{1}{N} \sum_{i=1}^{N} \lambda_{z,i}\lambda_{1,i}\sigma_f^2 + \sigma_{z1}} \equiv \frac{\Lambda_{z,2}\sigma_f^2 + \sigma_{z2}}{\Lambda_{z,1}\sigma_f^2 + \sigma_{z1}}. \tag{14}$$

The population value of the IV estimator from regressing $\lambda_{2,i} f_t$ on $\lambda_{1,i} f_t$ with $\lambda_{z,i} f_t$ as an instrument is:

$$\frac{\Lambda_{z,2}}{\Lambda_{z,1}}. \tag{15}$$

The population value of the IV estimator from regressing $e_{i,t}^{(2)}$ on $e_{i,t}^{(1)}$ with $e_{i,t}^{(z)}$ as an instrument is:

$$\frac{\sigma_{z2}}{\sigma_{z1}} \tag{16}$$

The Nakamura and Steinsson (2014) estimate corresponds to eq. (14), which is a convex combination of eq. (15) and eq. (16) with weights $\frac{\Lambda_{z,1}\sigma_f^2}{\Lambda_{z,1}\sigma_f^2+\sigma_{z1}}$ and $\frac{\sigma_{z1}}{\Lambda_{z,1}\sigma_f^2+\sigma_{z1}}$. Our estimator matches eq. (16), the

multiplier identified from idiosyncratic variation. Note that eq. (14) coincides with eq. (16) if and only if $\Lambda_{z,2}\sigma_{z1} = \Lambda_{z,1}\sigma_{z2}$, which holds whenever the instrument carries no common component ($\lambda_{z,i} = 0$ for all i) or its loadings are cross-sectionally uncorrelated with those of spending and output ($\Lambda_{z,1} = \Lambda_{z,2} = 0$). In addition, the equality also holds when the exposure is not heterogeneous – that is, $\lambda_{i,1} = \lambda_{i,2} = \lambda_i$ – and the covariances are symmetric – that is, $\sigma_{z,1} = \sigma_{z2} = \sigma_z$, – even though $\Lambda_{z,i}$ may differ from zero. Appendix B.4 further discusses these condition. In the shift-share design, the instrument loading is the exposure share itself, so the first condition fails by construction; in our design, instead, purging the variables of the common factors enforces it. Below, we explain in detail the procedure we use to purge the variables.

An additional advantage of eliminating the common factors is that our estimate is valid under quite general conditions and spatial correlation structures induced by the common components. In particular, Assumption 5 in Appendix C requires independent and identically distributed observations; however, this assumption is required only after purging the variables from the common factors.

In practice, we purge the variables as follows. Let $V_{i,t}$ denote the vector of variables entering the estimation: county real GDP per capita $X_{i,t}$, real DoD spending per capita $G_{i,t}$, the county-level shock $\varepsilon_{i,t}^{F,X}$ of eq. (1), and the county unemployment rate $U_{i,t}$ used in Section 4.4. We assume that each variable admits a common-factor representation with county-specific loadings:

$$V_{i,t} = a_i^V + \lambda_i^{V\prime} f_t + \check{V}_{i,t}, \qquad t = 2008, \ldots, 2023, \tag{17}$$

where f_t is a vector of latent common factors shared by all variables and all counties, λ_i^V is a county- and variable-specific loading, and $\check{V}_{i,t}$ is the idiosyncratic component. The common factor f_t is extracted jointly from the four variables.[23] The first component of the common factor f_t accounts for 40% of the variance of the stacked panel; it loads mainly on GDP and unemployment, has a correlation of 0.96 with aggregate real GDP per capita, and traces the post-2009 recovery, the 2020 recession and the subsequent expansion – i.e., it is the national business cycle. The second component (19%) is concentrated in 2008 and is essentially flat over our estimation period. In our main specification, we use the first principal component; results with two components are reported in Appendix B.3 and are very close for the pre-pandemic years, while for the pandemic years they differ only within the wide confidence bands of those estimates.

For each county and each variable we then estimate eq. (17) by least squares over 2008–2023 and

[23] We stack the four annual county panels, each demeaned within county and scaled by its pooled standard deviation so that no variable dominates by its units, into a matrix whose rows are county-variable series and whose columns are the years 2008–2023, and we take the principal components of its covariance matrix. This delivers a single factor space for the whole system, with a loading for every county-variable pair.

remove only the estimated common component, $\tilde{V}_{i,t} = V_{i,t} - \hat{\lambda}_i^{V\prime}\hat{f}_t$, keeping the county level a_i^V in the estimated purged vector, $\tilde{V}_{i,t}$, which includes the purged series $\tilde{X}_{i,t}$, $\tilde{G}_{i,t}$, $\tilde{\varepsilon}_{i,t}^{F,X}$ and $\tilde{U}_{i,t}$. Preserving the level is what allows the purged GDP to remain strictly positive and to serve as the Hall–Barro normalization in Section 4.1, as well as a control in the variable $w_{i,t-1}$ in levels, exactly as its raw (unpurged) variable counterpart does. This implies that the transformation and its interpretation are unchanged. Importantly, the purge is applied once to the annual levels, and it covers every horizon of the local projections: since the transformed variables of Section 4.1 are linear in the levels, the common component of the h-period growth rate $x_{i,t+h}$ is $\hat{\lambda}_i^{X\prime}(\hat{f}_{t+h} - \hat{f}_{t-1})/\check{X}_{i,t-1}$, a known function of the level factor and loading, so that no horizon-specific extraction is needed and the loadings are estimated on the full sixteen years rather than on samples whose size decreases with the horizon h.

The purge causes the average pairwise correlation of the county-level variables over the 2008–2023 sample to drop: that of county GDP falls from 0.36 to 0.15 and that of the unemployment rate from 0.84 to 0.47,[24] while the shock loses less than 1% of its variance over the 2017–2023 estimation years (about 9% when the factor is extracted from the shock panel alone, Appendix B.4). This indicates that the estimated shock series was already close to being idiosyncratic, as it was extracted from a residual that nets out lagged aggregates county by county. Indeed, this can explain why the purge has so little effect on the estimates, as shown in the specification results based on the raw variables reported in Appendix B.3. In what follows, we let $X_{i,t}$, $G_{i,t}$, $\varepsilon_{i,t}^{F,X}$ and $U_{i,t}$ denote the purged series $\tilde{X}_{i,t}$, $\tilde{G}_{i,t}$, $\tilde{\varepsilon}_{i,t}^{F,X}$ and $\tilde{U}_{i,t}$, in order to simplify the notation.

4.3 Construction of Time-Varying Cumulative Fiscal Multipliers

In terms of our notation in Section 3, $y_{i,t+h}^{(2)} = x_{i,t+h}$ is the transformed GDP and $y_{i,t+h}^{(1)} = g_{i,t+h}$ is DoD spending, as defined in eq. (9). The time-varying cumulative fiscal multiplier at horizon h at time t is defined as the cumulative response of GDP to a fiscal shock at time t over the cumulative response of spending to the same shock over the h periods following the shock.[25] That is,

$$m_{h,t+h} = \frac{\sum_{j=0}^{h} \Theta_{j,t+j}^{(2,1)}}{\sum_{j=0}^{h} \Theta_{j,t+j}^{(1,1)}}, \tag{18}$$

[24] For each variable, we compute the correlation between the 2008–2023 time series of every pair of counties with complete data, and average it over all pairs.

[25] Consistently with existing literature (e.g. Ramey (2011)), we focus on the cumulative (rather than the peak) multiplier.

where

$$\Theta_{h,t+h}^{(2,1)} = \frac{\partial x_{i,t+h}}{\partial \varepsilon_{i,t}^{F,X}} = \frac{\partial \left(\frac{X_{i,t+h} - X_{i,t-1}}{X_{i,t-1}} \right)}{\partial \varepsilon_{i,t}^{F,X}} = \frac{\partial X_{i,t+h}}{\partial \varepsilon_{i,t}^{F,X}} X_{i,t-1}^{-1}$$

$$\Theta_{h,t+h}^{(1,1)} = \frac{\partial g_{i,t+h}}{\partial \varepsilon_{i,t}^{F,X}} = \frac{\partial \left(\frac{G_{i,t+h} - G_{i,t-1}}{X_{i,t-1}} \right)}{\partial \varepsilon_{i,t}^{F,X}} = \frac{\partial G_{i,t+h}}{\partial \varepsilon_{i,t}^{F,X}} X_{i,t-1}^{-1}$$

and the last equalities are due to the fact that the variables in $(t-1)$ are not impacted by the shock at time t. This implies that the multipliers defined in terms of the original series, $X_{i,t+h}$, are the same as those defined in terms of the detrended variables, $x_{i,t+h}$. As discussed in Section 4.2, $m_{h,t+h}$ is an open economy relative multiplier in the sense of Nakamura and Steinsson (2014), identified from idiosyncratic county-level variation.

We estimate the cumulative multiplier at horizon h from the following equation:

$$\sum_{j=0}^{h} x_{i,t+j} = c_h + m_{h,t+h} \sum_{j=0}^{h} g_{i,t+j} + \gamma_{h,t+h}' w_{i,t-1} + v_{i,t+h}, \tag{19}$$

using $\varepsilon_{i,t}^{F,X}$ as an instrument. Note that the unit normalization that we impose throughout the estimation, i.e., $\Theta_{0,t}^{(1,1)} = 1$ for every t, allows a clearer interpretation of the cumulative multiplier. In fact, $m_{h,t+h}$ is the cumulative response of GDP, relative to that of government spending, h periods after a fiscal shock that increased spending by 1 unit on impact. Then, applying the estimator from eq. (A-28), we have

$$\hat{m}_{h,t+h} = \frac{\frac{1}{N} \sum_{i=1}^{N} \varepsilon_{i,t}^{F,X} \sum_{j=0}^{h} \tilde{x}_{i,t+j}}{\frac{1}{N} \sum_{i=1}^{N} \varepsilon_{i,t}^{F,X} \sum_{j=0}^{h} \tilde{g}_{i,t+j}}, \tag{20}$$

where $\tilde{x}_{i,t+j}$ and $\tilde{g}_{i,t+j}$ are the residuals of the regressions of $x_{i,t+j}$ and $g_{i,t+j}$ on the set of controls $w_{i,t-1}$, which contains a constant, $X_{i,t-1}$, $G_{i,t-1}$ and $\varepsilon_{i,t-1}^{F,X}$. Standard errors are computed directly using eq. (A-31). Appendix A shows that this one-step estimator coincides with a time-varying version of the three-step procedure of Ramey and Zubairy (2018).

4.4 Estimates of Time-varying, County-level Fiscal Multipliers

In Figure 2, we show the estimated multiplier on impact ($h = 0$), together with the associated 90% confidence intervals. The local projection equation to estimate the multiplier is obtained from eq. (19). Given the unit

normalization ($\Theta_{0,t}^{(1,1)} = 1$ for all t), the $h = 0$ multiplier corresponds to the $h = 0$ response coefficient, since

$$m_{0,t} = \frac{\Theta_{0,t}^{(2,1)}}{\Theta_{0,t}^{(1,1)}} = \Theta_{0,t}^{(2,1)} = \theta_{0,t}.$$

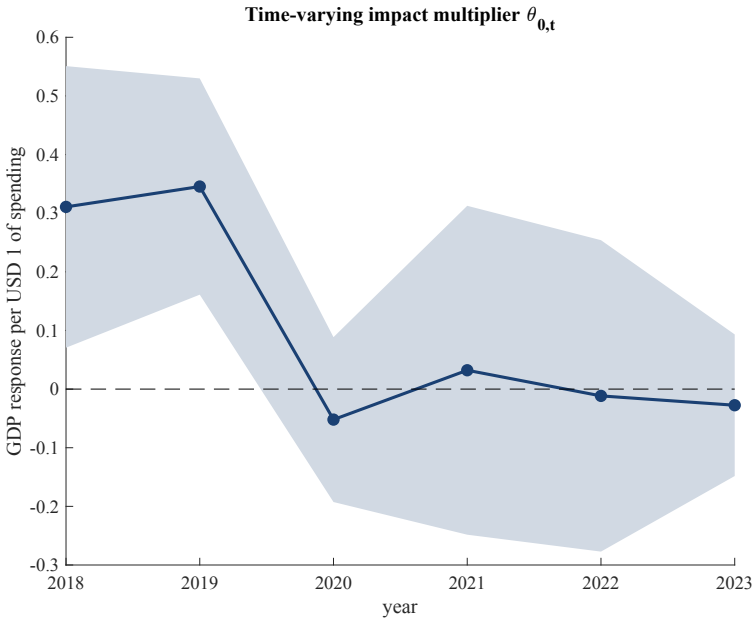
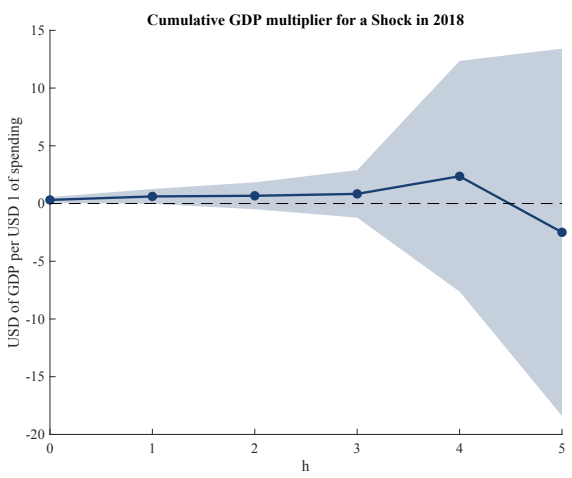


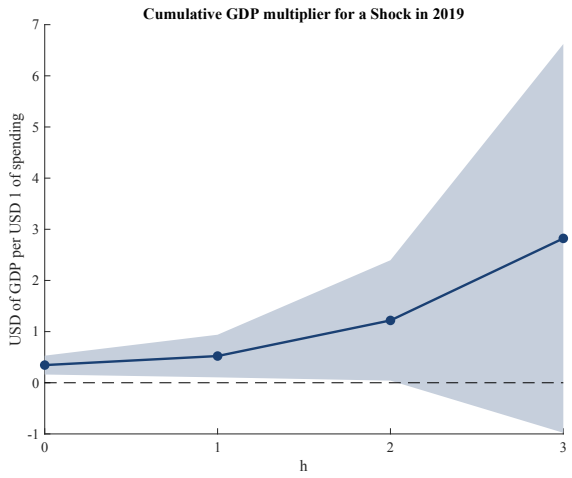
Figure 2: Fiscal multiplier on impact ($h = 0$), together with 90% confidence bands.

The figure reveals substantial time variation in the immediate response of GDP to a fiscal stimulus. The impact effects in 2018 and 2019 are both significantly positive, and equal 0.31 and 0.35 respectively, while those for 2020–2023 are all insignificantly different from zero. The 2023 shock is included in Figure 2 for completeness: since the county-level GDP data end in 2023, only its impact effect can be estimated, and it does not appear in the cumulative-multiplier and impulse-response figures below.

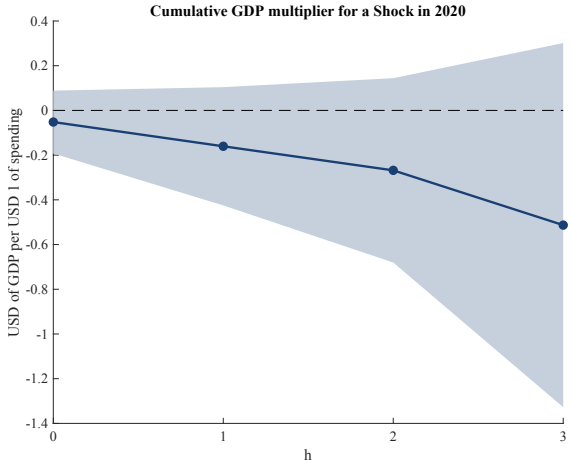
Figures 3a-3e plot the h-period-ahead cumulative multipliers for the shocks in 2018 to 2022. The cumulative multipliers of the 2018 and 2019 shocks are positive at all horizons but the last, and those of the 2019 shock increase with the horizon, whereas the multipliers of the 2020 and 2021 shocks are negative or essentially zero. The one-year-ahead cumulative multiplier is 0.61 for the 2018 shock and 0.52 for the 2019 shock, and the latter is significantly different from zero. The small multipliers of 2020 and 2021 are most likely due to the shutdowns associated with the COVID-19 pandemic, which started in March 2020: it is possible that the stimulating effects of DoD spending were delayed in those years, since economic activity, and production in general, was frozen for an extended period of time. Regarding the pre-pandemic years,



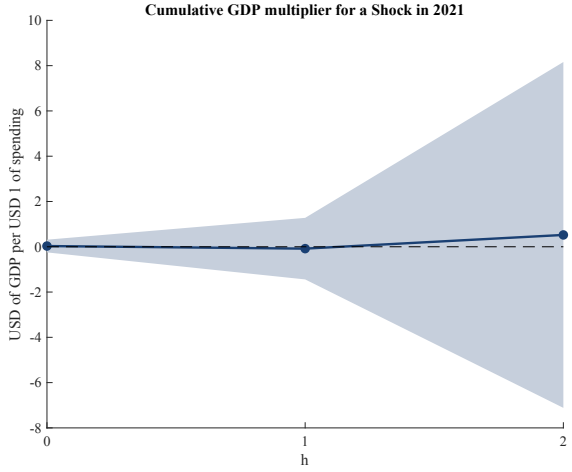
(a) $m_{h,2018+h}$



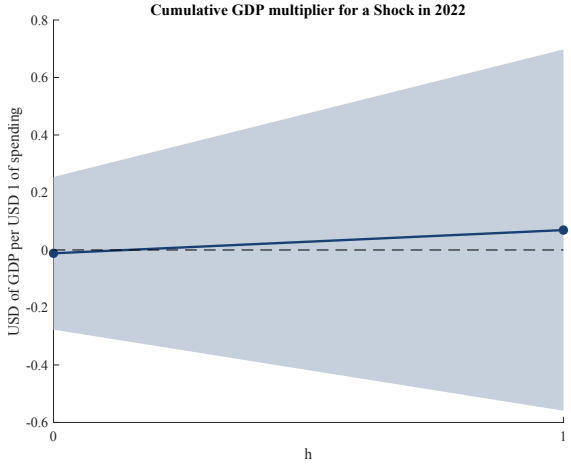
(b) $m_{h,2019+h}$



(c) $m_{h,2020+h}$



(d) $m_{h,2021+h}$



(e) $m_{h,2022+h}$

Figure 3: Time-varying cumulative multipliers, together with 90% confidence bands.

multipliers up to two years ahead are significantly different from zero in 2019 while those in 2018 are not statistically significant except for the impact multiplier. The uncertainty around the point estimates is large, in particular starting from 2021, again most likely due to the effect of the uncertainty associated with the COVID pandemic.

Turning to time-varying impulse-responses, they are estimated by the following local projection regression:

$$x_{i,t+h} = \alpha + \theta_{h,t+h}\, g_{i,t} + \gamma'_{h,t+h} w_{i,t-1} + v_{i,t+h}. \tag{21}$$

Figure 4 plots the impulse responses of real output per capita to a shock in military spending associated with a unitary increase in real per capita procurement spending ($\theta_{h,t+h}$) estimated at different times. Note that responses before the COVID pandemic (2018 and 2019) are similar to the usual responses to a fiscal shock, where output increases after an unexpected increase in fiscal spending. In particular, a fiscal shock in 2019 has a persistent and statistically significant effect on real output per capita: the response is 0.35 on impact and 0.55 three years later, both significantly different from zero, in line with the associated multipliers in Figure 3b. However, during the pandemic (2020 and 2021), our time-varying results show that the responses are very different. For example, responses become negative and insignificant, most likely due to the decrease in production associated with the pandemic. In 2022, the point estimates turn positive again one year after the shock, but they are not significantly different from zero; the instrument-only purge of Appendix B.4, which leaves output and spending unpurged, delivers a significant one-year response (Table A-3), so we regard the sign of the 2022 response as suggestive at best. Notice that, as discussed in the previous sections, the government spending variable from which we extract the fiscal shock is the number of signed procurement contracts. Therefore, the timing of the spending variables is associated with signature rather than actual payments, which usually happens after the work is performed. During the pandemic, procurement contracts continued to be signed, although actual production decreased, as it was delayed due to freeze in production. Hence, our measure allows us to empirically quantify the effects of the slowdown in real economic activity in a better and more timely way. Looking at the pre-pandemic results in Figures 3a - 3b and 4a-4b at longer horizons, we observe that fiscal shocks in 2018 and 2019 had a positive effect on GDP when the pandemic hit. This evidence suggests that the economic slowdown during those years could have been even more severe without this type of fiscal stimulus.

We analyze the strength of the purged instrument using both conventional first-stage F-statistics as well as the test of Lewis and Mertens (2022), year by year and horizon by horizon, with bias tolerance $\tau \in \{0.10, 0.15\}$ and significance levels $\alpha \in \{0.05, 0.10\}$. The conventional first-stage F statistic is above hundred in every

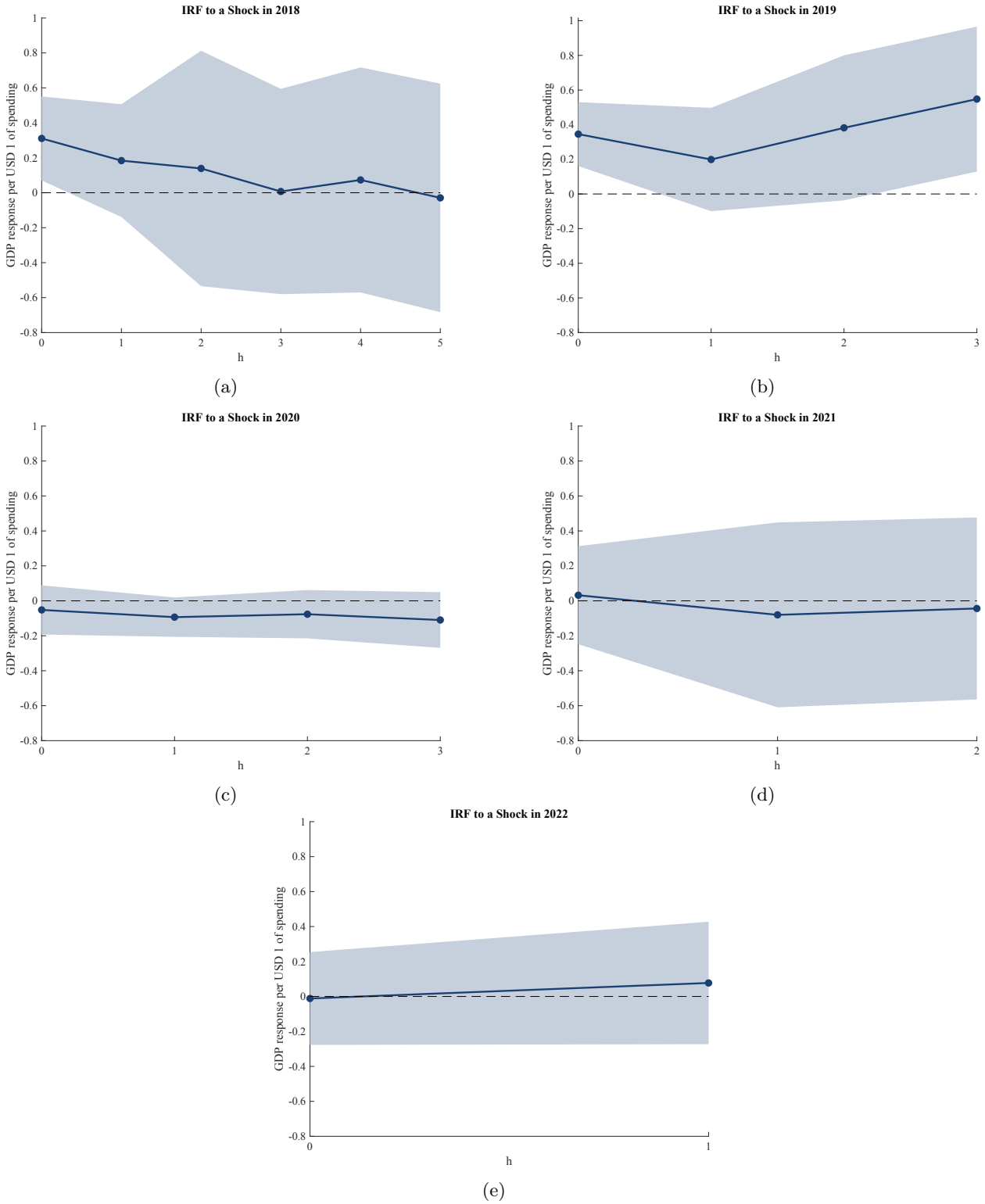


Figure 4: Time-varying response of GDP ($\theta_{h,t+h}$) at time t equal to: 2018 (top left panel), 2019 (top right panel), 2020 (bottom left panel), 2021 (bottom right panel), 2022 (bottom center panel). The shaded areas represent 90% confidence bands.

year, but the generalized statistic of Lewis and Mertens (2022), which is robust to heteroskedasticity and heavy tails, rejects the weak-instrument null at all conventional tolerances in 2019 and 2023, and in 2018 at a 15% bias tolerance, while in the pandemic years 2020–2021 and in 2022 it does not reject. For unemployment the instrument is strong in every year at a 15% bias tolerance.[26] We therefore read the estimates for 2018–2019 as our main evidence, and interpret the small and insignificant GDP responses to the 2020–2022 shocks with caution: they are consistent both with a genuinely small multiplier during the shutdowns and with a low signal-to-noise ratio of procurement shocks in those years, which the wide confidence bands also reflect.[27]

The estimates above are, by construction, free of the bias discussed in Section 4.2: the instrument $\varepsilon_{i,t}^{F,X}$ has no common component with county-specific loadings, so that the cross-sectional covariance between the instrument and county exposure to general-equilibrium effects associated with the aggregate cycle is zero at every horizon. Appendix B.4 states this condition and reports two further checks: the results without any purge, and the results in which only the instrument is purged, with the factor extracted from the shock panel itself rather than from the whole system. Both are essentially identical to the estimates above, which confirms that our disaggregated instrument carries virtually no common component to begin with. The results are also robust to how the county-level shock is constructed: Appendices B.5 and B.6 re-estimate the multipliers and impulse responses with a shock that controls for lagged county-level, rather than national, GDP in equation (1), and with a shock that does not control for GDP at all: the results are very similar.

Figure 5 shows the impulse response computed with the constant parameter specification, estimated on the same (purged) data. The response is obtained using a standard panel LP-IV estimator with Heteroskedasticity and Serial Correlation (HAC) robust standard errors. The figure shows that the response is small and insignificant at all horizons. Comparing Figure 4 with Figure 5, it is evident that, for a given horizon h, the constant parameter estimates lie between the time-varying ones: they average the positive and significant responses to the 2018 and 2019 shocks with the zero or negative responses to the pandemic-year shocks; thus, overall the estimate is close to zero. This result suggests that accounting for time variation can help us better understand the transmission of fiscal policy, especially around periods characterized by large shocks that are completely exogenous to the economy. The latter conclusion is not specific to our dataset: in Appendix B.7, we use the TVP-PLP estimator to estimate the open economy fiscal multiplier

[26] The pattern is very similar without the purge – see Appendix B.3.

[27] The results in detail are as follows: Weak IV tests for GDP as outcome: for 2019 and 2023 we reject the null hypothesis of weak IV at all horizons and for all the tolerances considered; for 2018 we reject the null at all horizons for a bias threshold of 0.15 (at both significance levels), but not for a bias threshold of 0.10; for 2020, 2021 and 2022 we cannot reject the null at the tolerances considered, at any horizon. Weak IV tests for unemployment as outcome: for 2018, 2019, 2020 and 2023 we reject the null at all horizons and for all the tolerances considered; for 2021 and 2022 we reject the null at all horizons for a significance level of 0.10 with a bias threshold of 0.10, and for both significance levels with a bias threshold of 0.15. The conventional first-stage F statistic exceeds 400 in every year and for both outcomes.

using state-level data, as in Nakamura and Steinsson (2014), and we find time variation in these multipliers as well, which a constant parameter estimator would disregard.

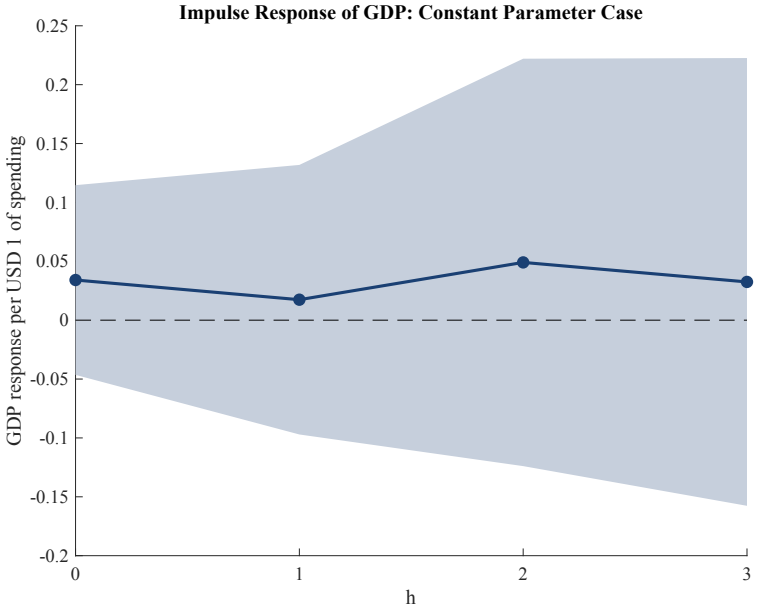


Figure 5: Response of GDP: Constant parameter case (θ_h), together with 90% confidence bands.

To deepen our understanding of the time-varying effects of fiscal policy in recent years, we also compute the impulse responses of unemployment to a military spending shock.[28] County-level unemployment data are obtained from the USDA Economic Research Service website. In terms of our notation in Section 3, $y^{(2)}_{i,t+h} = u_{i,t+h} - u_{i,t-1}$, where $u_{i,t}$ is the unemployment rate in county i and year t, and $y^{(1)}_{i,t} = G_{i,t} - G_{i,t-1}$[29]. Both series are purged from the common factor, as described in Section 4.2; the purge is particularly consequential for unemployment, whose common component is large. Figure 6 plots the response of unemployment to a shock in military spending associated with a unit increase in real per capita procurement spending at different points in time. As was the case for GDP, we uncover variation in the responses across different years. The fiscal shock of 2018 lowers unemployment significantly one and three years after the shock, and the point estimates remain negative up to five years ahead; the 2019 shock raises unemployment on impact, though not significantly, with an insignificant response at longer horizons. As for output, the responses to the shocks in the pandemic year are small and insignificant, with the exception of a significant fall in unemployment two years after the 2020 shock, and the 2022 shock lowers unemployment one-year-ahead, though not significantly.

[28] Here, we use a different data transformation, as unemployment is less persistent than output and we do not compute a multiplier, only impulse responses.

[29] We use the first difference in real per capita county-level DoD spending, rather than its percentage change, due to the presence of a few zeroes in our dataset.

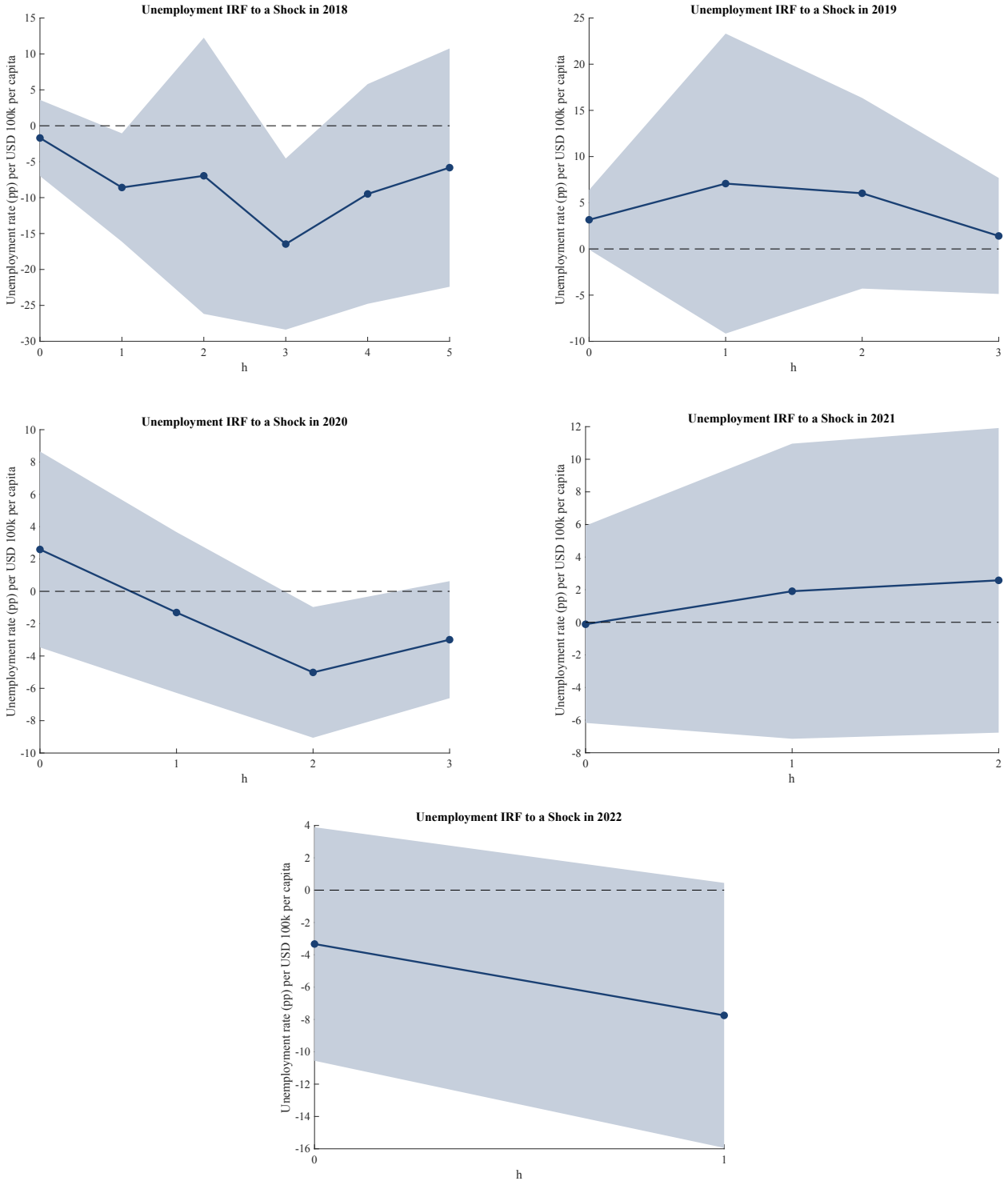


Figure 6: Time-varying response of unemployment ($\theta_{h,t+h}$) at time t equal to: 2018 (top left panel), 2019 (top right panel), 2020 (bottom left panel), 2021 (bottom right panel), 2022 (bottom center panel). The shaded areas represent 90% confidence bands.

For comparison, Figure 7 shows the impulse response under the constant parameter specification: the estimates in the latter case are small and insignificantly different from zero at all horizons. Again, ignoring time variation would lead to substantial under-estimation of the effects of fiscal policy shocks.

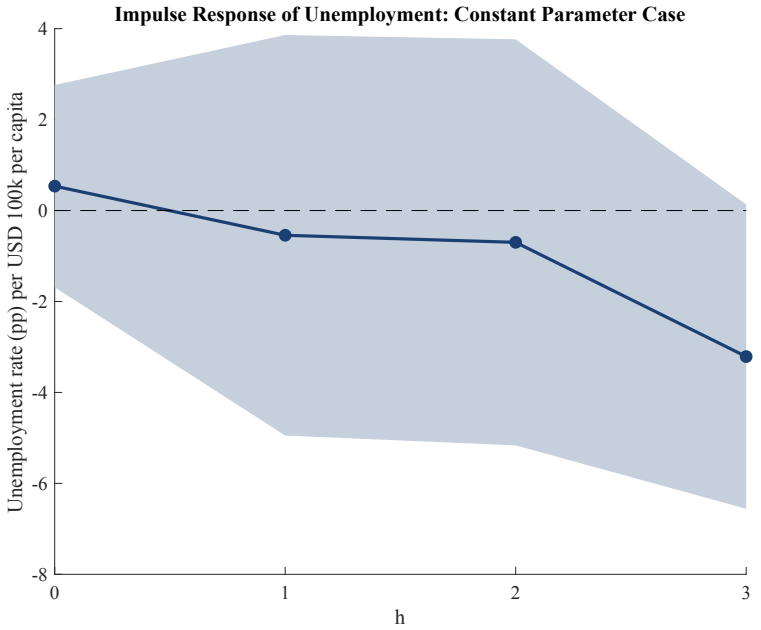


Figure 7: Response of unemployment: Constant parameter case (θ_h), together with 90% confidence bands.

5 High Frequency Fiscal Multipliers

In this section, we focus on the effects of fiscal policy shocks at high frequencies, using the shocks identified in Section 2.3. Our goal is to measure the effects of fiscal shocks on real activity at the daily frequency. As a measure of daily real activity, we use the real business conditions index of Aruoba et al. (2009). The Aruoba et al. (2009) index is a high frequency (daily) index summarizing information from a rich dataset of underlying (seasonally adjusted) economic indicators at various frequencies, including measures of unemployment (such as weekly initial jobless claims and monthly payroll employment), industrial production, sales, and real GDP.

Figure 8a plots the response of the ADS business conditions index to an increase of one billion dollars in obligations on government procurement contracts, together with 90% confidence bands. The horizon is measured in days. Increases in government spending lead to improvements in real business conditions, although the effects are not significant over the full sample we consider. However, a closer analysis uncovers significant time variation. In fact, focusing on the period of the financial crisis – up to and including the subsequent period when unemployment was above 6.7 percent (2007-2013) – the business condition index

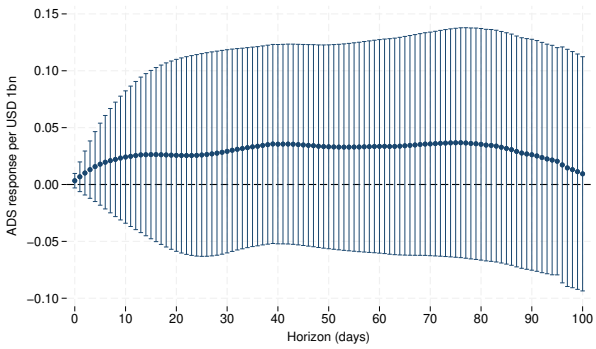
response is positive and significant starting two weeks after the initial shock and until roughly one month afterwards. The effect of the shock peaks about a month after the shock took place. In the 2017–2024 sub-sample (Figure 8c), the response is again positive and significant, but over a different, longer range of horizons, between 45 and 95 days, and the effects take longer to reach their peak. The results are robust to augmenting the shock regression (2) with the lagged effective federal funds rate, to control for the financial environment: Appendix B.2 shows that the resulting shock and the responses of the business conditions index are very close to the ones reported here, including the timing of the significant responses in each sub-sample.

It is important to note that widely-used, state-dependent models where the state is measured by unemployment would not necessarily uncover the time variation that we find in the data. In fact, our sample includes another period in which unemployment is particularly high, namely the COVID-19 pandemic. Figure 8d focuses on the sub-sample where unemployment is above six percent during the recent pandemic. The response becomes significant about ten days after the shock, peaks after roughly three weeks and remains positive and significant over the whole horizon we consider; it is therefore very different from the response in the high-unemployment period associated with the financial crisis, both in timing and in persistence. This evidence suggests that, while both samples in Figures 8b and 8d would fall into the same "high-unemployment state" category, they are very different periods in terms of fiscal policy transmission. In particular, the most recent high-unemployment period is characterized by a large shock that is completely exogenous to the state of the economy and that may cause economic dynamics to be very different compared to other crises.

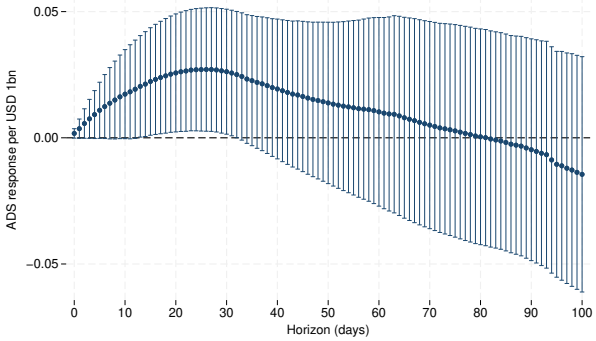
Our daily shock could also be used more broadly to investigate the effects of fiscal policy on other daily macroeconomic variables, such as consumer sentiment, using, for example, the Gallup US Daily Poll dataset or any other data available at a high frequency. We chose to focus on the business conditions index, consistent with the fiscal multiplier analysis in the rest of our paper.

5.1 The Importance of Identifying the Shock

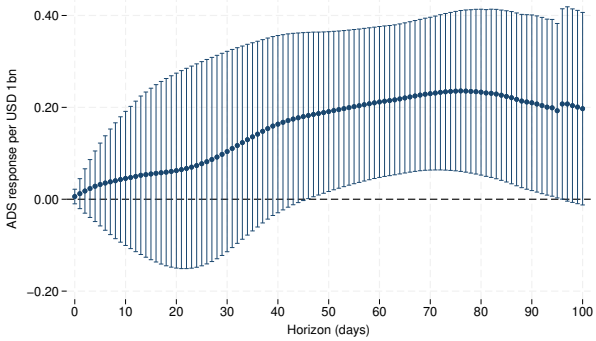
We are not the first to exploit daily data on defense procurement. Auerbach and Gorodnichenko (2016) use the daily volume of contract awards announced by the Department of Defense, together with daily Treasury payments to defense contractors, to study the response of the exchange rate and other financial variables to government spending, and Auerbach et al. (2020) use the same announcement data to study credit markets. Their empirical strategy uses the daily spending series directly as a regressor in the local projection, with twenty lags of its first difference and of the first difference of the outcome as controls, after



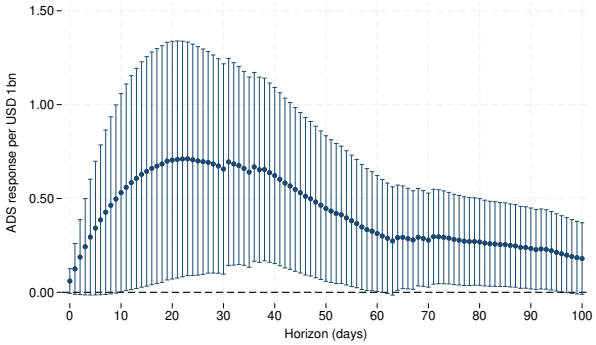
(a) Daily response of the business conditions index: Full sample estimates (2007-2024).



(b) Daily response of the business conditions index: the financial crisis and the subsequent high unemployment period (2007-2013).



(c) Daily response of the business conditions index: 2017-2024.



(d) Daily response of the business conditions index during the COVID-19 pandemic (2020:4-2021:4).

Figure 8: Daily response of the ADS business conditions index to an increase of one billion dollars in DoD obligations, using the identified shock of eq. (2) as instrument, and 90% confidence bands; horizon in days.

removing seasonal cycles;[30] the contemporaneous variation in spending is thereby cleaned from predictable movements based on its own lags and the lags of the outcome variable. Our approach differs from the latter in three respects: first, we use the universe of contract obligations recorded in the FPDS rather than announced awards above a reporting threshold, which is what makes the county-level analysis in the rest of the paper possible; second, we construct an explicitly identified shock series, equation (2), that removes the institutional end-of-fiscal-year surge rather than a generic seasonal pattern; and third, we validate the shock against identified monetary and central bank information shocks.

A natural and important question is whether identifying the shock like we do, rather than using the spending series itself, matters for estimating the effects of fiscal policy. To answer this question, we re-estimate the response of the business conditions index following the approach in Auerbach and Gorodnichenko (2016) with two alternative sources of identifying variation: the raw spending series, including only its own lags and the lags of the outcome as controls, exactly as in their approach; and our identified shock. To isolate the role of the shock, we hold the specification of the local projection fixed and change only what identifies the variation in spending. The specification follows Auerbach and Gorodnichenko (2016) and includes, besides the controls of our baseline, twenty lags of the first differences of obligations and of the business conditions index.[31]

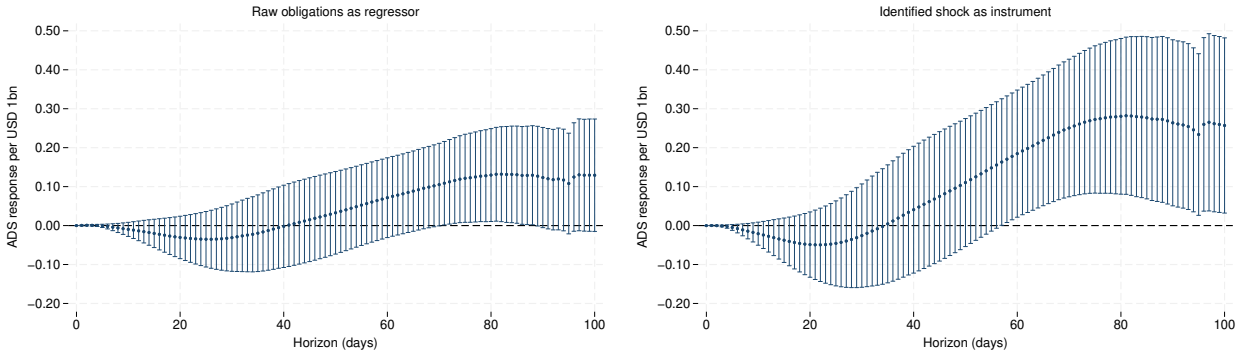
Figure 9 reports, for the same 2017-2024 sample of Figure 8c, the response of the business conditions index to an increase of one billion dollars in obligations when (a) the raw obligation series enters directly as the regressor, so that identification comes from the lags alone, as in a spending-based local projection; (b) obligations are instrumented by our identified shock.[32] The two responses have the same sign and the same shape: real business conditions improve gradually after the shock, and the response peaks around 75–80 days. What differs is the magnitude and the precision. Panel (a) shows that the peak response is 0.13 per billion dollars, and that the response is significantly different from zero only between 71 and 88 days after the shock. With our identified shock, instead, the peak response is 0.28, more than twice as large, and is significantly different from zero from 57 days after the shock to the longest horizon; it lies above the confidence bands of panel (a) for forty days.

The following conclusions emerge. The positive effects of daily procurement spending on real activity

[30] The seasonal cycles are removed from government spending using a trigonometric filter.

[31] The baseline of Section 2.3 controls for one lag of the index, of obligations and of the shock. Adding twenty lags of both first differences leaves the 2017–2024 response essentially unchanged: the peak is 0.28 per billion dollars with the lags against 0.24 without (Figure 8c), with the same shape and a range of significant horizons of 57–100 days against 45–95. The instrument is orthogonal to lagged spending by construction, so the lags only absorb residual variation in the outcome.

[32] In the full 2007–2024 sample, none of these specifications delivers a significant response at any horizon, as in Figure 8a, with peak responses between 0.03 and 0.04 per billion dollars.



(a) Raw obligations as regressor, lags only

(b) Identified shock as instrument (baseline)

Figure 9: Daily response of the business conditions index to an increase of one billion dollars in DoD obligations, 2017–2024, with 90% confidence bands; horizon in days. All panels use the specification of Figure 8c with the conditioning set of Auerbach and Gorodnichenko (2016), including as controls twenty lags of the first differences of obligations and of the index.

that we document are not a consequence of how the shock is identified: using the spending series directly and controlling for its own lags, as Auerbach and Gorodnichenko (2016) do, delivers a response of the same sign and timing as using our identified shock. However, identifying the shock matters in terms of both magnitude and inference.[33]

6 Conclusions

We make two main contributions to the literature. First, we construct a new dataset of county-level fiscal policy shocks at several frequencies, including annual, quarterly, and monthly, at the most geographically disaggregated level, as well as at the daily frequency for aggregate national data. The dataset will be extremely useful for practitioners studying the effects of fiscal policy shocks at the most disaggregated geographical level and at the highest possible frequency. Since our data are based on procurement contracts, whose timing is associated with the signature date rather than the actual payment, our identified shock measures exogenous changes in fiscal spending at the time the fiscal expense is committed, rather than after the payment has occurred, which, in turn, happens after production. For this reason, our dataset of fiscal shocks provides a better measure of fiscal policy, allowing us to empirically quantify in a timely fashion the

[33] We have also investigated whether isolating the effects of the inclusion of the end-of-fiscal-year obligations explains the difference in the results. It is useful to note that the end-of-fiscal-year obligations that we control for in our shock are large but are followed by little movement in real business conditions – consistent with their being anticipated, as the "use-it-or-lose-it" logic of Section 2.3 suggests – so that leaving them in the identifying variation attenuates the estimated response per dollar and widens the confidence bands, in the same way that measurement error in a regressor does. Removing them more than doubles the estimated response, which drastically improves their precisions over most horizons.

effects of a slowdown in real economic activity.

Second, we estimate novel time-varying responses to fiscal policy shocks using a new estimator, whose theoretical properties and small-sample performance are documented in the appendix. Our unique dataset and estimator confirm the intuition that the COVID-19 pandemic posed unprecedented challenges to governments' ability to stimulate the economy. Indeed, we find that the responses of economic variables to a government spending shock are in line with macroeconomic theory in the years before the pandemic. The COVID-19 years, by contrast, are characterized by a lack of significant responses to fiscal shocks. Regarding fiscal multipliers, we find that they are significantly positive on impact in 2018 and 2019 and, for the 2019 shock, that they increase with the horizon and remain significant up to two years ahead. Starting in 2020, the multipliers are never significantly different from zero, pointing to a potential jam in the fiscal policy transmission mechanism. A similar picture emerges from the analysis of the responses of GDP and unemployment, featuring (i) positive and significant responses for 2018 and 2019; (ii) insignificant responses during the pandemic, i.e. 2020 and 2021; and (iii) point estimates that turn positive again in 2022, after the end of the pandemic, but remain imprecisely estimated. A constant-parameter estimator applied to the same data would average these regimes and deliver small and insignificant responses throughout, which shows that ignoring time variation would lead to a substantial under-estimation of the effects of fiscal policy before the pandemic. These conclusions do not depend on how the common component of county-level variables is treated: they hold whether all variables, only the instrument, or none are purged of common factors, in line with the formal robustness analysis in Appendix B.4.

At the daily frequency, our identified national shock uncovers time variation that a state-dependent model would not: the response of real business conditions to a procurement shock is short-lived during the financial crisis and persistent during the pandemic – two episodes that a state-dependent model would not be able to distinguish, since it would associate them with the same state. In addition, we show that identifying the shock matters: using raw daily observations instead of our identified shock more than halves the estimated responses and adds lacks of precision, because the predictable end-of-fiscal-year surge dominates the raw variation. Hence, our estimates of both fiscal multipliers and responses show substantial time variation, in particular around the recent COVID-19 pandemic.

References

Almuzara, M., and V. Sancibrian. 2024. "Micro Responses to Macro Shocks." *Federal Reserve Bank of New York Discussion Papers* 1090.

Aruoba, S.B., F.X. Diebold, and C. Scotti. 2009. "Real-Time Measurement of Business Conditions." *Journal of Business and Economic Statistics* 27 417–427.

Auerbach, Alan, Yuriy Gorodnichenko, Peter B McCrory, and Daniel Murphy. 2022. "Fiscal multipliers in the COVID19 recession." *Journal of International Money and Finance* 126 102669.

Auerbach, Alan, Yuriy Gorodnichenko, and Daniel Murphy. 2020. "Local Fiscal Multipliers and Fiscal Spillovers in the USA: A. Auerbach et al.." *IMF Economic Review* 68 (1): 195–229.

Auerbach, Alan J, and Yuriy Gorodnichenko. 2016. "Effects of Fiscal Shocks in a Globalized World." *IMF Economic Review* 64 (1): 177–215.

Auerbach, Alan J, Yuriy Gorodnichenko, and Daniel Murphy. 2020. "Effects of Fiscal Policy on Credit Markets." *AEA Papers and Proceedings* 110 119–124.

Barro, R.J., and C.J Redlick. 2011. "Macroeconomic Effects from Government Purchases and Taxes." *The Quarterly Journal of Economics* 126 (1): 51–102.

Bianco, Timothy, Ana Herrera, and Elena Pesavento. 2026. "Heterogeneous Effects of Economic Policy Uncertainty on Corporate Credit Growth: Evidence from U.S. Firms." *mimeo.*

Blanchard, O., and R. Perotti. 2002. "An Empirical Characterization of the Dynamic Effects of Changes in Government Spending and Taxes on Output." *The Quarterly Journal of Economics* 117 (4): 1329–1368.

Briganti, E. 2023. "On the Effects of Government Purchases and Their Transmission Mechanism." *Mimeo.*

Briganti, E., and V. Sellemi. 2023. "Why Does GDP Move Before Government Spending? It's all in the Measurement." *Mimeo.*

Buda, G., V.M. Carvalho, G. Corsetti et al. 2025. "The Short Lags of Monetary Policy." *Mimeo.*

Canova, Fabio, and Fernando J Pérez Forero. 2015. "Estimating overidentified, nonrecursive, time-varying coefficients structural vector autoregressions." *Quantitative Economics* 6 (2): 359–384.

Chodorow-Reich, Gabriel. 2019. "Geographic cross-sectional fiscal spending multipliers: What have we learned?" *American Economic Journal: Economic Policy* 11 (2): 1–34.

Cloyne, James S, Oscar Jorda, and Alan M Taylor. 2020. "Decomposing the fiscal multiplier."Technical report, National Bureau of Economic Research.

Cogley, T., and T.J. Sargent. 2005. "Drifts and Volatilities: Monetary Policies and Outcomes in the Post WWII US." *Review of Economic Dynamics* 8 (2): 262–302.

Cogley, Timothy, and Thomas J Sargent. 2005. "Drifts and volatilities: monetary policies and outcomes in the post WWII US." *Review of Economic dynamics* 8 (2): 262–302.

Cruicini, M., and N. Vu. 2021. "Did the American Recovery and Reinvestment Act Help Counties Most Affected by the Great Recession?" *Review of Economic Dynamics* 42 264–282.

David, J., R. Giacomini, X. Jiao, and W. Wang. 2026. "Causal State-dependent Local Projections." *mimeo.*

Donaldson, Paula. 2026. "Cross-sectional Identification with Heterogeneous Exposure to General Equilibrium Effects."

Dube, A., D. Girardi, Ò. Jordà, and A.M. Taylor. 2023. "A Local Projections Approach to Difference-in-Differences Event Studies."Technical report, National Bureau of Economic Research Cambridge, Massachusetts.

Florens, Jean Pierre, and Anna Simoni. 2025. "Panel data models with randomly generated groups." *mimeo.*

Gargiulo, Valeria, Christian Matthes, and Katerina Petrova. 2025. "Monetary policy across inflation regimes." *European Economic Review* 105109.

Giannone, D., and L. Reichlin. 2006. "Trends and cycles in the euro area: how much heterogeneity and should we worry about it?" *European Central Bank Working Paper* 595.

Gonçalves, S., A.M. Herrera, L. Kilian, and E. Pesavento. 2021. "Impulse response analysis for structural dynamic models with nonlinear regressors." *Journal of Econometrics* 107–130.

Gonçalves, S., A.M. Herrera, L. Kilian, and E. Pesavento. 2024a. "Nonparametric Local Projections." *Federal Reserve Bank of Dallas 2414.*

Gonçalves, S., A.M. Herrera, L. Kilian, and E. Pesavento. 2024b. "State-dependent Local Projections." *Journal of Econometrics* 105702.

Guren, Adam, Alisdair McKay, Emi Nakamura, and Jón Steinsson. 2021. "What do we learn from cross-regional empirical estimates in macroeconomics?" *NBER Macroeconomics Annual* 35 (1): 175–223.

Gurkaynak, R.S., and J.H. Wright. 2013. "Identification and Inference Using Event Studies." *The Manchester School* 81 48–65.

Hall, R.E. 2009. "By How Much Does GDP Rise if the Government Buys More Output?"Technical report, National Bureau of Economic Research.

Inoue, A., B. Rossi, and Y. Wang. 2024. "Local Projections in Unstable Environments." *Journal of Econometrics* 244(2).

Inoue, A., B. Rossi, and Y. Wang. 2025. "Has the Phillips Curve Flattened and Why?" *Mimeo.*

Inoue, Atsushi, Barbara Rossi, and Yiru Wang. 2026. "Heterogeneous Local Projections." *mimeo.*

Jarociński, Marek, and Peter Karadi. 2020. "Deconstructing monetary policy surprises—the role of information shocks." *American Economic Journal: Macroeconomics* 12 (2): 1–43.

Jordà, Ò. 2005. "Estimation and Inference of Impulse Responses by Local Projections." *American Economic Review* 95 (1): 161–182.

Jordà, Ò. 2020. "Local Projections: New Developments." Mimeo.

Jordà, Ò., M. Schularick, and A.M. Taylor. 2015. "Leveraged Bubbles." *Journal of Monetary Economics* 76 S1–S20.

Jordà, Ò., and A.M. Taylor. 2025. "Local projections." *Journal of Economic Literature* 63 (1): 59–110.

Lewis, D.J., and K. Mertens. 2022. "A Robust Test for Weak Instruments with Multiple Endogenous Regressors." *Mimeo.*

Liebman, Jeffrey B, and Neale Mahoney. 2017. "Do expiring budgets lead to wasteful year-end spending? Evidence from federal procurement." *American Economic Review* 107 (11): 3510–3549.

Matthes, Christian, Naoya Nagasaka, and Felipe F Schwartzman. 2025. "Estimating the missing intercept."

Mertens, Karel, and Morten O Ravn. 2013. "The dynamic effects of personal and corporate income tax changes in the United States." *American economic review* 103 (4): 1212–1247.

Nakamura, E., and J. Steinsson. 2014. "Fiscal Stimulus in a Monetary Union: Evidence from US Regions." *American Economic Review* 104 (3): 753–92. 10.1257/aer.104.3.753.

Plagborg-Møller, M., and M. Kolesar. 2025. "Dynamic causal effects in a nonlinear world: the good, the bad and the ugly." *mimeo*.

Plagborg-Møller, M., and C.K. Wolf. 2021. "Local Projections and VARs Estimate the Same Impulse Responses." *Econometrica* 89 (2): 955–980.

Plagborg-Møller, M., and C.K. Wolf. 2022. "Instrumental Variable Identification of Dynamic Variance Decompositions." *Journal of Political Economy* 130 (8): 2164–2202.

Primiceri, G. 2005. "Time-Varying Structural Vector Autoregressions and Monetary Policy." *The Review of Economic Studies* 72 (3): 821–852.

Ramey, V.A. 2011. "Can Government Purchases Stimulate the Economy?" *Journal of Economic Literature* 49 (3): 673–85.

Ramey, V.A., and S. Zubairy. 2018. "Government Spending Multipliers in Good Times and in Bad: Evidence from US Historical Data." *Journal of Political Economy* 126 (2): 850–901.

Ruisi, G. 2019. "Time-varying Local Projections." Working Paper.

Stock, J.H., and M.W. Watson. 2018. "Identification and Estimation of Dynamic Causal Effects in Macroeconomics Using External Instruments." *The Economic Journal* 128 (610): 917–948.

Sørensen, B.E., Y.-T. Wu, O. Yosha, and Y. Zhu. 2007. "Home bias and international risk sharing: Twin puzzles separated at birth." *Journal of International Money and Finance* 26 (4): 587–605.

Wolf, Christian K. 2023. "The missing intercept: A demand equivalence approach." *American Economic Review* 113 (8): 2232–2269.

Online Appendix for

"A New Approach to Fiscal Multipliers: Time Variation and Highly Disaggregated Shocks"

A Alternative Estimates of the Cumulative Multiplier **A-3**

B Additional Results **A-4**

B.1 Robustness: A Single End-of-Fiscal-Year Dummy A-4

B.2 Alternative Specification for Daily National Spending Shock Regression A-4

B.3 Robustness: Results Without Removing Common Factors A-6

B.4 Robustness: Results Purging Only the Instrument A-15

B.5 Alternative Specification for the County-level Fiscal Spending Shock: Including County-level GDP in the Regression A-18

B.6 Alternative Specification for County-level Fiscal Spending Shock: Not Controlling for GDP A-21

B.7 Time Variation in State-level Fiscal Multipliers A-24

C Theoretical Results **A-29**

C.1 Impulse Responses: Definition A-29

C.2 The TVP-PLP Estimator A-30

C.3 Asymptotic Properties A-33

C.4 Orthogonality A-35

C.5 Proof of Theorem 1 . . . . . A-37

C.6 Proof of Theorem 2 . . . . . A-38

D Monte Carlo Simulations **A-38**

D.1 DGP with Constant Coefficients A-39

D.2 DGP with Smooth Time Variation A-40

D.3 DGP with Random Walk A-41

D.4 DGP With One-Time Structural Break A-42
D.5 DGP with One-time Structural Break and VMA(10) A-44
D.6 DGP with Multivariate Instruments A-46

A Alternative Estimates of the Cumulative Multiplier

Ramey and Zubairy (2018) show that, in the case of constant parameters, the multiplier can be equivalently calculated using a three-step procedure. We implemented a time-varying version of the three-step approach to estimate eq. (18) and have verified that it provides similar parameter estimates. The time-varying approach was obtained as follows:

1. Estimate $\Theta^{(2,1)}_{h,t+h}$ from eq. (A-28) as follows:

$$\hat{\Theta}^{(2,1)}_{h,t+h} = \frac{\frac{1}{N}\sum_{i=1}^{N} \varepsilon^{F,X}_{i,t}\tilde{x}_{i,t+h}}{\frac{1}{N}\sum_{i=1}^{N} \varepsilon^{F,X}_{i,t}\tilde{g}_{i,t}},$$

   where $\tilde{x}_{i,t+h}$ and $\tilde{g}_{i,t}$ are the residuals of the regression of $x_{i,t+h}$ and $g_{i,t}$ on the set of controls $w_{i,t-1}$, which contains a constant, $X_{i,t-1}$, $G_{i,t-1}$ and $\varepsilon^{F,X}_{i,t-1}$.

2. Estimate $\Theta^{(1,1)}_{h,t+h}$ from

$$\hat{\Theta}^{(1,1)}_{h,t+h} = \frac{\frac{1}{N}\sum_{i=1}^{N} \varepsilon^{F,X}_{i,t}\tilde{g}_{i,t+h}}{\frac{1}{N}\sum_{i=1}^{N} \varepsilon^{F,X}_{i,t}\tilde{g}_{i,t}}$$

3. Then [1]

$$\hat{m}_{h,t+h} = \frac{\sum_{j=0}^{h} \hat{\Theta}^{(2,1)}_{j,t+j}}{\sum_{j=0}^{h} \hat{\Theta}^{(1,1)}_{j,t+j}}. \tag{A-1}$$

This approach is analytically equivalent to the one-step estimator in eq. (20). The estimates in steps 1 and 2 share the same denominator, $D_t \equiv \frac{1}{N}\sum_{i=1}^{N} \varepsilon^{F,X}_{i,t}\tilde{g}_{i,t}$, so that

$$\sum_{j=0}^{h} \hat{\Theta}^{(2,1)}_{j,t+j} = \frac{1}{D_t}\frac{1}{N}\sum_{i=1}^{N} \varepsilon^{F,X}_{i,t}\sum_{j=0}^{h}\tilde{x}_{i,t+j}, \qquad \sum_{j=0}^{h} \hat{\Theta}^{(1,1)}_{j,t+j} = \frac{1}{D_t}\frac{1}{N}\sum_{i=1}^{N} \varepsilon^{F,X}_{i,t}\sum_{j=0}^{h}\tilde{g}_{i,t+j},$$

and taking the ratio, D_t cancels:

$$\frac{\sum_{j=0}^{h} \hat{\Theta}^{(2,1)}_{j,t+j}}{\sum_{j=0}^{h} \hat{\Theta}^{(1,1)}_{j,t+j}} = \frac{\frac{1}{N}\sum_{i=1}^{N} \varepsilon^{F,X}_{i,t}\sum_{j=0}^{h}\tilde{x}_{i,t+j}}{\frac{1}{N}\sum_{i=1}^{N} \varepsilon^{F,X}_{i,t}\sum_{j=0}^{h}\tilde{g}_{i,t+j}},$$

which is exactly the cumulative multiplier estimated in one step in eq. (20). The advantage of the one-step approach is that its standard errors can be computed directly.

[1] Ramey and Zubairy (2018) claim that, in the constant parameter case, estimating $\hat{m}_h$ directly from their equation (3) on p. 864 (using $\varepsilon^{F,X}_{i,t}$ as instrument for $\sum_{j=0}^{h} g_{t+j}$) is equivalent.

B Additional Results

B.1 Robustness: A Single End-of-Fiscal-Year Dummy

In the construction of the daily national spending shock in our main analysis (equation (2)), we remove the end-of-fiscal-year seasonality with a full set of daily dummies for September 19–30, including a different dummy for each day. As a more parsimonious alternative, we replace those indicators with a single dummy for the whole period:

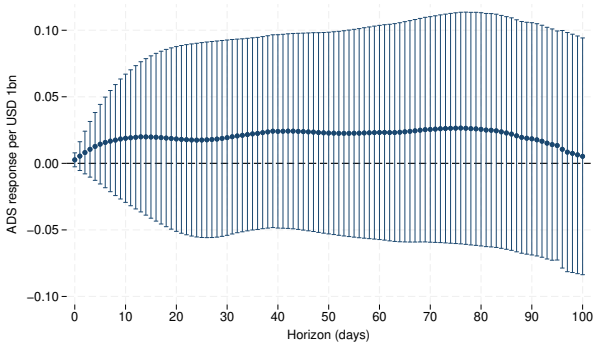
$$g_t^F = \alpha_g + \sum_{j=1}^{28} \phi_j g_{t-j}^F + \psi\, D_t + \epsilon_t^F, \qquad D_t = \mathbb{1}\{t \in [19,\ 20,\ \ldots\ 30 \text{ September}]\}. \tag{A-2}$$

This removes the average level of the window but not the predictable within-window variation, since obligations rise monotonically toward September 30. As a consequence, the resulting shock retains a forecastable end-of-fiscal-year component: a regression of the single-dummy shock on the dummy indicators $D_{d,t}$ of equation (2) still has an R^2 of about 0.07, against essentially zero for the day-specific shock used in the main text and 0.14 for the residual of the plain autoregression. The single-dummy and day-specific shocks have a correlation of 0.94.

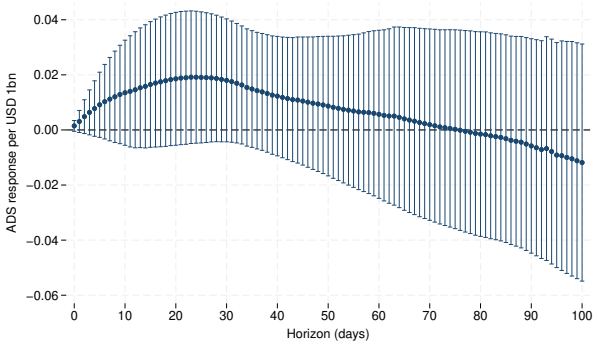
Figure A-1 reports the daily responses of the business conditions index when we consider the shock estimated as above. Over the full sample the response is insignificant, as in the main text. Across sub-samples, however, the estimates are less precise: the COVID (2020:4–2021:4) and 2017–2024 sub-samples display a significant response, whereas the response during the financial crisis (2007–2013) is insignificant. The reason is that the single dummy leaves a predictable component in the instrument—an anticipated component that adds noise without contributing valid identifying variation. Removing the end-of-the-year seasonality in full, as we do in the main text, eliminates this residual seasonality and delivers the sharper sub-sample responses reported there.

B.2 Alternative Specification for Daily National Spending Shock Regression

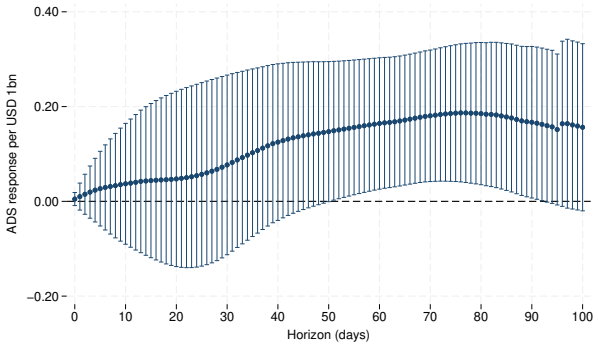
As a robustness exercise, we construct a second daily national spending shock. Starting from the same daily dataset used in section 2.3 (where contracts are aggregated by date of signature), we add the interest rate as an additional variable in the daily regression to control for the financial environment. In particular, we include the Effective Federal Funds Rate (EFFR), which we downloaded from FRED. The choice of the EFFR was driven by the fact that this is a short-term overnight rate, making it very sensitive to monetary



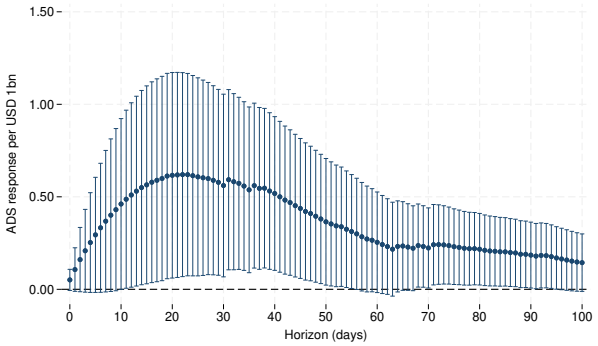
(a) Full sample estimates (2007–2024).



(b) Financial crisis and subsequent high-unemployment period (2007–2013).



(c) 2017–2024.



(d) COVID-19 pandemic (2020:4–2021:4).

Figure A-1: Daily response of the business conditions index under the single end-of-fiscal-year dummy of equation (A-2), with 90% confidence bands.

policy or liquidity conditions. Because the EFFR is available at the business-day frequency while the DoD spending is daily, we cover non-business-days by carrying the last observation forward (e.g. the value of the rate on Friday is carried over to Saturday and Sunday). Then, the national spending shock is constructed as the residual of the following regression[2]

$$g_t^F = \alpha_g + \sum_{j=1}^{28} \phi_j g_{t-j}^F + \gamma_g i_{t-1} + \epsilon_t^F \tag{A-3}$$

The shock is depicted in figure A-2. Its dynamics closely follows that in Figure 1. Then, we perform the same analysis as in Section 5, and uncover strikingly similar results, as shown in Figure A-3a-A-3d. The shapes of the impulse responses are extremely similar to the ones in the main paper, with the full-sample response remaining insignificant and the responses significant in all three sub-samples. Similarly to the results in the

[2]The number of lags is selected via BIC as in the paper.

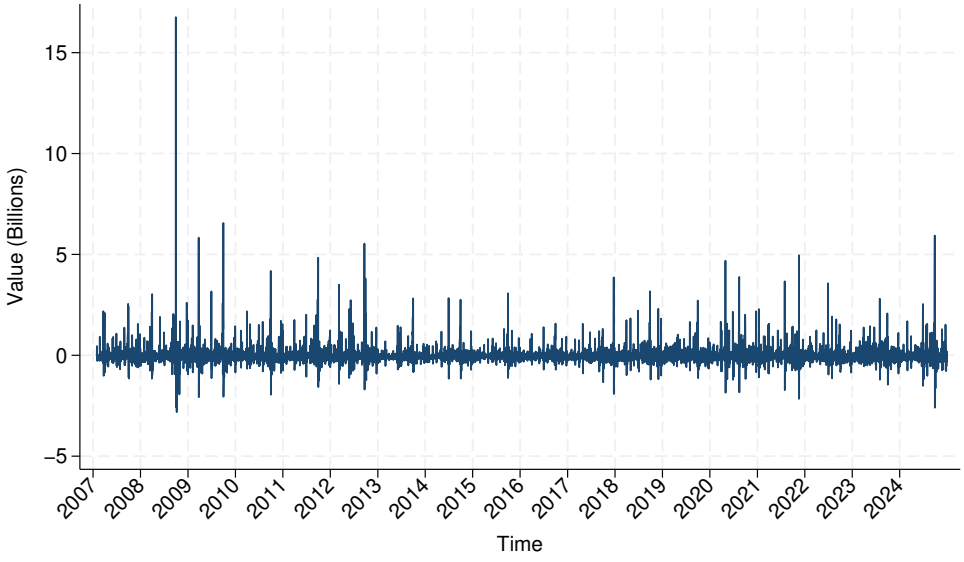


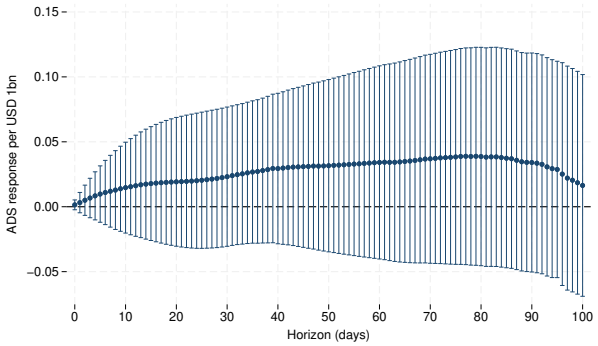
Figure A-2: Daily national fiscal spending shock, estimated using the alternative specification in eq. (A-3): 29/1/2007-31/12/2024.

main paper, the times when the responses are significant vary substantially across sub-samples: in the Great Financial Crisis period, the times are similar to those in Figure 8b – namely, roughly from two weeks to one month after the shock; in the 2017-2024 sub-sample, the response becomes significant between 50 and 90 days; and for the COVID sub-sample the response is only significant between 10 and 30 days. These results corroborate our intuition in the main text that the different responses cannot be rationalized under a state-dependent model.

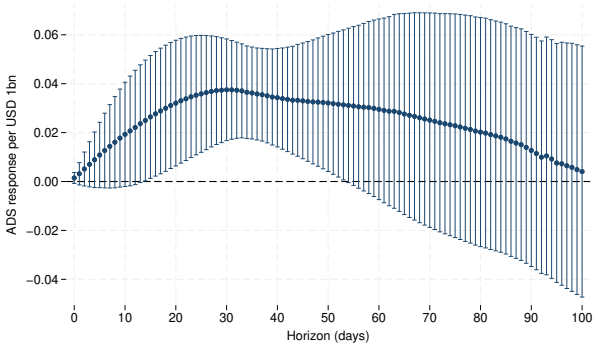
B.3 Robustness: Results Without Removing Common Factors

This section discusses how our main results would change if one used the raw variables – i.e., $X_{i,t}$, $G_{i,t}$, $U_{i,t}$ and $\varepsilon_{i,t}^{F,X}$ directly, –rather than their purged counterparts, as we do in Section 4.2. Table A-1 compares the impact and the one-year cumulative multipliers of the raw and the purged specifications – the latter with either one and two common factors. Figures A-4–A-7 report the full set of results for the raw variables in the same format as that in the main text, and Figures A-8–A-9 report the responses of GDP and unemployment in the constant parameter case, when using raw variables; the latter, similarly to the purged counterpart case in Figures 5 and 7 in the main text, are small and insignificant at every horizon.

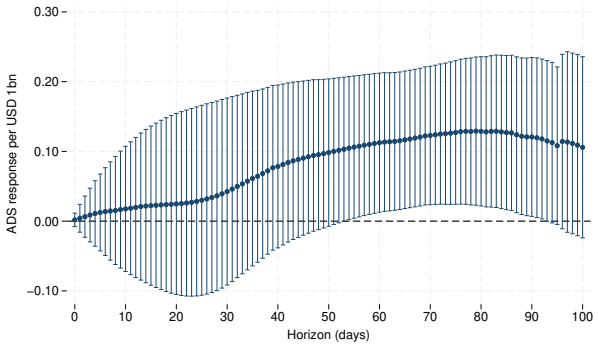
The strength of the raw instrument, assessed again via the test of Lewis and Mertens (2022) year by year



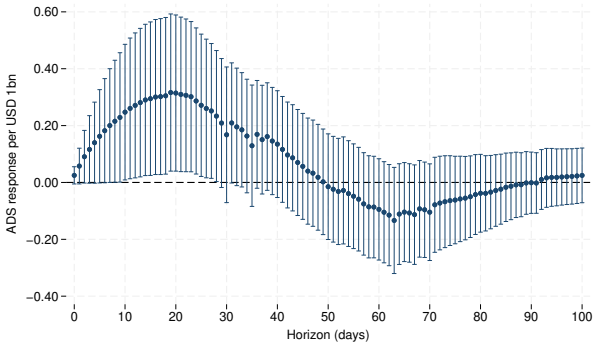
(a) Daily response of the business conditions index: Full sample estimates (2007-2024).



(b) Daily response of the business conditions index: the financial crisis and the subsequent high unemployment period (2007-2013).



(c) Daily response of the business conditions index: 2017-2024.



(d) Daily response of the business conditions index during the COVID-19 pandemic (2020:4-2021:4).

Figure A-3: Daily response of the ADS business conditions index to an increase of one billion dollars in DoD obligations, using the alternative shock of eq. (A-3) as instrument, with 90% confidence bands. Sub-samples as in Figure 8.

and horizon by horizon, is very similar to that of the purged one, reported in Section 4.4.[3]

Table A-1: Impact and Cumulative Multipliers: Raw versus Purged Specifications

	Raw	Purged: One Factor	Purged: Two Factors
Panel A: Impact Multiplier, $\theta_{0,t}$			
2018	0.28*	0.31*	0.29*
2019	0.35*	0.35*	0.37*
2020	−0.07	−0.05	−0.09
2021	−0.10	0.03	0.07
2022	0.10	−0.01	0.02
Panel B: Cumulative Multiplier, $m_{1,t+1}$			
2018	0.49	0.61	0.52
2019	0.62*	0.52*	0.55*
2020	−0.20	−0.16	−0.23
2021	−0.29	−0.08	0.11
2022	0.23	0.07	0.30

Notes: Results reported in the column labeled "Raw" are based on the variables described in Section 4 and the shock $\varepsilon_{i,t}^{F,X}$ from equation (1). Results reported in the column labeled "Purged" uses the variables and the instrument purged of one or two common factors extracted from the stacked system, with county-specific loadings estimated over 2008–2023 (Section 4.2). The column labeled "Purged: one-factor" is exactly the same as our baseline specification in the main text. Panel B reports the cumulative multiplier one period after the shock. $N = 2158$ in every year and specification. Asterisks denote significance at the 10% level.

The estimates in the raw specification are very close to the purged one in the main text. The impact multipliers of 2018 and 2019 are 0.28 and 0.35, both significantly different from zero, against 0.31 and 0.35 in the main specification; the pandemic-year multipliers are negative or zero and never significantly different from zero; the 2022 impact multiplier is 0.10 and insignificantly different from zero, against −0.01. The one-year cumulative multiplier of the 2019 shock is 0.62 in the raw specification, and 0.52 in the main one, and both are significantly different from zero. The unemployment responses (Figure A-7) show the same pattern as in the main text: the 2018 shock lowers unemployment significantly one and three years ahead, and the pandemic-year shocks have no significant effect. Adding a second common factor leaves the purged estimates essentially unchanged.

[3]Here are the results in detail. Weak IV tests for the raw specification, GDP as outcome: for 2019 and 2022 (and 2023) we reject the null hypothesis of weak IV at all horizons and for all the tolerances considered ($\tau \in \{0.10, 0.15\}$, $\alpha \in \{0.05, 0.10\}$); for 2018 we reject the null at all horizons only for a bias threshold of 0.15 and a significance level of 0.10; for 2020 and 2021 we cannot reject the null at these tolerances. Unemployment as outcome: we reject the null at all horizons and for all the tolerances considered in every year. The only difference with the purged instrument is 2022, which passes the test without the purge and not with it (generalized statistic 42.3 against 13.9, critical value 23.1 at $\alpha = 0.05$, $\tau = 0.10$); the conventional first-stage F statistic exceeds 400 in every year in both cases.

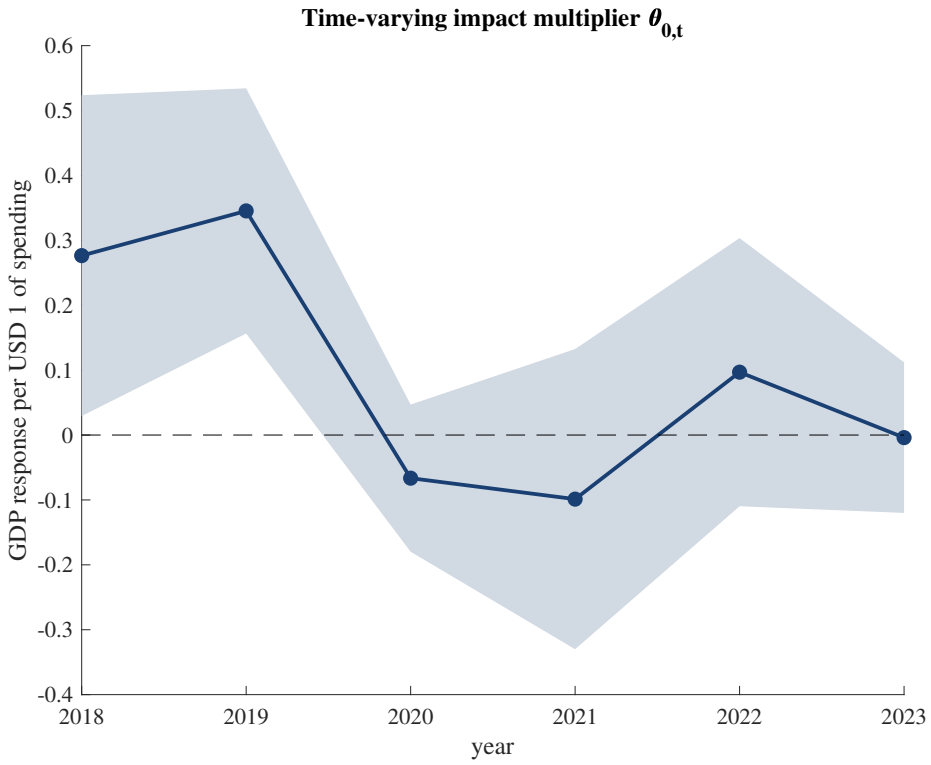
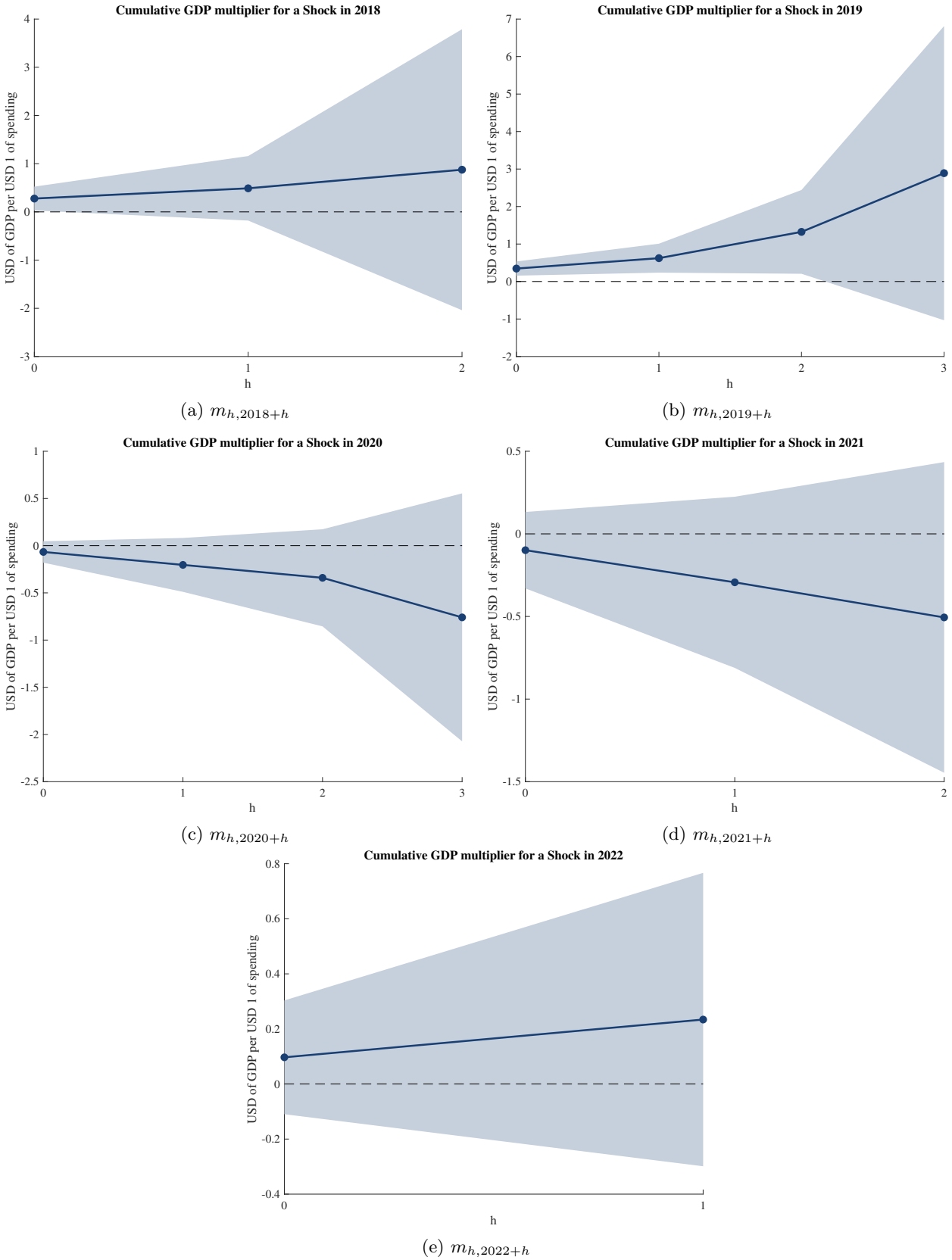


Figure A-4: Fiscal multiplier on impact ($h = 0$), together with 90% confidence bands. The regression uses raw time series variables.



(a) $m_{h,2018+h}$

(b) $m_{h,2019+h}$

(c) $m_{h,2020+h}$

(d) $m_{h,2021+h}$

(e) $m_{h,2022+h}$

Figure A-5: Time-varying cumulative multipliers, together with 90% confidence bands. The regression uses raw time series variables.

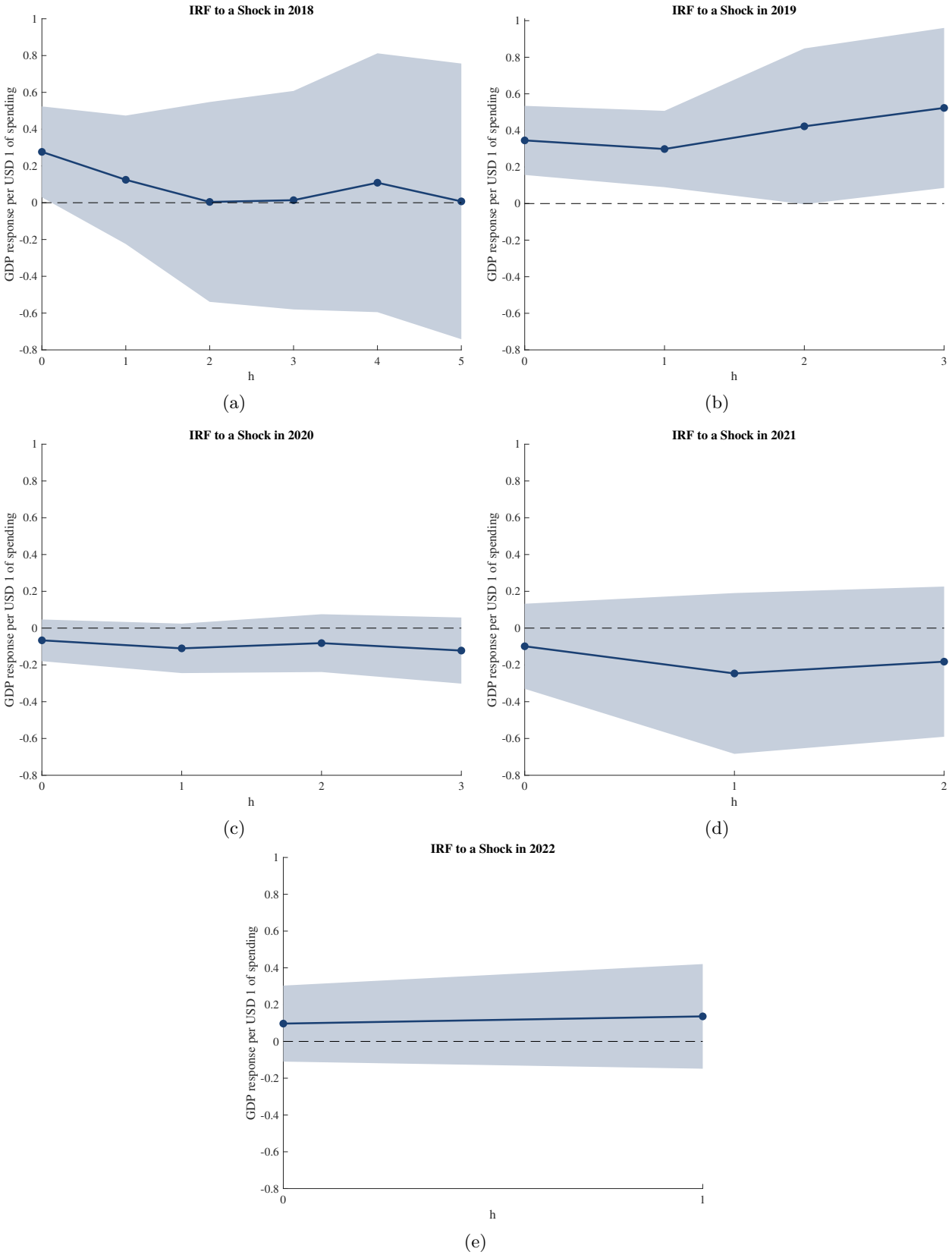


Figure A-6: Time-varying responses of GDP ($\theta_{h,t+h}$), for time t equal to: 2018 (top left panel), 2019 (top right panel), 2020 (bottom left panel), 2021 (bottom right panel), 2022 (bottom center panel). Shaded areas represent 90% confidence bands. The regression uses raw time series variables.

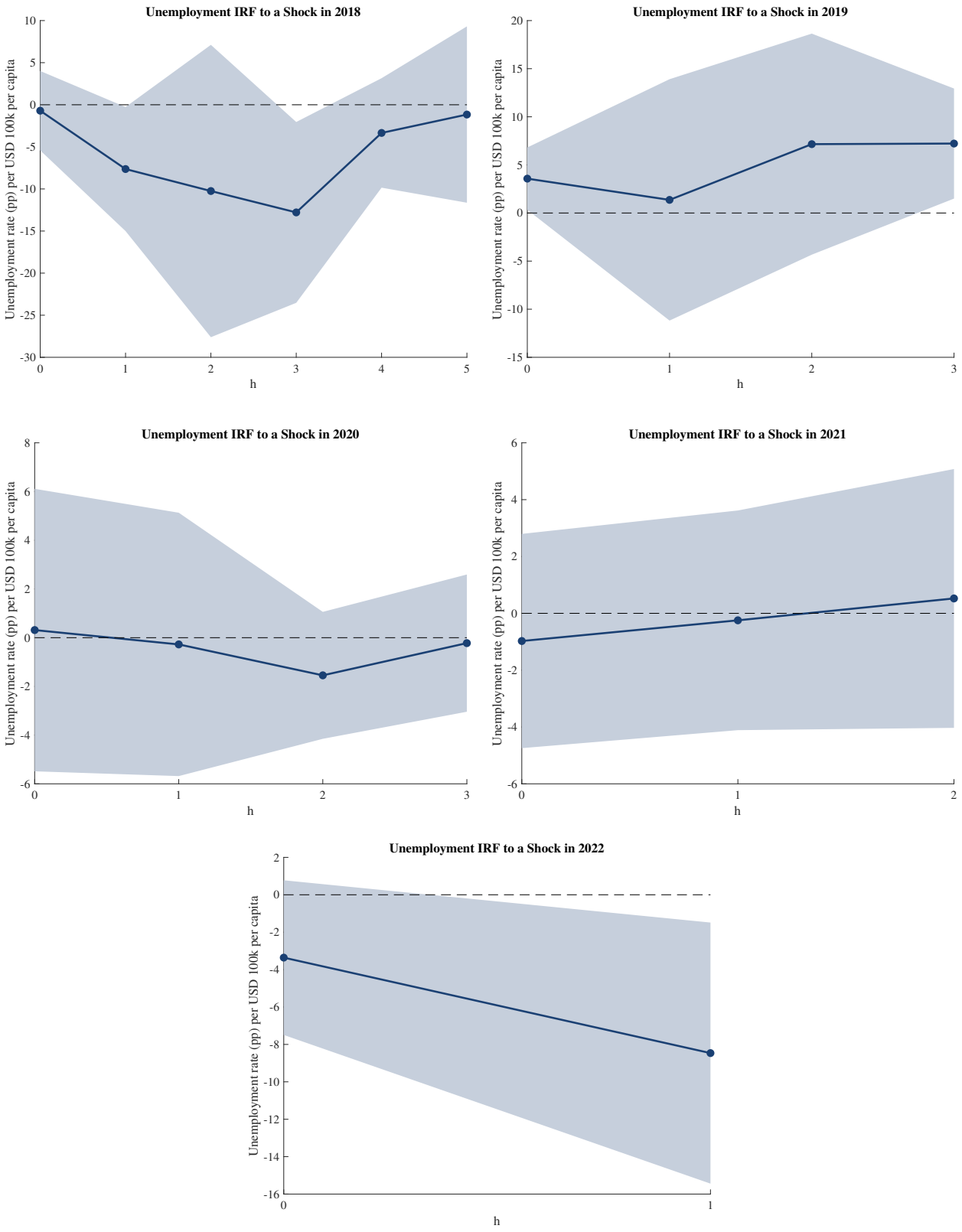


Figure A-7: Time-varying responses of unemployment ($\theta_{h,t+h}$), for time t equal to: 2018 (top left panel), 2019 (top right panel), 2020 (bottom left panel), 2021 (bottom right panel), 2022 (bottom center panel). Shaded areas represent 90% confidence bands. The regression uses raw time series variables.

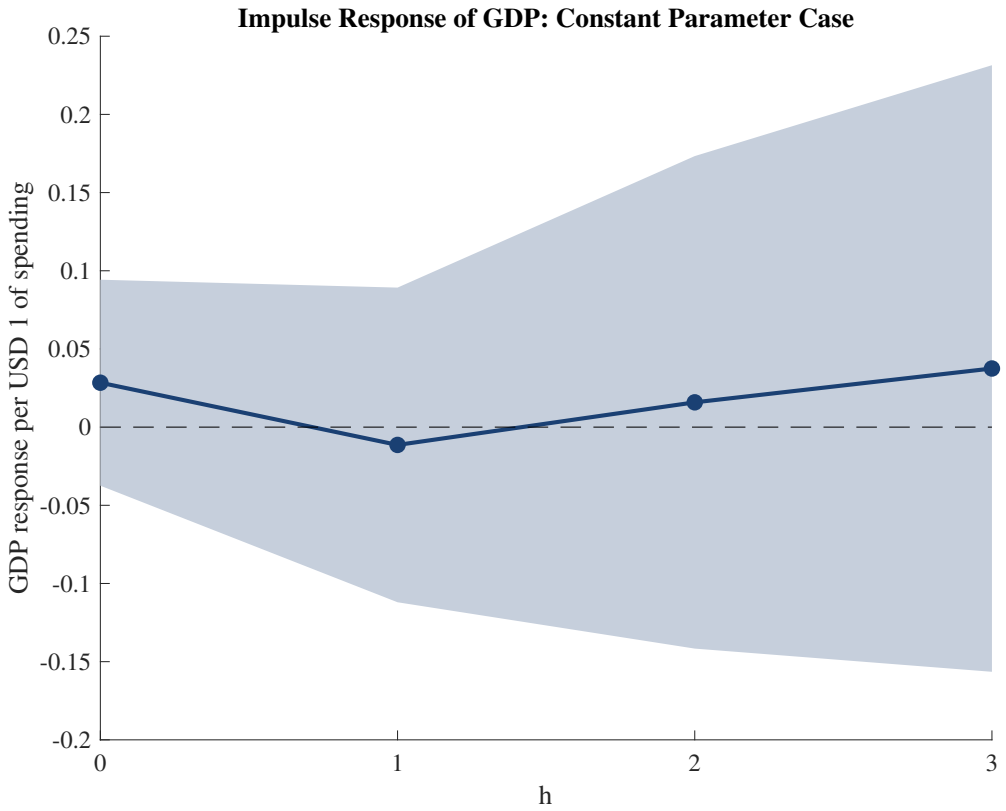


Figure A-8: Response of GDP: Constant parameter case (θ_h), together with 90% confidence bands. The regression uses raw time series variables.



Figure A-9: Response of unemployment: Constant parameter case (θ_h), together with 90% confidence bands. The regression uses raw time series variables.

B.4 Robustness: Results Purging Only the Instrument

As discussed in our main text, Donaldson (2026) shows that cross-sectional designs that exploit heterogeneous exposure to an aggregate shock are biased when units are also heterogeneously exposed to the general-equilibrium response of other aggregate variables to that shock. This section states the condition under which our year-by-year design is robust to this critique, and verifies it empirically for the unpurged specification of Appendix B.3 by purging the instrument alone. The exercise complements the main specification, which purges all variables: if the bias is absent when only the instrument is corrected, the correction of the outcome and spending variables in Section 4.2 cannot be what drives the results.

Section 4.2 shows that the IV estimand (eq. 14) equals the idiosyncratic multiplier (eq. 16) if and only if $\Lambda_{z,2}\sigma_{z1} = \Lambda_{z,1}\sigma_{z2}$, and that this holds whenever: either the instrument carries no common component, $\lambda_{z,i} = 0$ for all i, which we refer to as condition (C1); or its loadings are cross-sectionally uncorrelated with those of spending and of the outcome, $\Lambda_{z,1} = \Lambda_{z,2} = 0$, which we refer to as condition (C2). Alternatively, the equality also holds when the exposure is not heterogeneous – that is, $\lambda_{i,1} = \lambda_{i,2} = \lambda_i$ – and the covariances are symmetric – that is, $\sigma_{z,1} = \sigma_{z2} = \sigma_z$, – even though $\Lambda_{z,i} \neq 0$ (C3). Four remarks are in order. First, in the language of Donaldson (2026), $\lambda_{2,i}$ is county i's exposure to the general-equilibrium response of aggregate conditions, $\lambda_{z,i}$ is the loading of the identifying variation on the aggregate shock, and $\Lambda_{z,2}$ is the cross-sectional covariance of exposures that drives her bias term; in a shift-share design $\lambda_{z,i}$ is the exposure share itself, so (C1) fails by construction and robustness requires (C2), which cannot be verified from the design alone. Second, either condition works at every horizon: replacing $y^{(2)}_{i,t}$ by $y^{(2)}_{i,t+h} = \lambda_{2,i} f_{t+h} + e^{(2)}_{i,t+h}$ changes only the numerator of eq. 14, which becomes $\Lambda_{z,2}E(f_t f_{t+h}) + E(e^{(z)}_{i,t} e^{(2)}_{i,t+h})$, and under (C1) or (C2) the first term vanishes for every h. Third, the purge of Section 4.2, and the instrument-only purge below, enforce (C1) with respect to the estimated factors, up to estimation error in $\hat{f}_t$ and $\hat{\lambda}_i$, under the maintained assumption that the loadings are stable over the 2008–2023 sample on which they are estimated. Finally, in the year-by-year regressions we actually estimate, f_t is a constant across counties, and, because every regression includes an intercept, $\Lambda_{z,k}$ is replaced by the cross-sectional covariance of the loadings; the condition is otherwise identical, year by year. Because the bias operates only through the instrument, the natural empirical check is to purge the instrument alone, which we do next. Note that purging all the time series variables and not solely the instrument, as we do in the main text, ensures not only that we avoid the bias discussed in Donaldson (2026), but also ensures that our estimate is valid under quite general conditions and spatial correlation structures induced by the common components, as we discuss in the main text.

We arrange the estimated shocks of the counties in the estimation sample as a balanced matrix of

$N = 2158$ counties by $T = 16$ years (2008–2023) and extract principal components from its covariance matrix.[4] We then project each county's shock on the estimated factors, allowing the loadings to differ across counties,

$$\varepsilon_{i,t}^{F,X} \;=\; a_i \;+\; \lambda_i' \hat{f}_t \;+\; \breve{\varepsilon}_{i,t}^{F,X}, \qquad t = 2008, \ldots, 2023, \tag{A-4}$$

and use the residual $\breve{\varepsilon}_{i,t}^{F,X}$ as the instrument in place of $\varepsilon_{i,t}^{F,X}$, with all other variables unpurged. We consider one and two estimated factors. The loadings are county-specific and estimated over the full sample of data we have available for the shocks (2008–2023), rather than over the shorter sample on which we estimate the local projections. We restrict attention to at most two factors because, with county-by-county loadings, the number of factors is limited by the number of time-series observations available for each county. The purge removes about 9% of the variance of the shock over the estimation years with one factor (against less than 1% with the system-wide factor of Section 4.2, which is extracted mainly from GDP and unemployment), indicating that the instrument is already almost entirely idiosyncratic.

Table A-2 reports the impact and cumulative multipliers and Table A-3 the impulse responses, next to the estimates based on the raw variable specification. The estimates are essentially unchanged. The impact multiplier in 2018 is 0.29 and 0.27 with one and two factors respectively, against 0.28 without the purge, and that in 2019 is 0.35 and 0.31 against 0.35, all significant. For comparison, our main specification, which purges all variables of the system-wide factor, gives 0.31 and 0.35 (Table A-1); for the pre-pandemic years the three sets of estimates are within 0.03 of each other on impact, so that none of the small differences across specifications comes from the instrument. For the pandemic years the specifications differ by more (up to 0.16 on impact in 2021–2022), but all estimates are insignificant and well within each other's confidence bands. The one-year cumulative multiplier of the 2019 shock is 0.69 and 0.79 with one and two factors, against 0.62 without the purge, all significant. The multipliers in 2020 and 2021 are negative or essentially zero, and are never significantly different from zero, while the multipliers in 2022 are positive and insignificant on impact. The impulse responses tell the same story: real output per capita responds positively and significantly to the 2019 shock at all horizons and does not respond significantly to the shocks of 2020 and 2021 at any horizon. The one difference relative to the main specification concerns 2022: when we only purge the instrument, the one-year-ahead response of output is 0.26–0.27 and significantly different from zero (Table A-3), whereas in the main specification it is positive but insignificant (Figure 4e); given the weak instrument problem in 2022 (cfr. Section 4.4), such difference is not unexpected. As for unemployment, the responses are very close to

[4] We restrict the factor extraction to the counties that enter the estimation. Counties without county-level GDP – notably a number of Alaska boroughs with very large and volatile procurement – are not used in any regression, and including them in the extraction lets a handful of extreme series dominate the covariance principal components.

Table A-2: Impact and Cumulative Multipliers: Purging Only the Instrument

	Raw	Purged: One Factor	Purged: Two Factors
Panel A: Impact Multiplier $\theta_{0,t}$			
2018	0.28*	0.29*	0.27*
2019	0.35*	0.35*	0.31*
2020	−0.07	−0.06	−0.06
2021	−0.10	−0.04	0.01
2022	0.10	0.15	0.15
Panel B: Cumulative Multiplier $m_{1,t+1}$			
2018	0.49	0.54	0.47
2019	0.62*	0.69*	0.79*
2020	−0.20	−0.19	−0.17
2021	−0.29	−0.18	−0.03
2022	0.23	0.58	0.57

Notes: The results reported in the column labeled "Raw" are based on regressions where the shock is $\varepsilon_{i,t}^{F,X}$ from eq. (1) and the time series variables are raw; the results reported in the columns labeled "Purged: One Factor" and "Purged: Two Factors" are based on regressions where the shock $\tilde{\varepsilon}_{i,t}^{F,X}$ is estimated from eq. (A-4) with one and two factors respectively, where the latter are estimated from the panel of shocks from the 2158 counties, with county-specific loadings estimated over 2008–2023, and the time series outcome and spending variables are raw. Panel B reports the cumulative multiplier one period after the shock. Asterisks denotes significance at the 10% level.

those from the raw variable specification, with the 2018 shock lowering unemployment significantly three years ahead and the 2022 shock one year ahead. We conclude that the common component in the instrument is not crucial for our results and that the identification from idiosyncratic county-level variation ensured by the disaggregated shock plays a major role, as expected.

Table A-3: Impulse Responses: Purging Only the Instrument

		GDP Response			Unemployment Response		
		Raw	Purged: One Factor	Purged: Two Factors	Raw	Purged: One Factor	Purged: Two Factors
2018	$h=0$	0.28*	0.29*	0.27*	−0.70	−0.35	−0.27
	$h=1$	0.12	0.17	0.13	−7.64*	−6.04	−5.95
	$h=2$	0.00	0.13	0.11	−10.25	−10.92	−10.98
	$h=3$	0.01	0.11	0.10	−12.79*	−14.19*	−13.90*
2019	$h=0$	0.35*	0.35*	0.31*	3.57*	4.11*	4.15*
	$h=1$	0.30*	0.33*	0.32*	1.36	0.84	−1.02
	$h=2$	0.42	0.44*	0.51*	7.16	4.94	4.36
	$h=3$	0.52*	0.54*	0.67*	7.22*	6.60*	6.02
2020	$h=0$	−0.07	−0.06	−0.06	0.31	0.34	−0.17
	$h=1$	−0.11	−0.11	−0.09	−0.28	0.43	0.31
	$h=2$	−0.08	−0.09	−0.05	−1.55	−1.26	−1.45
	$h=3$	−0.12	−0.12	−0.08	−0.22	−0.06	−0.25
2021	$h=0$	−0.10	−0.04	0.01	−0.97	0.55	0.69
	$h=1$	−0.25	−0.15	−0.05	−0.25	0.31	−0.18
	$h=2$	−0.18	−0.06	0.04	0.52	1.08	0.61
2022	$h=0$	0.10	0.15	0.15	−3.36	−4.73	−4.82
	$h=1$	0.14	0.26*	0.27*	−8.46*	−11.01*	−11.14*

Notes: See Table A-2. Response coefficients ($\theta_{h,t+h}$) are reported up to horizon $h = 3$, or to the latest horizon available given the end of the sample. Asterisks denote significance at the 10% level.

B.5 Alternative Specification for the County-level Fiscal Spending Shock: Including County-level GDP in the Regression

The robustness exercises in this sub-section and the next use the raw variables as in Appendix B.3. In this sub-section, we estimate a second county-level fiscal policy shock, $\varepsilon_{i,t}^{F,X_i}$, from a regression that includes county-level GDP as an additional control variable. The shock is estimated as the residual of the following regression:

$$G_{i,t} = \alpha_{g,i} + \sum_{j=1}^{p_i^*} \beta_j G_{i,t-j} + \gamma_i G_{t-1} + \delta_i X_{i,t-1} + \varepsilon_{i,t}^{F,X_i}, \tag{A-5}$$

where $G_{i,t}$ is county-level fiscal spending, $G_t = \sum_i G_{i,t}$ is aggregate fiscal spending at the national level, and $X_{i,t}$ is county-level GDP. All variables are in real, per capita terms, as in the main paper. Note that, as county-level GDP is only available annually, this specification can only be estimated at the yearly frequency. County-level GDP data are available since 2007, and thus it can be included in our sample. As in the main specification, the lag length p_i^* is selected using the BIC criterion. Note that, since the analysis in this section and the next use the raw variables of Appendix B.3, the relevant comparison is with Figures A-4–A-7.

We again perform a series of weak instruments tests by Lewis and Mertens (2022) to verify the strength

of our instrument for most of the sample, similarly to those for the main shock in Section 4.4.[5]

Then, we use this alternative shock to replicate the analysis performed in Section 4 in the main text. Figure A-10 shows the estimated multiplier on impact, estimated from eq. (19). The values are very similar to those in Figure A-4, the main difference being that, under this specification, the impact multiplier is slightly higher in 2018 and 2019.

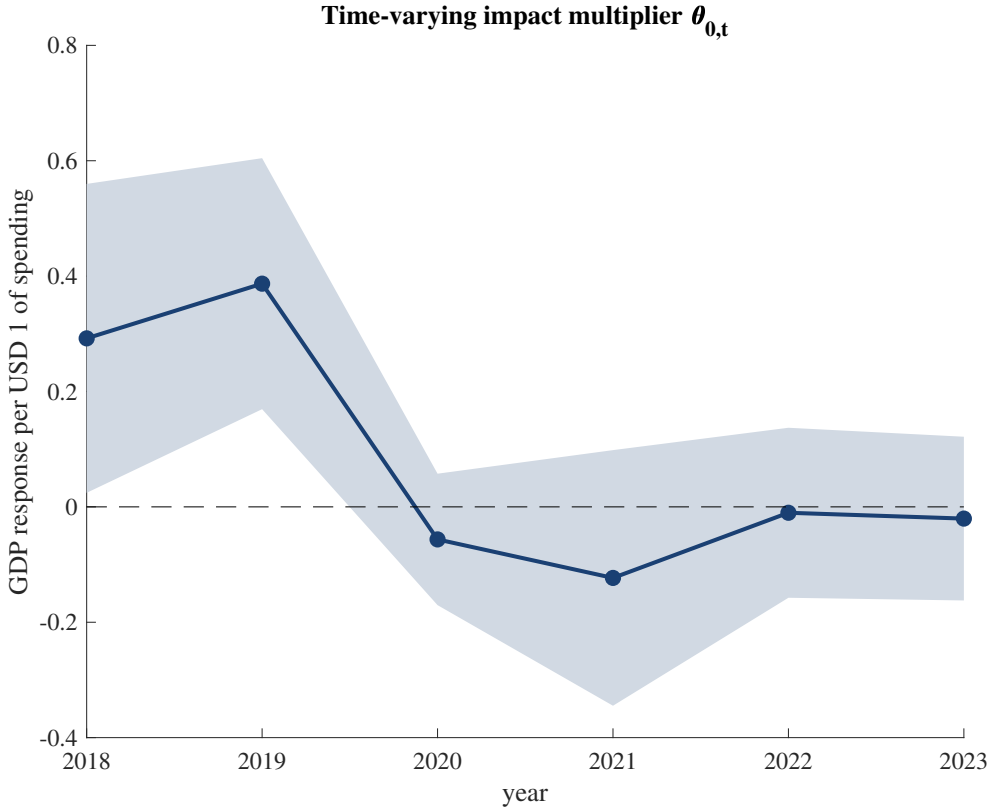
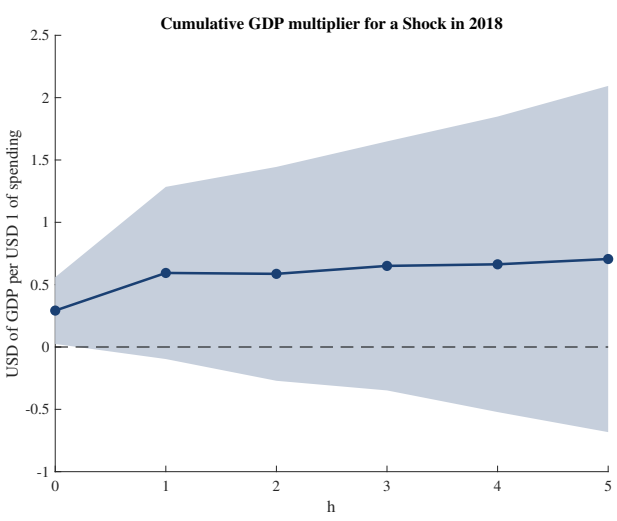


Figure A-10: Fiscal multiplier on impact ($h = 0$), together with 90% confidence bands.

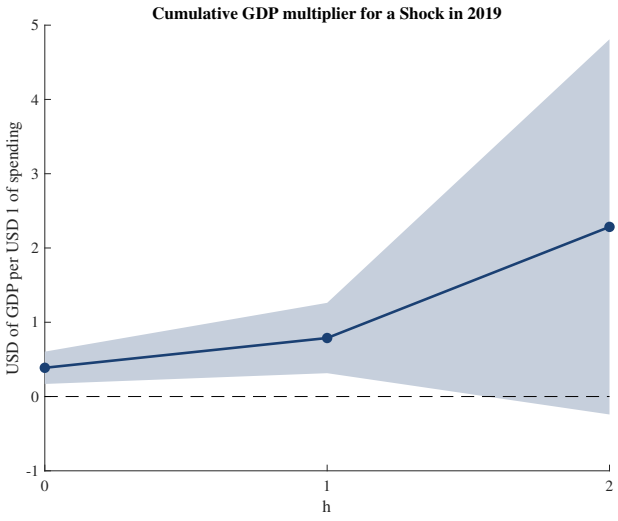
Figures A-11a-A-11e plot the h-period-ahead cumulative multipliers, together with the associated 90% confidence intervals. The results are again in line with the main specification, with multipliers being positive and increasing in the years prior to the COVID pandemic.[6] The only difference is the impact multiplier for a shock in 2018, as already seen in Figure A-10, as well as a flatter shape for the multiplier in 2022. However, the latter is not significant in either specification.

[5]Weak IV tests for GDP as outcome: for 2023 we reject the null hypothesis of weak IV at all horizons and for all the tolerances considered ($\tau \in \{0.10, 0.15\}$, $\alpha \in \{0.05, 0.10\}$); for 2018 we reject the null at all horizons except for a bias threshold of 0.10 at a significance level of 0.05; for 2020 and 2021 we reject the null at all horizons only for a bias threshold of 0.15 and a significance level of 0.10; for 2019 and 2022 we cannot reject the null at these tolerances. Unemployment as outcome: for 2018, 2020, 2021 and 2023 we reject the null at all horizons and for all the tolerances considered; for 2022 we reject the null at all horizons except for a bias threshold of 0.10 at a significance level of 0.05; for 2019 we cannot reject the null at these tolerances.

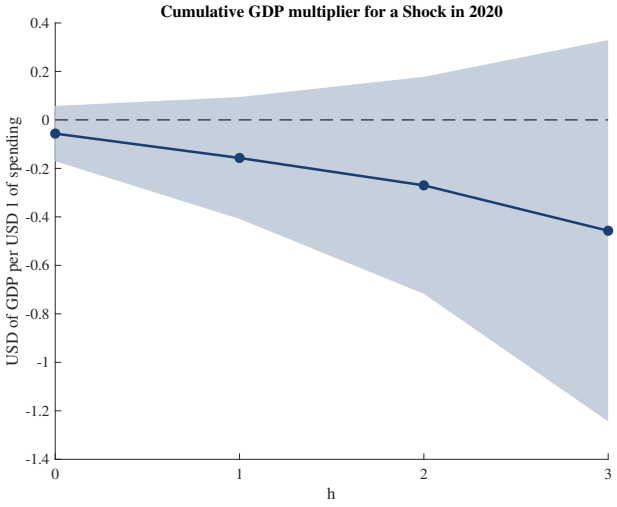
[6]Note that, differently from Figure 3b, we only display the results for the 2019 shock up to $h = 2$, because the large uncertainty around the estimate at longer horizons in this case.



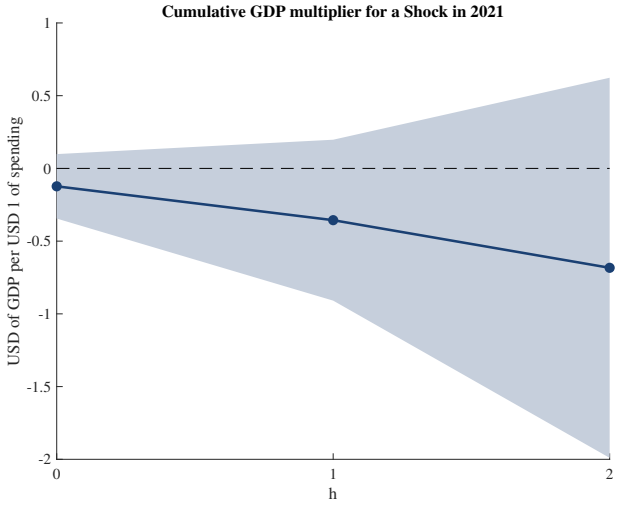
(a) $m_{h,2018+h}$



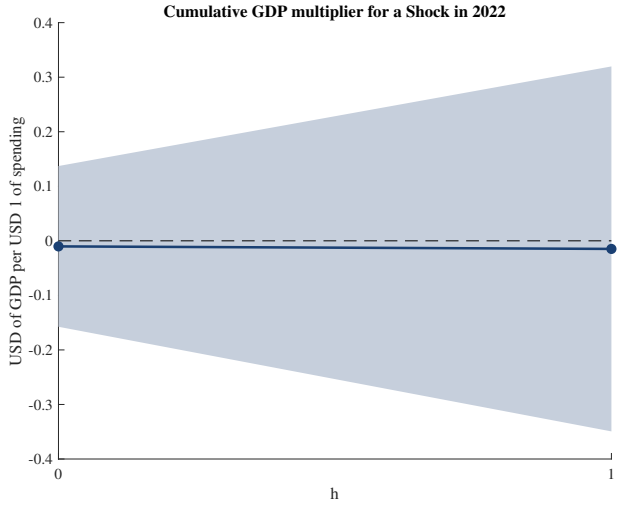
(b) $m_{h,2019+h}$



(c) $m_{h,2020+h}$



(d) $m_{h,2021+h}$



(e) $m_{h,2022+h}$

Figure A-11: Time-varying cumulative multipliers, together with 90% confidence bands.

Turning to time-varying impulse responses, we estimate them using eq.(21), both for GDP and unemployment. Figure A-12 plots the response of per capita, real GDP to a shock in military spending associated with a unitary increase in real per capita procurement spending.[7] Figure A-13 instead shows the unemployment response. Comparing figure A-13 with its raw specification counterpart in Figures A-6 and A-7, we again confirm very similar results, with the exception of the responses for 2022.

B.6 Alternative Specification for County-level Fiscal Spending Shock: Not Controlling for GDP

As a second robustness exercise for our county-level fiscal spending shock, we estimate a third shock, $\varepsilon_{i,t}^{F,0}$, without including aggregate GDP as a control. That is, the shock is the residual of the following regression:

$$G_{i,t} = \alpha_{g,i} + \sum_{j=1}^{p_i^*} \beta_j G_{i,t-j} + \gamma_i G_{t-1} + \varepsilon_{i,t}^{F,0} \tag{A-6}$$

where, differently from eq. (1), we do not control for aggregate GDP, X_{t-1}. This exercise, too, uses the raw variables, as in Appendix B.3. For comparison purposes, this is also estimated at the yearly frequency. Again, we perform weak instruments tests by Lewis and Mertens (2022) to verify the strength of our instrument.[8] Similarly to Section B.5, we perform the same analysis as in Section 4. Figure A-14 displays the impact GDP multiplier, estimated from eq.(19), which is very similar to the one reported in Figures A-4 and A-10. Figures A-15a-A-15e plot the h-year-ahead cumulative multipliers and their 90% confidence bands. The estimates are, once again, extremely similar to the previous specifications, with the main difference being the 2018 multiplier, which displays much less uncertainty.

Next, we estimate the time-varying impulse responses using eq.(21), for GDP and unemployment. Figure A-16 plots the impulse response of real GDP per capita to a shock in military spending that is associated with a unit increase in DoD spending per capita in different years. Figure A-17 plots the response of unemployment to the same shock. Once again, the plots confirm the robustness of our results.

[7] Again, as in the main text, the normalization is done separately at each point in time, due to the time-varying nature of our estimator.

[8] Detailed results are as follows: Weak IV tests for GDP as outcome: for 2018, 2019, 2022 and 2023 we reject the null hypothesis of weak IV at all horizons and for all the tolerances considered ($\tau \in \{0.10, 0.15\}$, $\alpha \in \{0.05, 0.10\}$); for 2020 we reject the null at all horizons for a bias threshold of 0.15 (at both significance levels), but not for a bias threshold of 0.10; for 2021 we cannot reject the null at these tolerances, although narrowly (generalized statistic 14.5 against a critical value of 14.8 at $\tau = 0.15$, $\alpha = 0.10$). Unemployment as outcome: we reject the null at all horizons and for all the tolerances considered in every year from 2018 to 2023.

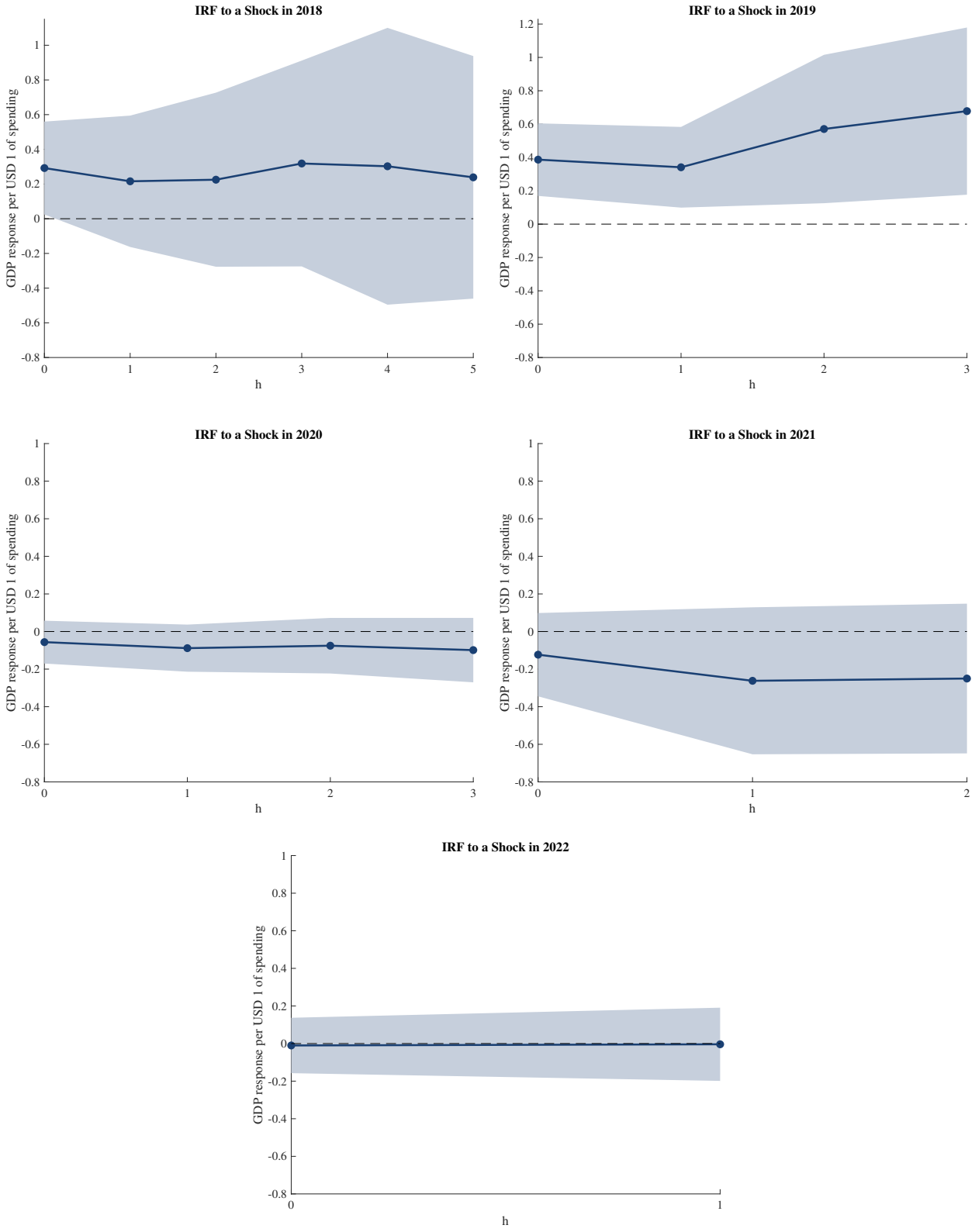


Figure A-12: Time-varying responses of GDP ($\theta_{h,t+h}$) for time t equal to: 2018 (top left panel), 2019 (top right panel), 2020 (bottom left panel), 2021 (bottom right panel), 2022 (bottom center panel). Shaded areas represent 90% confidence bands.

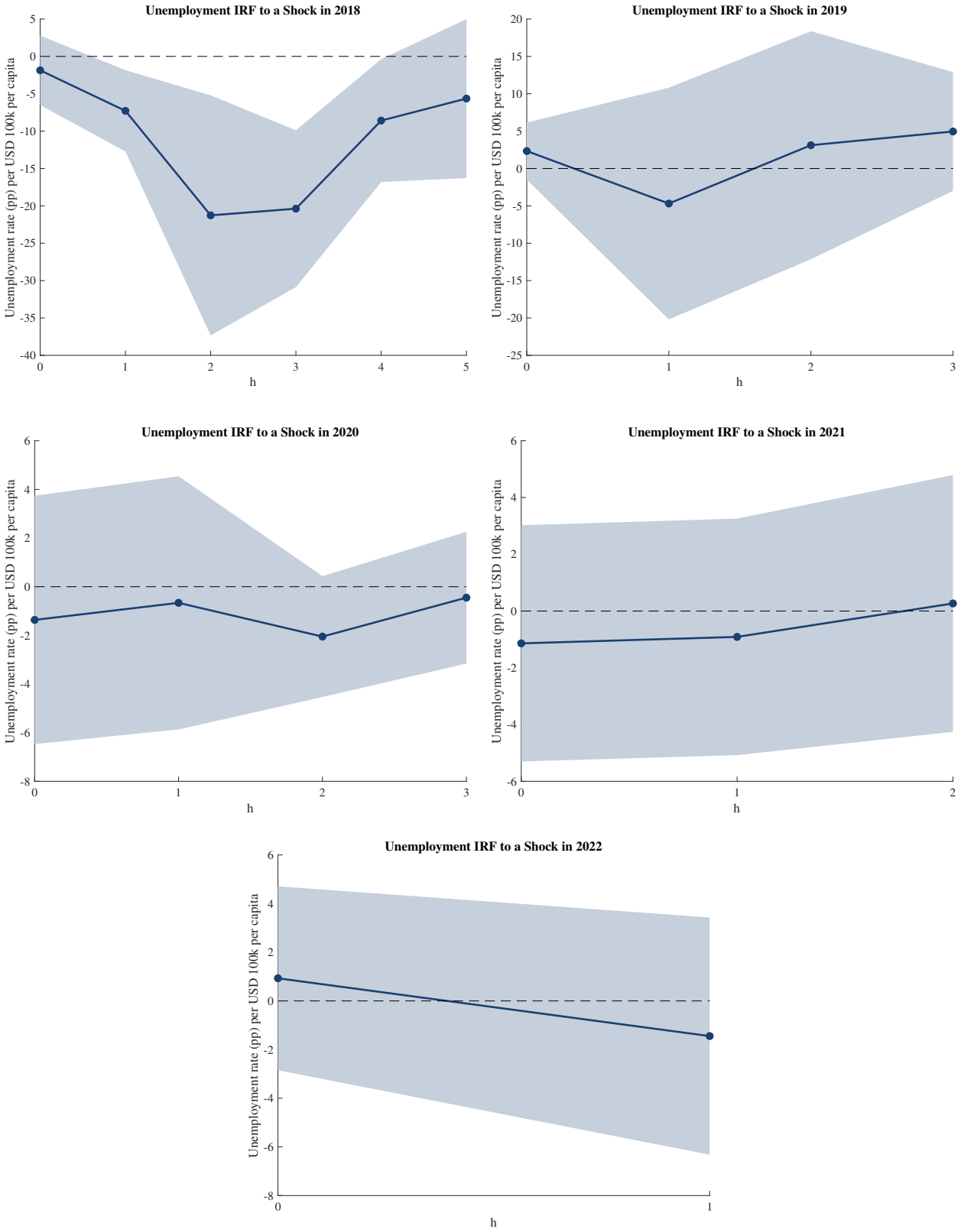


Figure A-13: Time-varying responses of unemployment ($\theta_{h,t+h}$) for time t equal to: 2018 (top left panel), 2019 (top right panel), 2020 (bottom left panel), 2021 (bottom right panel), 2022 (bottom center panel). Shaded areas represent 90% confidence bands.

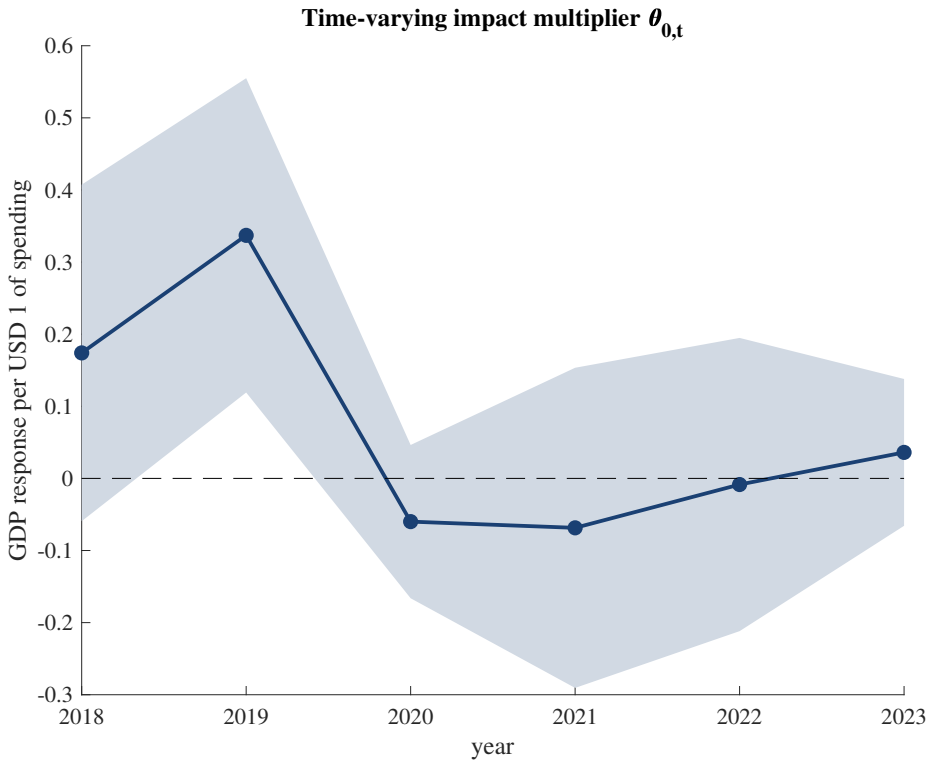
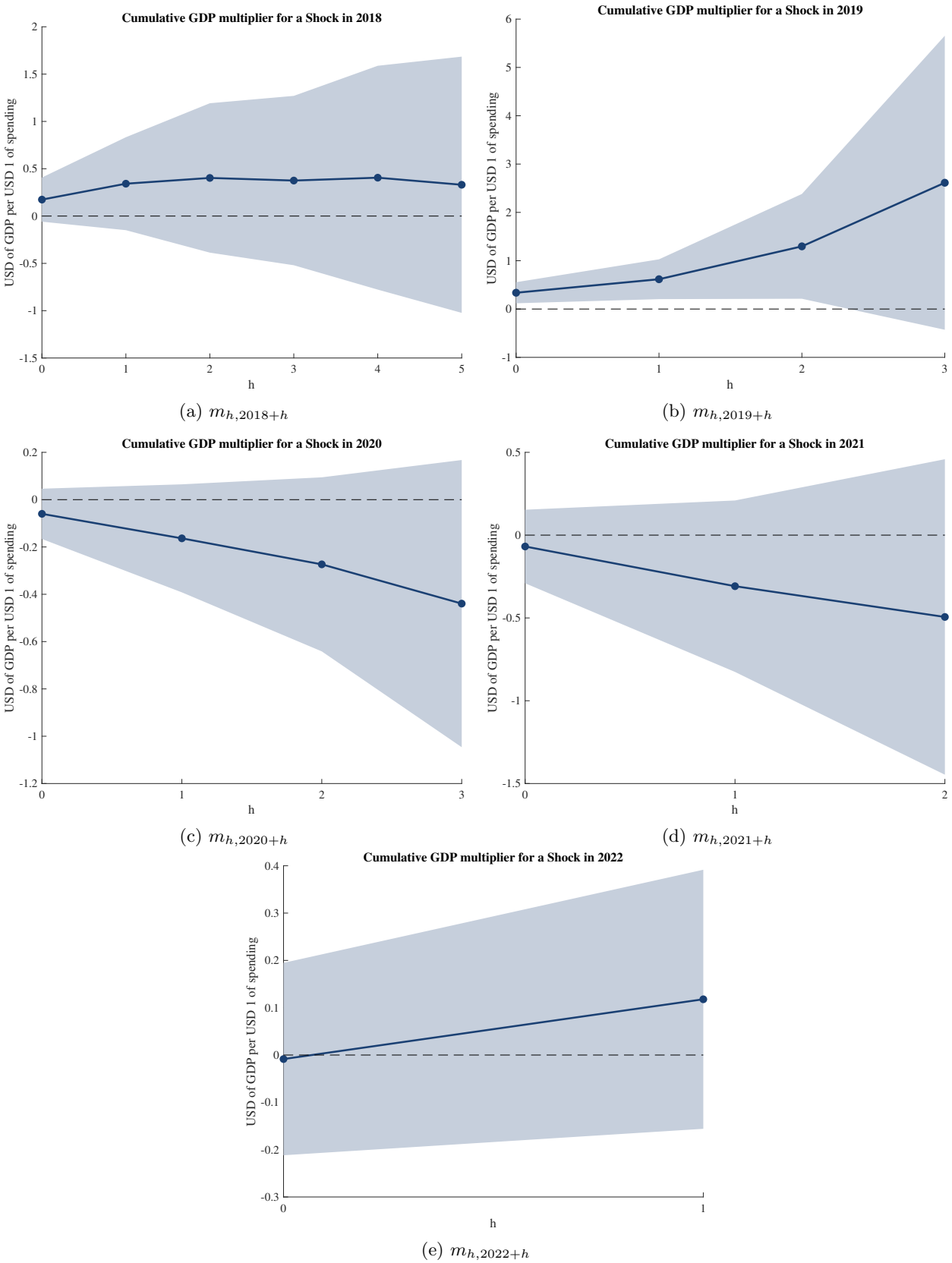


Figure A-14: Fiscal multiplier on impact ($h = 0$), together with 90% confidence bands.

B.7 Time Variation in State-level Fiscal Multipliers

To show that allowing for time variation can shed new light on traditional estimates, consider the analysis of the effects of government spending in a fiscal and monetary union in Nakamura and Steinsson (2014). Nakamura and Steinsson (2014) exploit cross-sectional variation in regional military procurement data associated with military buildups to estimate an open economy fiscal multiplier. Their multiplier, which can be interpreted as the relative per capita output increase in a region when the relative per capita government purchases in the same region rise by one percent, is estimated to be 1.43 percent. However, their analysis assumes that the multiplier is constant over time.

We estimate a time-varying open economy fiscal multiplier with our TVP-PLP estimator, using exactly the same annual data and variables as in Nakamura and Steinsson (2014). Consider eq. (4), where we let $y^{(2)}_{i,t+h}$ be the two-year change rate in per capita output in region i in year t and $y^{(1)}_{i,t}$ be the two-year change rate in per capita military procurement spending in the same region and year. The instrument set in Nakamura and Steinsson (2014) includes time fixed-effects, state (or individual) fixed-effects, and the interaction between the aggregate government spending and the individual states. In our setup, however,



(a) $m_{h,2018+h}$

(b) $m_{h,2019+h}$

(c) $m_{h,2020+h}$

(d) $m_{h,2021+h}$

(e) $m_{h,2022+h}$

Figure A-15: Time-varying cumulative multipliers, together with 90% confidence bands.

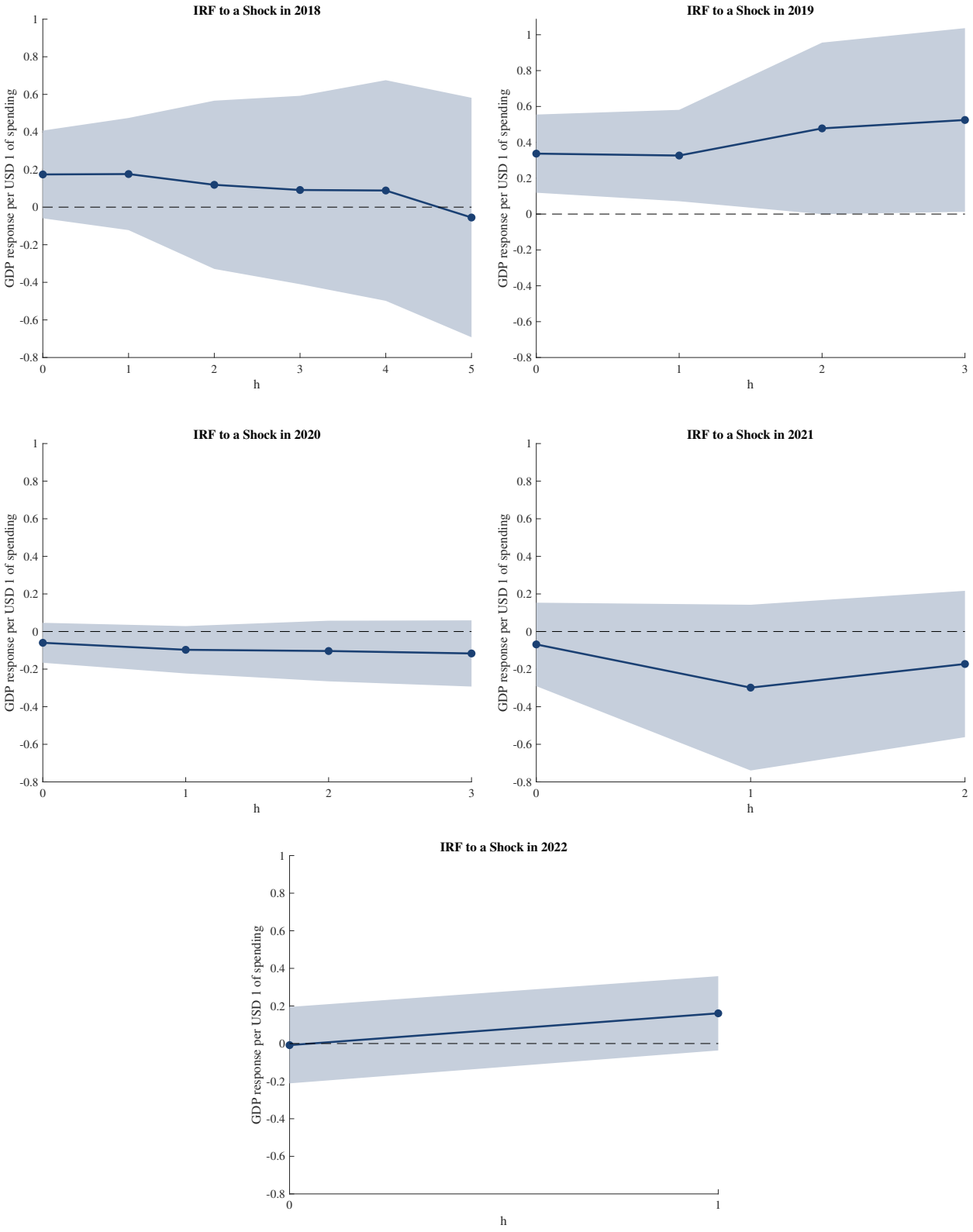


Figure A-16: Time-varying responses of GDP ($\theta_{h,t+h}$) for time t equal to: 2018 (top left panel), 2019 (top right panel), 2020 (bottom left panel), 2021 (bottom right panel), 2022 (bottom center panel). Shaded areas represent 90% confidence bands.

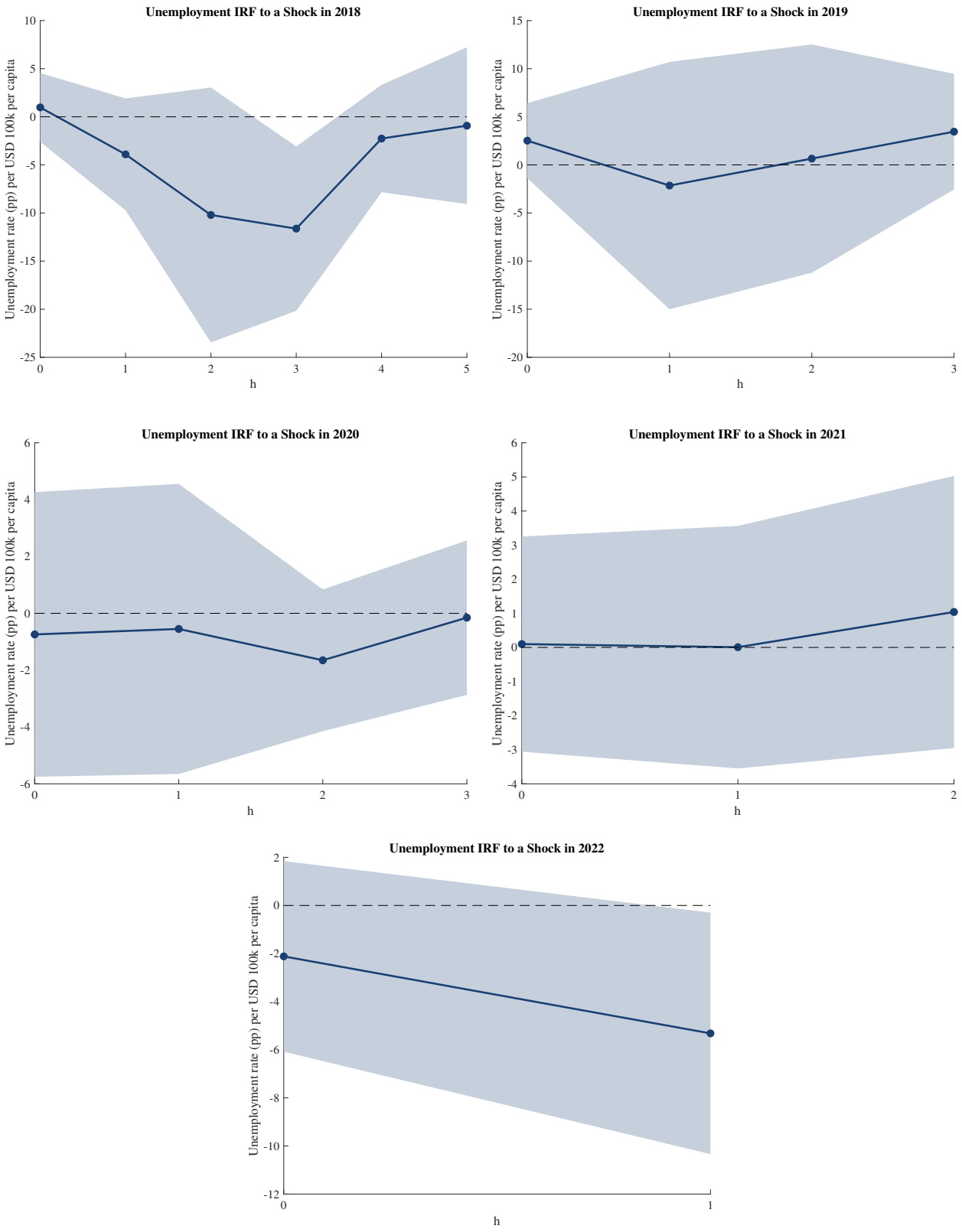


Figure A-17: Time-varying responses of unemployment ($\theta_{h,t+h}$) for time t equal to: 2018 (top left panel), 2019 (top right panel), 2020 (bottom left panel), 2021 (bottom right panel), 2022 (bottom center panel). Shaded areas represent 90% confidence bands.

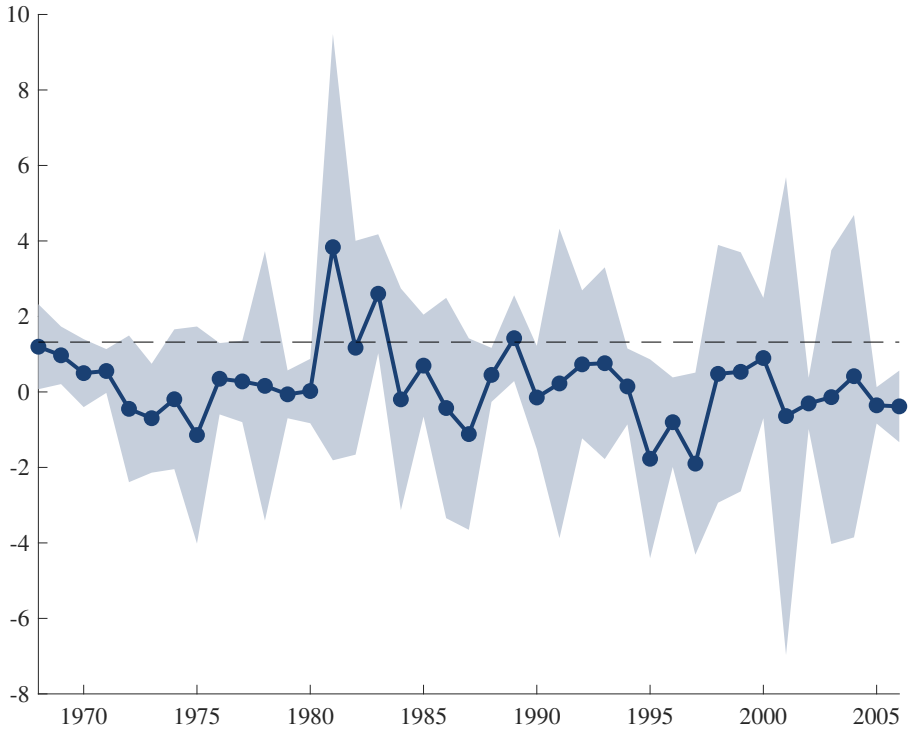


Figure A-18: The figure depicts the time-varying fiscal multiplier (solid line with circles), together with 90% confidence bands (shaded areas), and the constant multiplier (dashed line). Annual data.

the time dimension of the instruments is crucial, as each time period estimates a coefficient at that point in time t; therefore, at each point in time, the set of instruments include: an intercept (to capture the time fixed-effects); the interaction between time-t government spending and the individual states (that is, time-t government spending multiplied by an identity matrix); and state/individual fixed-effects (that is, an identity matrix). Since our estimator employs a unit normalization for the impact of the shock on the endogenous regressor $y_{i,t}^{(1)}$ in every period, we cannot compare it with the value of the estimated multiplier in Nakamura and Steinsson (2014). For this reason, we use their dataset to estimate both the time-varying fiscal multiplier, as well as the constant-parameter one, imposing the unit normalization at each point in time.[9]

Figure A-18 reports our time-varying estimate of the state-level output multiplier of prime military contracts, together with 90% (pointwise) confidence bands, as well as the corresponding constant parameter estimated multiplier, depicted by the dashed line. The picture indeed shows substantial time variation in

[9]For the constant parameter case, the unit normalization is not period-specific.

the multiplier estimates even at the state level.[10]

C Theoretical Results

Let $\mathbf{y}_{i,t}$ be an $(m \times 1)$ dimensional vector of time series variables, where $i = 1, 2, ..., N$ denotes the number of cross-sectional units, and $t = 1, 2, ..., T$ denotes the number of time periods. Also, let $j = 0, 1, 2, ...$ denote the lags of the moving average parameters; $k = 1, ..., m$ denote the indices of the structural shocks; and $s, t = 1, ..., T$ be indices to denote time periods; h denotes the horizon and is fixed throughout the following discussion. Finally, recall that the symbol $x^{\perp}$ denotes the population linear projection residuals, whereas the symbol $\tilde{x}$ denotes the least squares residuals.

C.1 Impulse Responses: Definition

In this section, we assume that the time-variation is deterministic in nature. The assumption is formalized as follows:

Assumption 1 (Deterministic Time Variation)
The time variation in the parameters is deterministic in the sense that $E(\mathbf{\Theta}_{j,t}|\boldsymbol{\varepsilon}_{i,s}, \forall s \leq t, c_i) = \mathbf{\Theta}_{j,t}, \forall s, t.$

If Assumption 1 is satisfied, even stochastic time variation is allowed. For example, the smooth time variation and one-time structural break cases we consider in the simulations of Appendix D do satisfy Assumption 1 provided the time of the break is known, as in our case. When Assumption 1 is not satisfied, the model becomes nonlinear in $\varepsilon_{i,s}$ and the local projection estimand does not coincide with the impulse response defined below – see Gonçalves et al. (2021), Gonçalves et al. (2024b) and Plagborg-Møller and Kolesar (2025). Importantly, note that this assumption still allows the parameters to evolve stochastically, as in, for example, a random walk fashion similar to Primiceri (2005); however, it requires that heterogeneity in impulse responses over time is exogenous (as in, e.g., Primiceri (2005), Cogley and Sargent (2005) and Canova and Pérez Forero (2015)) and unrelated to the state of the economy.[11] When the time variation is endogenous to the state of the economy and its determinants are known, one could instead use the approach proposed

[10] Note that the estimate based on aggregate data (depicted by the dashed line) may potentially be affected by Donaldson's (2026) bias, while the estimate based on disaggregated data (depicted by the continuous line with dots) is robust to to the bias. Hence, it is not surprising that the latter is not necessarily centered around the former.

[11] The simulation section shows that our method performs well when parameters evolve according to random walks.

by Cloyne et al. (2020).

The m-dimensional random vectors $\{\mathbf{y}_{i,t}\}$ are generated from a panel vector moving average process with individual fixed effects:

$$\mathbf{y}_{i,t} \;=\; \mathbf{c}_i + \boldsymbol{\mu} + \sum_{j=0}^{\infty} \boldsymbol{\Theta}_{j,t}\boldsymbol{\varepsilon}_{i,t-j},\ for\ i = 1, ..., N\ and\ t = 1, ..., T, \tag{A-7}$$

where $\boldsymbol{\mu}$ is a vector of constants; $\mathbf{c}_i$ is the fixed effect with finite fourth moments; $\boldsymbol{\varepsilon}_{i,t} \sim (0, \boldsymbol{\Sigma})$ are martingale difference sequences over time and independent across i with finite fourth moments; $\boldsymbol{\Sigma}$ is an $(m \times m)$ diagonal matrix with positive diagonal elements; and $\{\boldsymbol{\varepsilon}_{i,t}\}_{t=1}^{T}$ is mean independent of $\mathbf{c}_i$[12].

Without loss of generality, the parameter of interest is the h-step-ahead response of the second element of $\mathbf{y}_{i,t}$, $y_{i,t}^{(2)}$, to the first element of $\boldsymbol{\varepsilon}_{i,t}$, $\varepsilon_{i,t}^{(1)}$:

$$\begin{aligned} \theta_{h,t+h} \quad \equiv \quad & E(y_{i,t+h}^{(2)} | \varepsilon_{i,t}^{(1)} = \varepsilon + \delta, \varepsilon_{i,t}^{(k)}, \forall k = 2, ..., m, \{\boldsymbol{\varepsilon}_{i,s}\}_{s<t}, \mathbf{c}_i) \\ & - E(y_{i,t+h}^{(2)} | \varepsilon_{i,t}^{(1)} = \varepsilon, \varepsilon_{i,t}^{(k)}, \forall k = 2, ..., m, \{\boldsymbol{\varepsilon}_{i,s}\}_{s<t}, \mathbf{c}_i). \end{aligned} \tag{A-8}$$

Note that this definition is different from that in Stock and Watson (2018) in that the information set includes $\mathbf{c}_i$ and $\boldsymbol{\Theta}_{j,s}$ for all $j = 0, 1, ...$ and $s \leq t$. Note that, as in the example, $\theta_{h,t+h}$ measures the effect on the dependent variable of an exogenous increase in $\varepsilon_{i,t}^{(1)}$ at time t after h periods; that is, the time when the exogenous increase takes place is obtained as the difference of the second and the first subscripts in $\theta_{h,t+h}$.

Because (A-7) is a linear process, it follows that

$$\theta_{h,t+h} = \Theta_{h,t+h}^{(2,1)} \delta. \tag{A-9}$$

C.2 The TVP-PLP Estimator

This paper considers linear projection estimators (Jordà 2005) with instrumental variables (Jordà et al. 2015 and Ramey and Zubairy 2018). Without loss of generality, we are interested in the impulse response of $y_{i,t+h}^{(2)}$ to $\varepsilon_{i,t}^{(1)}$ using equation (4). To that end, we estimate eq. (4) using external instruments. Let $\mathbf{z}_{i,t}$ denote a vector of instrumental variables and $\mathbf{z}_{i,t}^{\perp}$ denote the residual from a linear projection of $\mathbf{z}_{i,t}$ on $\mathbf{w}_{i,t}$ in population. Suppose that the following three conditions are satisfied:

[12] Our arguments below will carry through even if we include time effects, provided that the time effects are mean independent of $\boldsymbol{\varepsilon}_{i,s}$ and $\mathbf{c}_i$.

Assumption 2 (Instrument Properties)

(i) $E(\mathbf{z}_{i,t}^{\perp}\varepsilon_{i,t}^{(1)\perp}) = \boldsymbol{\kappa}_t \neq \mathbf{0}$ *(relevance)*;

(ii) $E(\mathbf{z}_{i,t}^{\perp}\varepsilon_{i,t}^{(k)\perp}) = \mathbf{0}$ *for* $k = 2, ..., m$ *(contemporaneous exogeneity)*;

(iii) $E(\mathbf{z}_{i,t}^{\perp}\boldsymbol{\varepsilon}_{i,s}^{\perp\prime}) = \mathbf{0}$ *for all* $t \neq s$ *(lead and lag exogeneity).*

We follow Stock and Watson (2018) and impose the relevance condition as well as the contemporaneous and lead and lag exogeneity conditions on the external instruments in conditions (i), (ii) and (iii). Also, again following Stock and Watson (2018), we make the following assumption to identify the *relative* impulse response:

Assumption 3: Normalization

The unit effect normalization condition is $\Theta_{0,t}^{(1,1)} = 1$.

Assumption 3 ensures the comparability of the impulse responses over time, in the sense that impulse responses at different times will all measure the effect of a unitary increase in the instrumented regressor on the outcome variable. Although empirically the size of the shock might be different at each point in time, this normalization assumption allows us to systematically compare the impulse responses in terms of observable macroeconomic variables and economically interpret the effects. Note that assumptions of this type are routinely used in non-linear VARs (Primiceri (2005) for the TVP case, Gargiulo et al. (2025) for the regime-dependent one).

Because we assume that T is fixed, we cannot include infinitely many lags, which rules out internal instruments; we formally state this as Assumption 4 below. The assumption is stated in the first difference of $\mathbf{y}_{i,t}$ to eliminate the unobserved heterogeneity, $\mathbf{c}_{i,t}$. Assumptions 2 and 4, in turn, imply that $\mathbf{z}_{i,t}^{\perp} = \mathbf{z}_{i,t}$ *in population*, which makes it easier to interpret the instrumental relevance and exogeneity conditions.

Assumption 4: Control variables

$\mathbf{w}_{i,t}$ *consists of 1 and at most finitely many lags of* $\Delta\mathbf{y}_{i,t}$.

Given the exogeneity condition and the unit effect normalization condition, we can write the moment con-

ditions as:[13]

$$E\left[\mathbf{z}_{i,t}^{\perp}(\Delta y_{i,t+h}^{(2)\perp} - \beta_{h,t+h}'\Delta y_{i,t}^{(1)\perp})\right] \;=\; \mathbf{0}, \tag{A-10}$$

where $\mathbf{z}_{i,t}^{\perp}$, $\Delta y_{i,t+h}^{(2)\perp}$ and $\Delta y_{i,t}^{(1)\perp}$ are the residuals from linearly projecting $\mathbf{z}_{i,t}$, $\Delta y_{i,t+h}^{(2)}$ and $\Delta y_{i,t}^{(1)}$ on $\mathbf{w}_{i,t}$ in population, respectively. Here we take the first differences to eliminate the unobserved heterogeneity, $\mathbf{c}_{i,t}$.

Our Time-Varying Parameter Panel Local Projection (*TVP PLP*) estimator is a generalized method of moments (GMM) estimator based on the moment conditions in eq. (A-10). Using eq. (A-7), write $\Delta y_{i,t}^{\perp}$ and $\Delta y_{i,t+h}^{\perp}$ as

$$\Delta y_{i,t}^{\perp} \;=\; \mathbf{\Theta}_{0,t}\boldsymbol{\varepsilon}_{i,t}^{\perp} + f\{\boldsymbol{\varepsilon}_{i,s}^{\perp}\,; s<t\}, \tag{A-11}$$

$$\Delta y_{i,t+h}^{\perp} \;=\; \mathbf{\Theta}_{0,t+h}\boldsymbol{\varepsilon}_{i,t+h}^{\perp} + \sum_{j=1}^{h-1}\Delta\mathbf{\Theta}_{j,t+h}\boldsymbol{\varepsilon}_{i,t+h-j}^{\perp} + \Delta\mathbf{\Theta}_{h,t+h}\boldsymbol{\varepsilon}_{i,t}^{\perp} + f\{\boldsymbol{\varepsilon}_{i,s}^{\perp}\,; s<t\}, \tag{A-12}$$

respectively, where $f\{\boldsymbol{\varepsilon}_{i,s}^{\perp}\,; s<t\}$ denotes random variables spanned by $\boldsymbol{\varepsilon}_{i,s}^{\perp}$ $s<t$ and $\Delta\mathbf{\Theta}_{j,t+h} = (\mathbf{\Theta}_{j,t+h} - \mathbf{\Theta}_{j-1,t+h-1})$ for $j = 1, 2, ..., h$. It follows from equations (A-10), (A-11) and (A-12), and Assumptions 1–4 that

$$\beta_{h,t+h} \;=\; \begin{cases} \mathbf{\Theta}_{h,t+h}^{(2,1)} - \mathbf{\Theta}_{h-1,t+h-1}^{(2,1)} & \text{if h>0} \\ \mathbf{\Theta}_{0,t}^{(2,1)} & \text{if h=0} \end{cases} \tag{A-13}$$

Thus,

$$\begin{aligned} \mathbf{\Theta}_{h,t+h}^{(2,1)} &= \beta_{h,t+h} + \beta_{h-1,t+h-1} + \cdots + \beta_{0,t} = \boldsymbol{\ell}'\widehat{\boldsymbol{\beta}}_{t+h} \\ &= (\mathbf{\Theta}_{h,t+h}^{(2,1)} - \mathbf{\Theta}_{h-1,t+h-1}^{(2,1)}) + (\mathbf{\Theta}_{h-1,t+h-1}^{(2,1)} - \mathbf{\Theta}_{h-2,t+h-2}^{(2,1)}) + \cdots + \mathbf{\Theta}_{0,t}^{(2,1)}, \end{aligned} \tag{A-14}$$

where $\boldsymbol{\ell}$ is the $(h+1)\times 1$ vector of ones.

In other words, we estimate $\mathbf{\Theta}_{h,t+h}^{(2,1)}$ by estimating $\beta_{j,t+j}$ from moment conditions (A-10) and summing them up:

[13] The moment condition can also be written in terms of orthogonalized variables as:

$$E\left[\mathbf{z}_{i,t}(y_{i,t+h}^{(2)} - \mathbf{x}_{i,t}'\boldsymbol{\zeta}_{h,t+h})\right] \;=\; \mathbf{0}.$$

$$\widehat{\boldsymbol{\beta}}_{t+h} = (\widehat{\mathbf{A}}_t^{(1)\prime}\widehat{\mathbf{W}}_{h,t+h}\widehat{\mathbf{A}}_t^{(1)})^{-1}\widehat{\mathbf{A}}_t^{(1)\prime}\widehat{\mathbf{W}}_{h,t+h}\left(\frac{1}{N}\sum_{i=1}^{N}(\widetilde{\Delta\Upsilon}_{i,t+h}^{(2)}\otimes\tilde{\mathbf{z}}_{i,t})\right), \tag{A-15}$$

$$\widehat{\mathbf{A}}_t^{(1)} = \mathbf{I}_{h+1}\otimes\left(\frac{1}{N}\sum_{i=1}^{N}\tilde{\mathbf{z}}_{i,t}\widetilde{\Delta y}_{i,t}^{(1)}\right), \tag{A-16}$$

where $\widetilde{\Delta\Upsilon}_{i,t+h}^{(2)} = [\widetilde{\Delta}y_{i,t}^{(2)}, \widetilde{\Delta}y_{i,t+1}^{(2)}, ..., \widetilde{\Delta}y_{i,t+h}^{(2)}]'$, $\tilde{x}$ denotes the residual from regressing x on $\mathbf{w}_t$ by ordinary least squares (OLS), and $\widehat{\mathbf{W}}_{h,t+h}$ is a weighting matrix [14].

Note that it is often the case that local projections are estimated using long differences and/or cumulative regressions (as, for example, in the multiplier regressions in eq. (19)), in which case the differencing approach is already implemented by construction. In particular, with long differencing (as in eq. (21)), the coefficient of interest is estimated directly, without having to sum the $\beta_{h,t+h}$. This is because the long difference is usually defined as: $\Delta_h y_{i,t+h}^{\perp} = y_{i,t+h}^{\perp} - y_{i,t-1}^{\perp}$, and the outcome variable at $(t-1)$ is not affected by the shock that occurs at time t.

Being based on period-by-period cross-sectional regressions rather than pooling, our estimator is more robust to arbitrary time variation in the impulse responses compared to a constant-parameter estimator, while the latter are more efficient in the absence of time variation.

If the instruments are orthogonal to the unobserved heterogeneity, then our estimator simplifies. We discuss this as a special case in Section C.4 below.

C.3 Asymptotic Properties

This sub-section presents the main theoretical result in the paper, namely consistency and asymptotic normality of the TVP-PLP estimator. We make the following additional assumption:

Assumption 5 (Consistency and Asymptotic Normality)

(a) $\mathbf{y}_{i,t}$ *is generated from a VMA process in equation* (A-7) *where* $\mathbf{c}_i$ *is independently and identically distributed (i.i.d.) with finite fourth moments,* $\{\boldsymbol{\varepsilon}_{i,t}\}_{t=1}^{T}$ *are independent across* i *with finite fourth moments and satisfy* $E(\boldsymbol{\varepsilon_{i,t}}|\boldsymbol{\varepsilon_{i,t-1}}, \boldsymbol{\varepsilon}_{i,t-2}, ...) = \mathbf{0}$, $Var(\boldsymbol{\varepsilon_{i,t}}|\boldsymbol{\varepsilon_{i,t-1}}, \boldsymbol{\varepsilon}_{i,t-2}, ...) = \boldsymbol{\Sigma}$, $\boldsymbol{\Sigma}$ *is an* $(m \times m)$

[14] Note that $\Delta y_{i,t}^{(1)\perp}$ and $\widehat{\Delta y}_{i,t}^{(1)}$ are, respectively, the projection and least squares residuals of the first difference $\Delta y_{i,t}^{(1)}$, not the first differences of the projection and least squares residuals.

diagonal matrix with positive diagonal elements, $E(\boldsymbol{\varepsilon}_{i,t}|\mathbf{c}_i) = 0$ *and* $\sum_{j=0}^{\infty} \|\boldsymbol{\Theta}_{j,t}\|^2 < \infty$ *for all* t;

(b) $\{\mathbf{z}_{i,t}\}_{t=1}^{T}$ *is cross-sectionally i.i.d. and has finite fourth moments*;

(c) $\widehat{\mathbf{W}}_{h,t+h}$ *is positive definite with* $plim_{N\to\infty}\widehat{\mathbf{W}}_{h,t+h} = \mathbf{W}_{h,t+h}$, *where* $\mathbf{W}_{h,t+h}$ *is positive definite*;

(d) $E((\mathbf{v}^{\perp}_{i,t+h}\mathbf{v}^{\perp\prime}_{i,t+h})\otimes(\mathbf{z}^{\perp}_{it}\mathbf{z}^{\perp\prime}_{it}))$ and $E(\mathbf{z}^{\perp}_{i,t}\mathbf{z}^{\perp\prime}_{i,t})$ *are positive definite, where* $\mathbf{v}^{\perp}_{i,t+h} = [v^{\perp}_{i,h,t+h}\ v^{\perp}_{i,h-1,t+h-1}\ \cdots\ v^{\perp}_{i,0,t}]'$, *and* $v^{\perp}_{i,h,t+h} = y^{(2)\perp}_{i,t+h} - \beta_{h,t+h}y^{(1)\perp}_{i,t}$.

The asymptotic properties of our proposed TVP-PLP estimator are described in the following result.

Theorem 1: The TVP-PLP Estimator. Suppose that Assumptions 1-5 hold. Then, as $N \to \infty$ *and* T *is fixed, we have:*

(i) $\widehat{\Theta}^{(2,1)}_{h,t+h}$ *is consistent for* $\Theta^{(2,1)}_{h,t+h}$ *and is asymptotically normal:*

$$\sqrt{N}(\widehat{\Theta}^{(2,1)}_{h,t+h} - \Theta^{(2,1)}_{h,t+h}) \xrightarrow{d} N(0, \omega_{h,t+h}), \tag{A-17}$$

as $N \to \infty$ *while* T *is fixed, where*

$$\begin{aligned}
\omega_{h,t+h} &= \boldsymbol{\ell}'\left(\mathbf{A}^{(1)\prime}_t\mathbf{W}_{h,t+h}\mathbf{A}^{(1)}_t\right)^{-1}\mathbf{A}^{(1)\prime}_t\mathbf{W}_{h,t+h}E\left[(\mathbf{v}^{\perp}_{i,t+h}\mathbf{v}^{\perp\prime}_{i,t+h})\otimes(\mathbf{z}^{\perp}_{it}\mathbf{z}^{\perp\prime}_{it})\right]\mathbf{W}_{h,t+h}\mathbf{A}^{(1)}_t\left(\mathbf{A}^{(1)\prime}_t\mathbf{W}_{h,t+h}\mathbf{A}^{(1)}_t\right)^{-1}\boldsymbol{\ell}, \\
\mathbf{A}^{(1)}_t &= \mathbf{I}_{h+1}\otimes E(\mathbf{z}^{\perp}_{i,t}\Delta y^{(1)\perp}_{i,t}).
\end{aligned} \tag{A-18}$$

(ii) *Let*

$$\widehat{\boldsymbol{\omega}}_{h,t+h} = \boldsymbol{\ell}'\left(\widehat{\mathbf{A}}^{(1)\prime}_t\widehat{\mathbf{W}}_{h,t+h}\widehat{\mathbf{A}}^{(1)}_t\right)^{-1}\widehat{\mathbf{A}}^{(1)\prime}_t\widehat{\mathbf{W}}_{h,t+h}\left(\frac{1}{N}\sum_{i=1}^{N}(\widetilde{\mathbf{v}}_{i,t+h}\widetilde{\mathbf{v}}'_{i,t+h})\otimes(\tilde{\mathbf{z}}_{it}\tilde{\mathbf{z}}'_{it})\right)\widehat{\mathbf{W}}_{h,t+h}\widehat{\mathbf{A}}^{(1)}_t\left(\widehat{\mathbf{A}}^{(1)\prime}_t\widehat{\mathbf{W}}_{h,t+h}\widehat{\mathbf{A}}^{(1)}_t\right)^{-1}\boldsymbol{\ell}, \tag{A-19}$$

where $\widetilde{\mathbf{v}}_{i,t+h} = [\tilde{v}_{i,h,t+h}\ \tilde{v}_{i,h-1,t+h-1}\ \cdots\ \tilde{v}_{i,0,t}]'$, and $\tilde{v}_{i,h,t+h} = \widetilde{\Delta y}^{(2)}_{i,t+h} - \widehat{\beta}_{h,t+h}\widetilde{\Delta y}^{(1)}_{i,t}$.

Then: $\widehat{\omega}_{h,t+h} \xrightarrow{p} \omega_{h,t+h}$.

(iii) *The optimal weighting matrix is given by*

$$\widehat{\mathbf{W}}_{h,t+h} = \left(\frac{1}{N}\sum_{i=1}^{N}(\widetilde{\mathbf{v}}_{i,t+h}\widetilde{\mathbf{v}}'_{i,t+h})\otimes(\widetilde{\mathbf{z}}_{it}\widetilde{\mathbf{z}}'_{it})\right)^{-1}. \tag{A-20}$$

C.4 Orthogonality

There are a few other cases that are useful to highlight. For ease of exposition, we will begin with discussing an estimator that can be used if an orthogonality assumption holds, then we will consider the cases of conditional homoskedasticity and OLS while maintaining the orthogonality assumption.
Suppose the following orthogonality assumption holds:

Assumption 6 (Orthogonality) $E(\mathbf{z}_{i,t}^{\perp}\boldsymbol{c}_i^{\perp\prime}) = \mathbf{0}$.

If Assumption 6 holds, then the following moment condition is valid:

$$E\left[\mathbf{z}_{i,t}^{\perp}(y_{i,t+h}^{(2)\perp} - \beta_{h,t+h}' y_{i,t}^{(1)\perp})\right] \;=\; \mathbf{0}. \tag{A-21}$$

Thus, researchers can use the following *Exogenous* Time-Varying Parameter Panel LP (Exogenous TVP PLP) estimator:

$$\widehat{\boldsymbol{\Theta}}_{h,t+h}^{(2,1)} = \widehat{\beta}_{h,t+h} \;=\; (\widehat{\boldsymbol{A}}_t^{(1)\prime}\widehat{\mathbf{W}}_{h,t+h}\widehat{\boldsymbol{A}}_t^{(1)})^{-1}\widehat{\boldsymbol{A}}_t^{(1)\prime}\widehat{\mathbf{W}}_{h,t+h}\widehat{\mathbf{A}}_{t+h}^{(2)}, \tag{A-22}$$

$$\widehat{\boldsymbol{A}}_t^{(1)} \;=\; \frac{1}{N}\sum_{i=1}^{N}\tilde{\mathbf{z}}_{i,t}\widetilde{y}_{i,t}^{(1)}, \tag{A-23}$$

$$\widehat{\mathbf{A}}_{t+h}^{(2)} \;=\; \frac{1}{N}\sum_{i=1}^{N}\mathbf{z}_{i,t}y_{i,t+h}^{(2)}, \tag{A-24}$$

and $\widehat{\mathbf{W}}_{h,t+h}$ is a weighting matrix.

Theorem 2: The TVP-PLP Estimator under Exogeneity. Suppose that Assumptions 1-6 hold. Then, as $N \to \infty$ and T is fixed, $\widehat{\beta}_{h,t+h}$ in eq. (A-22) *is consistent for $\beta_{h,t+h}$ and asymptotically normal:*

$$\sqrt{N}(\widehat{\beta}_{h,t+h} - \beta_{h,t+h}) \;\xrightarrow{d}\; N\left(0, (\boldsymbol{A}_t^{(1)\prime}\mathbf{W}_{h,t+h}\boldsymbol{A}_t^{(1)})^{-1}(\boldsymbol{A}_t^{(1)\prime}\mathbf{W}_{h,t+h}\boldsymbol{\Omega}_{h,t+h}\mathbf{W}_{h,t+h}\boldsymbol{A}_t^{(1)})(\boldsymbol{A}_t^{(1)\prime}\mathbf{W}_{h,t+h}\boldsymbol{A}_t^{(1)})^{-1}\right), \tag{A-25}$$

as $N \to \infty$ while T is fixed, where

$$\mathbf{\Omega}_{h,t+h} \;=\; E\left[\frac{1}{N}\sum_{i=1}^{N}(y_{i,t+h}^{(2)^{\perp}} - \beta_{h,t}\tilde{y}_{i,t}^{(1)\perp})^2\mathbf{z}_{i,t}^{\perp}\mathbf{z}'^{\perp}_{i,t}\right], \tag{A-26}$$

and *the optimal weighting matrix is given by* $\widehat{\mathbf{W}}_{h,t+h} = \widehat{\mathbf{\Omega}}_{h,t+h}^{-1}$, where

$$\widehat{\mathbf{\Omega}}_{h,t+h} \;=\; \frac{1}{N}\sum_{i=1}^{N}\left[(\tilde{y}_{i,t+h}^{(2)} - \hat{\beta}_{h,t+h}\tilde{y}_{i,t}^{(1)})^2\tilde{\mathbf{z}}_{i,t}\tilde{\mathbf{z}}'_{i,t}\right]. \tag{A-27}$$

Special Case 1: Scalar Instrument (The Scalar TVP PLP-IV): When $\mathbf{z}_{i,t}$ is scalar ($\mathbf{z}_{i,t}$=$z_{i,t}$), then

$$\widehat{\Theta}_{h,t+h}^{(2,1)} \;= \frac{\frac{1}{N}\sum_{i=1}^{N} z_{i,t}^{\perp} y_{i,t+h}^{(2)\perp}}{\frac{1}{N}\sum_{i=1}^{N} z_{i,t}^{\perp} y_{i,t}^{(1)\perp}} \tag{A-28}$$

where

$$\sqrt{N}(\widehat{\Theta}_{h,t+h}^{(2,1)} - \Theta_{h,t+h}^{(2,1)}) \to_d \mathcal{N}(0, V_{h,t+h}) \tag{A-29}$$

with (A-30)

$$\hat{V}_{h,t+h} = \left(\frac{1}{N}\sum_{i=1}^{N}\tilde{z}_{i,t}\tilde{y}_{i,t}^{(1)'}\right)^{-1}\left(\frac{1}{N}\sum_{i=1}^{N}\left[\tilde{z}_{i,t}\left(\tilde{y}_{i,t+h}^{(2)} - \hat{\Theta}_{h,t+h}^{(2,1)}\tilde{y}_{i,t}^{(1)}\right)\right]^2\right)\left(\frac{1}{N}\sum_{i=1}^{N}\tilde{z}_{i,t}\tilde{y}_{i,t}^{(1)'}\right)^{-1}. \tag{A-31}$$

Special Case 2: Conditional Homoskedasticity: Note that, under conditional homoskedasticity, the estimator in eq. (A-30)-(A-31) is the two-stage least squares (2SLS) estimator.

Special Case 3: OLS estimation (TVP PLP-OLS):

When the shock is observable and exogenous, the responses can be estimated in a local projection by OLS as follows:

$$\widehat{\Theta}_{h,t+h}^{(2,1)} \;= \left(\frac{1}{N}\sum_{i=1}^{N}\tilde{y}_{i,t}^{(1)}\tilde{y}_{i,t}^{(1)'}\right)^{-1}\left(\frac{1}{N}\sum_{i=1}^{N}\tilde{y}_{i,t}^{(1)}\tilde{y}_{i,t+h}^{(2)}\right), \tag{A-32}$$

where

$$\sqrt{N}(\widehat{\Theta}^{(2,1)}_{h,t+h} - \Theta^{(2,1)}_{h,t+h}) \to_d \mathcal{N}(0, V_{h,t+h}), \quad \text{with} \tag{A-33}$$

$$\hat{V}_{h,t+h} = \left(\frac{1}{N}\sum_{i=1}^{N} \tilde{y}^{(1)}_{i,t}\tilde{y}^{(1)'}_{i,t}\right)^{-1} \left(\frac{1}{N}\sum_{i=1}^{N}\left[\tilde{y}^{(1)}_{i,t}\left(\tilde{y}^{(2)}_{i,t+h} - \hat{\Theta}^{(2,1)}_{h,t+h}\tilde{y}^{(1)}_{i,t}\right)\right]^2\right)\left(\frac{1}{N}\sum_{i=1}^{N} \tilde{y}^{(1)}_{i,t}\tilde{y}^{(1)'}_{i,t}\right)^{-1}. \tag{A-34}$$

In fact, this is a special case of our results in which $y^{(1)}_t$ is identical to the shock of interest (i.e., the moving average coefficients are all zero and the impact matrix coefficients on the other shocks are all zero) and it is used as an instrument.

C.5 Proof of Theorem 1

Proof of Theorem 1: Under the stated assumption, $\beta_{h,t+h}$, $\beta_{h-1,t+h-1}$,...,$\beta_{0,t}$ are point-identified through the moment conditions:

$$E\left\{\begin{bmatrix} \mathbf{z}_t\Delta y^{(2)}_{i,t+h} \\ \mathbf{z}_t\Delta y^{(2)}_{i,t+h-1} \\ \vdots \\ \mathbf{z}_t\Delta y^{(2)}_{i,t} \end{bmatrix} - \begin{bmatrix} \mathbf{z}_t\Delta y^{(1)}_{i,t} & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & \mathbf{z}_t\Delta y^{(1)}_{i,t} & \cdots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \cdots & \mathbf{z}_t\Delta y^{(1)}_{i,t} \end{bmatrix}\begin{bmatrix} \beta_{h,t+h} \\ \beta_{h-1,t+h-1} \\ \vdots \\ \beta_{0,t} \end{bmatrix}\right\} = \mathbf{0}. \tag{A-35}$$

Because $y^{(2)}_{i,t+h}$, $y^{(1)}_{i,t}$, $\mathbf{w}_{i,t}$, and $\mathbf{z}_{i,t}$ are cross-sectionally i.i.d. with finite fourth moments by Assumption 5, and since $\mathbf{w}_{i,t}$ is finite-dimensional by Assumption 4, the GMM estimator is consistent and asymptotically normal:

$$\begin{aligned} &\sqrt{N}\begin{bmatrix} \widehat{\beta}_{h,t+h} - \beta_{h,t+h} \\ \widehat{\beta}_{h-1,t+h-1} - \beta_{h-1,t+h-1} \\ \vdots \\ \widehat{\beta}_{0,t} - \beta_{0,t} \end{bmatrix} \\ \xrightarrow{d} \; & N\Big(\mathbf{0}, \left[(\mathbf{I}_{h+1}\otimes E(y^{(1)\perp}_{i,t}\mathbf{z}'_{i,t}))\mathbf{W}_{h,t+h}(\mathbf{I}_{h+1}\otimes E(\mathbf{z}_{i,t}y^{(1)\perp}_{i,t}))\right]^{-1}(\mathbf{I}_{h+1}\otimes E(y^{(1)\perp}_{i,t}\mathbf{z}'_{i,t}))\mathbf{W}_{h,t+h}E(\mathbf{v}^{\perp}_i\mathbf{v}^{\perp\prime}_i\otimes \mathbf{z}_{it}\mathbf{z}'_{it})\mathbf{W}_{h,t+h} \\ & (\mathbf{I}_{h+1}\otimes E(\mathbf{z}_{i,t}y^{(1)}_{i,t}))\left[(\mathbf{I}_{h+1}\otimes E(y^{(1)}_{i,t}\mathbf{z}'_{i,t}))\mathbf{W}_{h,t+h}(\mathbf{I}_{h+1}\otimes E(\mathbf{z}_{i,t}y^{(1)}_{i,t}))\right]^{-1}\Big), \end{aligned} \tag{A-36}$$

where $\mathbf{v}^{\perp}_i = [v^{\perp}_{i,h,t+h}\ v^{\perp}_{i,h-1,t+h-1}\ \cdots\ v^{\perp}_{i,0,t}]'$. Because $\beta_{h,t+h} + \beta_{h-1,t+h-1} + \cdots + \beta_{0,t}$ equals the impulse response of interest, $\mathbf{\Theta}^{(2,1)}_{h,t+h}$, the asymptotic normality of our estimator follows from the delta method. Note that including lags of $\Delta y_{i,t}$ among the controls $\mathbf{w}_{i,t}$ ensures that the orthogonality condition $E(\boldsymbol{\varepsilon}_{i,t}|\mathbf{c}_i) = 0$ in Assumption 5(a) is satisfied.

C.6 Proof of Theorem 2

Proof of Theorem 2: Under the stated assumptions, $y_{i,t+h}^{(2)}$, $y_{i,t}^{(1)}$, $\mathbf{w}_{i,t}$, and $\mathbf{z}_{i,t}$ are cross-sectionally i.i.d. with finite fourth moments. Thus, consistency and asymptotic normality follow from the standard results for GMM estimators based on i.i.d. observations, exactly as in the proof of Theorem 1. It remains to show that $\zeta_{h,t+h}^{(1)}$ that satisfies

$$E\left[\mathbf{z}_{i,t}^{\perp}(y_{i,t}^{(2)\perp} - \zeta_{h,t+h}^{(1)} y_{i,t}^{(1)})\right] \;=\; \mathbf{0}, \tag{A-37}$$

equals the impulse response of interest, $\mathbf{\Theta}_{h,t+h}^{(2,1)}$, where

$$\begin{aligned} y_{i,t}^{(1)\perp} &= y_{i,t}^{(1)} - E(y_{i,t}^{(1)}\mathbf{w}_{i,t})[E(\mathbf{w}_{i,t}\mathbf{w}_{i,t}')]^{-1}\mathbf{w}_{i,t}, && \text{(A-38)}\\ y_{i,t+h}^{(2)\perp} &= y_{i,t+h}^{(2)} - E(y_{i,t+h}^{(2)}\mathbf{w}_{i,t})[E(\mathbf{w}_{i,t}\mathbf{w}_{i,t}')]^{-1}\mathbf{w}_{i,t}, && \text{(A-39)}\\ \mathbf{z}_{i,t}^{\perp} &= \mathbf{z}_{i,t} - E(\mathbf{z}_{i,t}\mathbf{w}_{i,t})[E(\mathbf{w}_{i,t}\mathbf{w}_{i,t}')]^{-1}\mathbf{w}_{i,t}. && \text{(A-40)} \end{aligned}$$

Because $\mathbf{z}_{i,t}$ is lead-lag exogenous (Assumption 2) and $\mathbf{w}_{i,t}$ is predetermined (Assumption 4), $\mathbf{z}_{i,t}^{\perp}$ is identical to $\mathbf{z}_{i,t}$. Because $\mathbf{w}_{i,t}$ includes 1 and is predetermined,

$$\begin{aligned} y_{i,t}^{\perp} &= \mathbf{\Theta}_{0,t}\boldsymbol{\varepsilon}_{i,t} + \{\mathbf{c}_i, \boldsymbol{\varepsilon}_{i,s}\ s < t\}, && \text{(A-41)}\\ y_{i,t+h}^{\perp} &= \sum_{j=0}^{h} \mathbf{\Theta}_{j,t+h}\boldsymbol{\varepsilon}_{i,t+h-j} + \{\mathbf{c}_i, \boldsymbol{\varepsilon}_{s}\ s < t\}, && \text{(A-42)} \end{aligned}$$

where $\{\mathbf{c}_i, \boldsymbol{\varepsilon}_{i,s}\ s < t\}$ denotes the random variables spanned by $\mathbf{c}_i, \boldsymbol{\varepsilon}_{i,s}\ s < t$. Thus,

$$E\left[\mathbf{z}_{i,t}^{\perp}(y_{i,t+h}^{(2)\perp} - \beta_{h,t+h} y_{i,t}^{(1)})\right] \;=\; \boldsymbol{\kappa_t}(\Theta_{h,t+h}^{(2,1)} - \Theta_{0,t}^{(1,1)}\beta_{h,t+h}) \;=\; \mathbf{0}. \tag{A-43}$$

Because $\mathbf{z}_{i,t}$ is relevant ($\boldsymbol{\kappa_t} \neq \mathbf{0}$) and because of the unit effect normalization ($\Theta_{0,t}^{(1,1)} = 1$), we have

$$\beta_{h,t+h} \;=\; \Theta_{h,t+h}^{(2,1)}. \tag{A-44}$$

D Monte Carlo Simulations

This section evaluates the performance of our proposed TVP-PLP-IV estimator in finite samples. We consider six data generating processes (DGPs): (i) a benchmark constant parameter DGP; (ii) a DGP with smooth,

deterministic time variation; (iii) a DGP in which the parameter follows a random walk; (iv) a one-time structural break in a vector moving average of order one, where we consider the case of constant correlation between the shock and the instrument as well as the case of time variation in the same correlation; (v) a one-time structural break model in a vector moving average of order 10 to verify the performance of our proposed estimator at medium horizons. In all the DGPs above, the instrument is univariate. Finally, (vi) we consider a realistic DGP inspired by our empirical analysis in Section 4 with multiple instruments.

We consider two criteria to evaluate the performance of the estimator: the root mean squared error of the estimator (RMSE) and the empirical rejection frequency of the (two-sided) test statistic. In the first five DGPs, we evaluate the performance of the TVP PLP-IV estimator in eq. (A-28) as well as its asymptotic distribution in eqs. (A-29)-(A-31); the last DGP evaluates the performance of the TVP PLP estimator in eq. (A-22) as well as its asymptotic distribution, eq. (A-25). In all Monte Carlo simulation experiments, the root mean squared error and the empirical rejection frequency are reported as average over time, and the nominal size of the test is 5%.

D.1 DGP with Constant Coefficients

First, we consider a benchmark data generating process with constant parameters. We let:

$$\begin{bmatrix} y_{i,t} \\ z_{i,t} \end{bmatrix} = \Theta_0 \varepsilon_{i,t} + \Theta_1 \varepsilon_{i,t-1}$$

$$\begin{bmatrix} y_{i,t}^{(1)} \\ y_{i,t}^{(2)} \\ z_{i,t} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ \theta_0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \varepsilon_{i,t} + \begin{bmatrix} 0 & 0 & 0 \\ \theta_1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \varepsilon_{i,t-1}$$

with

$$z_{i,t} = \varepsilon_{i,t}^{(3)} \quad \text{and} \quad E\left[\varepsilon_{i,t}^{(1)} \varepsilon_{i,t}^{(3)}\right] = \kappa. \tag{A-45}$$

We set $\theta_0 = 0.8$, $\theta_1 = 0.5$ and $\kappa = 0.6$[15]. We let $N = [10, 50, 100, 200, 1000]$ and $T = 50$. The number of Monte Carlo replications is 1000. Table 1 shows that the performance of our estimator, in terms of both RMSE and empirical rejection rates, improves as the sample size increases. The empirical rejection rates are very close to the nominal size in finite samples provided that the cross-section contains at least 50

[15]The above relationships (eq (A-45)) and the value for κ are imposed throughout the following subsections.

observations.

Table 1. Monte Carlo Simulations:
Constant Parameter Case

	Panel A. Average RMSE				
	N = 10	N = 50	N = 100	N = 200	N = 1000
h = 0	1.8388	0.2232	0.1532	0.1072	0.0473
h = 1	2.9647	0.2561	0.1757	0.1217	0.0538

	Panel B. Empirical Rejection Rates				
	N = 10	N = 50	N = 100	N = 200	N = 1000
h = 0	0.1038	0.0582	0.0558	0.0530	0.0523
h = 1	0.1008	0.0574	0.0525	0.0512	0.0477

Notes: The Table reports, for various horizons h and cross-section dimensions N, the average Root Mean Squared Error (RMSE) (panel A) and the average empirical rejection rates for tests with 5% nominal size (panel b).

D.2 DGP with Smooth Time Variation

The second DGP introduces a simple form of time variation in the coefficients of interest, while keeping the unit normalization in the (1,1) element of the matrix $\Theta_{0,t}$ for all t and maintaining the one-to-one relation between $z_{i,t}$ and $\varepsilon_{i,t}^{(3)}$ as in eq. (A-45). We let:

$$\Theta_{0,t} = \begin{bmatrix} 1 & 0 & 0 \\ \theta_0(T-t)^x & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\Theta_{1,t} = \begin{bmatrix} 0 & 0 & 0 \\ \theta_1(T-t)^x & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix},$$

where $\theta_0 = 0.8$, $\theta_1 = 0.5$, T is the size of the time dimension, and $x = 1/2$. In this DGP, the time-variation appears only in the coefficients of interest of the matrices $\Theta_{h,t}$, that is, the (2,1)-th element. Let:

$$\begin{bmatrix} y_{i,t} \\ z_{i,t} \end{bmatrix} = \Theta_{0,t}\varepsilon_{i,t} + \Theta_{1,t}\varepsilon_{i,t-1}$$

and estimate the coefficients of interest following the expression in equation (A-22). Table 2 reports the

average RMSE across 1000 simulations for both coefficients, for sample sizes given by $N = [10, 50, 100, 200]$ and $T = 50$.

Table 2. Monte Carlo Simulations: Smooth Time-Varying Parameter Case

	Panel A. Average RMSE				
	N = 10	N = 50	N = 100	N = 200	N = 1000
h = 0	3.8271	0.5142	0.3478	0.2416	0.1072
h = 1	5.1370	0.7637	0.5233	0.3634	0.1601

	Panel B. Empirical Rejection Rates				
	N = 10	N = 50	N = 100	N = 200	N = 1000
h = 0	0.1044	0.0577	0.0541	0.0519	0.0495
h = 1	0.2237	0.1011	0.0723	0.0641	0.0496

Notes: The Table reports, for various horizons h and cross-section dimensions N, the average Root Mean Squared Error (RMSE) (panel A) and the average empirical rejection rates for tests with nominal size of 5% (panel b).

D.3 DGP with Random Walk

The third DGP considers a random walk process for the (2,1)-th element in $\Theta_{i,t}$, while keeping the other features unchanged with respect to the previous one. We let:

$$\Theta_{0,t} = \begin{bmatrix} 1 & 0 & 0 \\ \theta_{0,t} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad \Theta_{1,t} = \begin{bmatrix} 0 & 0 & 0 \\ \theta_{1,t} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix},$$

where:

$$\theta_{0,t} = \theta_{0,t-1} + \eta_t \qquad \eta_t \sim \mathcal{N}(0,1)$$
$$\theta_{1,t} = \theta_{1,t-1} + \nu_t \qquad \nu_t \sim \mathcal{N}(0,1)$$

with $\theta_{0,0} = 1$ and $\theta_{1,0} = 0.3$. The total sample size is $T = 50$. We generate the DGP as:

$$\begin{bmatrix} y_{i,t} \\ z_{i,t} \end{bmatrix} = \Theta_{0,t}\varepsilon_{i,t} + \Theta_{1,t}\varepsilon_{i,t-1}$$

and estimate the coefficients of interest ($\Theta_{i,t}^{(2,1)}$ for $i = 0, 1$) using equation (A-28) for $N = [10, 50, 100, 200, 1000]$. Panel A in Table 3 reports the average RMSE across 1000 Monte Carlo simulations. As in the previous cases, the RMSE decreases with the sample size. Panel B in Table 3 instead reports empirical rejection rates of test statistic for a 5% significance level test[16]. The results show that the performance is satisfactory for cross-sectional samples of at least 1000 observations.

Table 3. Monte Carlo Simulations: Random Walk

	Panel A. Average RMSE				
	N = 10	N = 50	N = 100	N = 200	N = 1000
h = 0	8.5380	1.2639	0.7389	0.4121	0.1959
h = 1	5.5092	1.7240	0.4370	0.3691	0.1457

	Panel B. Empirical Rejection Rates				
	N = 10	N = 50	N = 100	N = 200	N = 1000
h = 0	0.1061	0.0581	0.0553	0.0528	0.0484
h = 1	0.1049	0.0596	0.0532	0.0503	0.0499

Notes: The Table reports, for various horizons h and cross-section dimensions N, the average Root Mean Squared Error (RMSE) (panel A) and the average empirical rejection rates for tests with nominal size of 5% (Panel B).

D.4 DGP With One-Time Structural Break

The fourth DGP considers a one-time structural break for the (2,1)-th element in $\Theta_{i,t}$, where the break date is known. In particular, following the setup of the previous sub-sections, we set the coefficient matrices as follows:

- If $t \leq 20$

$$\Theta_{0,t} = \begin{bmatrix} 1 & 0 & 0 \\ \theta_{0,1} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad \Theta_{1,t} = \begin{bmatrix} 0 & 0 & 0 \\ \theta_{1,1} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

  with $\theta_{0,1} = 1$ and $\theta_{1,1} = 0.5$;

[16]Note that the empirical rejection frequencies we report in the tables are the average over time of the rejections at each point in time, since for each horizon h we have T estimates.

- If $t > 20$

$$\Theta_{0,t} = \begin{bmatrix} 1 & 0 & 0 \\ \theta_{0,2} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad \Theta_{1,t} = \begin{bmatrix} 0 & 0 & 0 \\ \theta_{1,2} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

with $\theta_{0,2} = 0.5$ and $\theta_{1,2} = -0.5$.

The total sample size is $T = 50$. We generate the DGP as:

$$\begin{bmatrix} y_{i,t} \\ z_{i,t} \end{bmatrix} = \Theta_{0,t}\varepsilon_{i,t} + \Theta_{1,t}\varepsilon_{i,t-1}$$

and estimate the coefficients of interest ($\Theta_{i,t}^{(2,1)}$ for $i = 0, 1$) using equation (A-28) for $N = [10, 50, 100, 200, 1000]$. Panel A in Table 4 reports the average RMSE across 1000 Monte Carlo simulations. Again, the RMSE decreases as the sample size increases. Panel B in Table 4 instead reports empirical rejection rates of the test statistic for a nominal 5% significance level test. The results show that the performance is satisfactory for cross-sectional samples of at least 1000 observations.

Table 4. Monte Carlo Simulations:
One-Time Structural Break

	Panel A. Average RMSE				
	N = 10	N = 50	N = 100	N = 200	N = 1000
h = 0	1.4400	0.2205	0.1504	0.1042	0.0463
h = 1	2.5295	0.2365	0.1625	0.1128	0.0498

	Panel B. Empirical Rejection Rates				
	N = 10	N = 50	N = 100	N = 200	N = 1000
h = 0	0.1057	0.0585	0.0549	0.0518	0.0505
h = 1	0.1030	0.0556	0.0539	0.0518	0.0501

Notes: The Table reports, for various horizons h and cross-section dimensions N, the average Root Mean Squared Error (RMSE) (panel A) and the average empirical rejection rates for tests with nominal size of 5% (Panel B).

In most applications, imposing a constant relationship between the shock and the instrument is quite restrictive. Therefore, we perform the same exercise as above, allowing for a structural break also in the correlation between the shock of interest and the instrument. That is,

- For $t \leq 20$:

$$E\left[z_{i,t}\varepsilon_{i,t}^{(1)}\right] = \kappa_1$$

- For $t > 20$

$$E\left[z_{i,t}\varepsilon_{i,t}^{(1)}\right] = \kappa_2$$

with $\kappa_1 = 0.6$ and $\kappa_2 = -0.6$. The rest of the characteristics of the DGP are left unchanged. Again, we estimate the coefficients of interest using the IV formula in equation (A-28) for $N = [10, 50, 100, 200, 1000]$. Panel A in Table 5 reports the average RMSE across 1000 Monte Carlo simulations while Panel B reports the empirical rejection rates of the test statistic for a nominal 5% significance level test. The results show that the performance is satisfactory for cross-sectional samples of at least 1000 observations.

Table 5. Monte Carlo Simulations:
One-Time Structural Break in $\Theta_{i,t}^{(2,1)}$ and κ_t

	Panel A. Average RMSE				
	N = 10	N = 50	N = 100	N = 200	N = 1000
$h = 0$	1.7257	0.2207	0.1495	0.1047	0.0465
$h = 1$	2.1637	0.2376	0.1616	0.1125	0.0496

	Panel B. Empirical Rejection Rates				
	N = 10	N = 50	N = 100	N = 200	N = 1000
h = 0	0.1076	0.0583	0.0549	0.0534	0.0514
h = 1	0.1032	0.0584	0.0519	0.0507	0.0480

Notes: The Table reports, for various horizons h and cross-section dimensions N, the average Root Mean Squared Error (RMSE) (panel A) and the average empirical rejection rates for tests with nominal size of 5% (Panel B).

D.5 DGP with One-time Structural Break and VMA(10)

A fifth DGP considers a Vector Moving Average process with 10 lags to investigate the performance of the proposed procedures at relatively longer horizons. Let:

$$\begin{bmatrix} y_{i,t} \\ z_{i,t} \end{bmatrix} = \sum_{j=0}^{9} \Theta_{j,t}\varepsilon_{i,t-j}$$

where each of the matrices $\Theta_{j,t}$ display a one-time structural break in their (2,1) coefficient. As in the previous subsections, $T = 50$, $N = [10, 50, 100, 200, 1000]$, and break date (which is 20) is known. The pattern of time variation in $\Theta_{j,t}$ is as follows:

- For $t \leq 20$:

$$\Theta_{0,t} = \begin{bmatrix} 1 & 0 & 0 \\ \theta_{0,1} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad \Theta_{j,t} = \begin{bmatrix} 0 & 0 & 0 \\ \theta_{j,1} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \text{for} \quad j = \{1, \ldots, 9\}$$

  with $\theta_{j,1} = [1.5, 1.2, 0.8, 1, 1.4, 1, 0.7, 0.5, 0.8, 0.2]'$ for $j = \{0, \ldots, 9\}$;

- For $t > 20$:

$$\Theta_{0,t} = \begin{bmatrix} 1 & 0 & 0 \\ \theta_{0,2} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad \Theta_{j,t} = \begin{bmatrix} 0 & 0 & 0 \\ \theta_{j,2} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \text{for} \quad j = \{1, \ldots, 9\}$$

  with $\theta_{j,2} = [0.8, 0.7, 0.2, -0.2, 0.5, 1, 1.3, 0.9, 0.2, -0.2]'$ for $j = \{0, \ldots, 9\}$.

Panel A in Table 6 reports the average RMSE of the TVP-PLP estimator across 1000 simulations, computed period by period, then averaged over time. Again, the RMSE decreases substantially as the cross section dimension increases. Panel B in Table 6 reports empirical rejection rates for tests of nominal size 5% and shows that the empirical rejection rates are very close to the nominal level as the cross section dimension is at least 50 observations.

Table 6. Monte Carlo Simulations:
One-time Parameter Break at Longer Horizons

	Panel A. Average RMSE				
	N = 10	N = 50	N = 100	N = 200	N = 1000
h = 0	3.0179	0.4987	0.3419	0.2395	0.1052
h = 1	3.1657	0.5076	0.3474	0.2425	0.1072
h = 2	3.1717	0.5282	0.3581	0.2502	0.1102
h = 3	3.0177	0.5255	0.3530	0.2482	0.1095
h = 4	3.2031	0.5062	0.3431	0.2406	0.1065
h = 5	2.8937	0.4819	0.3279	0.2297	0.1008
h = 6	2.9879	0.4543	0.3106	0.2145	0.0946
h = 7	3.0727	0.4850	0.3279	0.2287	0.1007
h = 8	3.1238	0.5031	0.3444	0.2397	0.1061
h = 9	3.0969	0.4995	0.3429	0.2389	0.1056

	Panel B. Empirical Rejection Rates				
	N = 10	N = 50	N = 100	N = 200	N = 1000
h = 0	0.1040	0.0576	0.0572	0.0546	0.0519
h = 1	0.1045	0.0576	0.0553	0.0534	0.0524
h = 2	0.1061	0.0586	0.0538	0.0531	0.0508
h = 3	0.1047	0.0580	0.0528	0.0523	0.0502
h = 4	0.1055	0.0595	0.0542	0.0535	0.0526
h = 5	0.1036	0.0586	0.0548	0.0536	0.0520
h = 6	0.1040	0.0589	0.0559	0.0523	0.0517
h = 7	0.1052	0.0589	0.0529	0.0527	0.0499
h = 8	0.1089	0.0553	0.0539	0.0534	0.0506
h = 9	0.1051	0.0584	0.0564	0.0534	0.0504

Notes: The Table reports, for various horizons h and cross-section dimensions N, the average Root Mean Squared Error (RMSE) (Panel A) and the average empirical rejection rates of tests with 5% nominal size (panel B).

D.6 DGP with Multivariate Instruments

The last Monte Carlo experiment we consider is inspired by the empirical analysis of Section 4. We allow the instrument $z_{i,t}$ to be multi-dimensional. We consider the same sample size as in our empirical application, which starts in 2018 and ends in 2023; thus $T = 6$ and $N = 2000$, close to the size of our cross-section. The

DGP is as follows:

$$\begin{bmatrix} \mathbf{y}_{i,t} \\ \mathbf{z}_{i,t} \end{bmatrix} = \Theta_{0,t}\varepsilon_{i,t} + \Theta_{1,t}\varepsilon_{i,t-1} \tag{A-46}$$

$$\begin{bmatrix} y_{i,t}^{(1)} \\ y_{i,t}^{(2)} \\ \mathbf{z}_{i,t} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ \theta_{0,t} & 1 & 0 \\ 0 & 0 & \mathbf{I}_3 \end{bmatrix} \varepsilon_{i,t} + \begin{bmatrix} 0 & 0 & 0 \\ \theta_{1,t} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \varepsilon_{i,t-1},$$

where, for simplicity, the vector of endogenous variables $\mathbf{y}_{i,t}$ is of dimension (2x1), the vector of instruments $\mathbf{z}_{i,t}$ is of dimension (3x1) and $\mathbf{I}_3$ is an identity matrix of dimension 3.[17] We focus our analysis on the parameter $\Theta_{h,t+h}^{(2,1)}$, which we allow to be time-varying. We choose the time of the break to match the start of the COVID-19 pandemic:

$$\Theta_{0,t}^{(2,1)} = \theta_{0,t} = \begin{cases} 1.5 \quad \text{if} \quad t \leq 2019 \\ 1 \quad \text{otherwise} \end{cases}$$

$$\Theta_{1,t+1}^{(2,1)} = \theta_{1,t+1} = \begin{cases} 0.5 \quad \text{if} \quad t \leq 2019 \\ 0 \quad \text{otherwise} \end{cases}$$

The vector of instruments $\mathbf{z}_{i,t}$ is constructed as follows. Let:

$$\varepsilon_{i,t} = \begin{bmatrix} \varepsilon_{i,t}^{(1)} \\ \varepsilon_{i,t}^{(2)} \\ \varepsilon_{i,t}^{(3)} \\ \varepsilon_{i,t}^{(4)} \\ \varepsilon_{i,t}^{(5)} \end{bmatrix} \overset{iid}{\sim} \mathcal{N}\left(\begin{pmatrix} 0 \\ 0 \\ 1 \\ -1.33e-05 \\ 0.02 \end{pmatrix}, \begin{pmatrix} 1 & 0 & -0.8 & 0.5 & -0.65 \\ 0 & 1 & 0 & 0 & 0 \\ -0.8 & 0 & 2 & 0 & 0 \\ 0.5 & 0 & 0 & 1.3 & 0 \\ -0.65 & 0 & 0 & 0 & 2.1 \end{pmatrix} \right),$$

then $z_{i,t}^{(1)} = \varepsilon_{i,t}^{(3)}$, $z_{i,t}^{(2)} = \varepsilon_{i,t}^{(4)}$ and $z_{i,t}^{(3)} = \varepsilon_{i,t}^{(5)}$. We set

$$\mathbb{E}(\mathbf{z}_{i,t}\varepsilon_{i,t}^{(1)}) = \boldsymbol{\kappa} = [-0.8, 0.5, -0.65]'$$

.

[17] The data generating process is inspired by Nakamura and Steinsson (2014). In their first stage, they use national spending interacted with a state dummy, state and year fixed effects, resulting in 139 instruments. We use one for each category of their instruments.

Then, we simulate the DGP as in equation (A-46), where:

$$\Theta_{0,t} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ \theta_{0,t} & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \qquad \Theta_{1,t} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ \theta_{1,t} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

This way, we impose the unit normalization (for every t) in $\Theta_{0,t}^{(1,1)}$. The estimate of $\theta_{h,t}$ is constructed using the optimal weighting matrix; the optimal weighting matrix is obtained from a first-step estimate of $\theta_{h,t}$ using an identity weighting matrix. Table 7 shows that the performance of the estimator is satisfactory even in this more realistic and complex DGP, yet simple enough to implement in a Monte Carlo simulation, with empirical rejection frequencies very similar to the nominal 5% rate.

Table 7

	Average RMSE	**Empirical Rejection Rates**
h = 0	0.0217	0.0477
h = 1	0.0254	0.0518

Notes: The Table reports, for horizons $h = 0, 1$, $N = 2000$ and $T = 6$, the average Root Mean Squared Error (RMSE) (second column) and the average empirical rejection rates for tests with nominal size of 5% (third column).